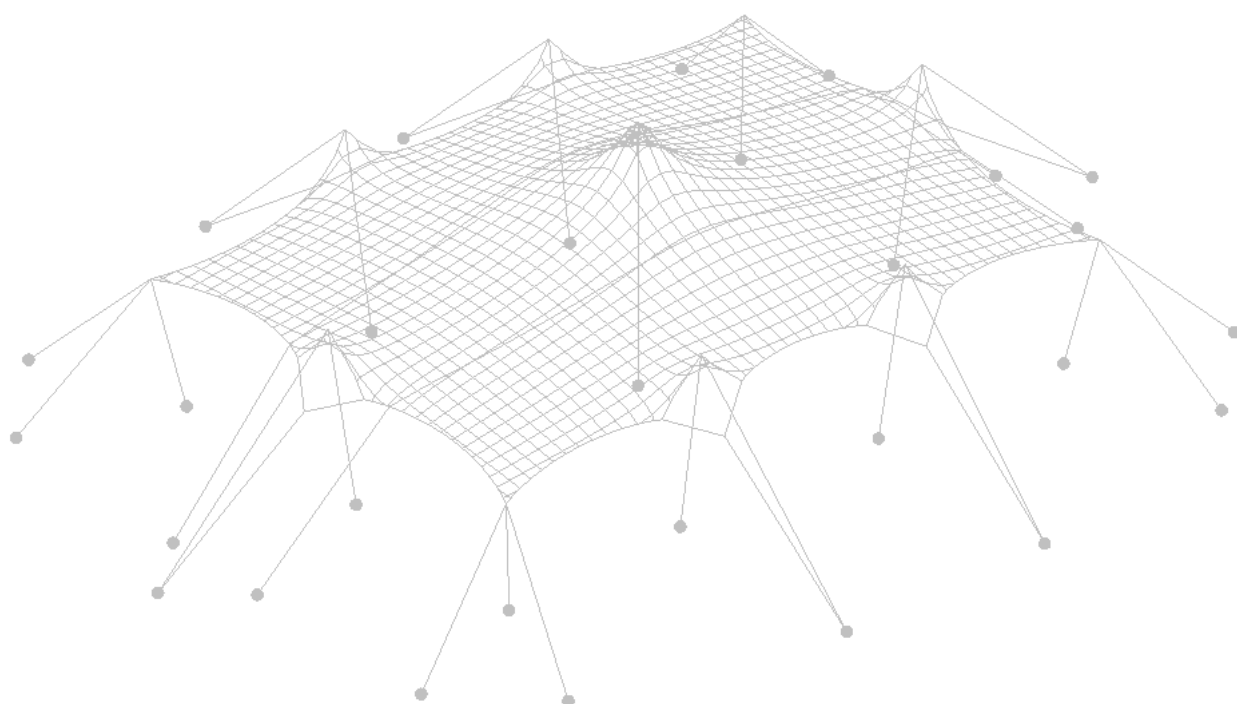




Content: **Tentbook (according to EN 13782)**

Object: **7.5m stretchtent and alternatives**

Owner object: **Eurostretch B.V.**

Document code: **26.02.00026.2**

Author: **S. Banga**

Date: **10.08.2026**

Object: 7.5m stretchtent and alternatives

Document code: 26.02.00026.2

Owner object: Eurostretch B.V.
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Valid date: 10.08.2031

A handwritten signature in black ink, appearing to read 'S. Banga'.

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Authorized by: ir. R. Houtman

A. Introduction

Eurostretch B.V. has commissioned Tentech B.V. to perform a static analysis of a stretch tent. Eurostretch B.V. is a Dutch company that specializes in rental and selling of tent structures made of a stretchable membrane, so called Stretch Tents or Bedouin tents. The stretchable membrane enables a freedom of form as there is not a pre-described shape necessary. Depending on the location, the number of poles, length of the poles, and the placement of the poles, number and type of ties can be varied. This results in a custom canopy at each location.

The freedom of form is enabled by a stretchable fabric, as the desired form is stretched from a rectangular fabric. The drawback of this flexibility in shape is the difficulty of investigating all the different possibilities and generalizing them for a static analysis.

This report presents the static analysis of a 7.5m stretch tent series which consists of different sizes – **7.5x10, 7.5x15, and 6.5x10m**. The report shows the static analysis of the decisive sizes – 7.5x10m and 7.5x15m. The static analysis of tent 7.5x10m is valid for a smaller size of 6.5x10m due to the application of the **scaling principle**. All models and variants are investigated in a “floating” set-up – membrane corners and sides are connected above ground, primarily to poles.

Scaling principle: A tent with a similar arrangement of poles and ties but smaller in size or having a more favourable shape is automatically verified when the calculation of the less favourable or bigger tent has been approved, provided that the spacing between the poles and ties of the scaled tent is similar or smaller than the original tent. While constructing smaller tent sizes of different shapes, it is significant to use similar (or more favourable) arrangements of poles and ties based on ratio of distances (to ensure correct pole and tie spacing).

This document contains the data required for a tent book, according to EN 13782, for the 7.5m tent series of Eurostretch B.V., including:

- Ownership data;
- Drawings of the different variants of the tent, including dimensions, indications of elements and required anchoring/ballast;
- Permitted live load;
- Maximum wind speeds (according to EN 1991-1-4:2005);
- Structural analysis (according to EN 13782:2015);
- Material certificates (strength properties and fire properties).

Utrecht, S. Banga, 10.08.2026

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C. Codes and standards

- EN 1990 Eurocode, Basis of structural Design
- EN 1991 Eurocode 1, Part 1-4: General actions - wind actions.
- EN 1995 Eurocode 5, Design of wooden structures
- EN 13782 Temporary Structures – Tents - Safety
- EN 12195-2 Belts made from synthetic fibres
- ISO 1141 Fibre ropes – Polyester
- CTS French standard - Chapiteaux, Tentes, Structures

D. Summary

Manufacturer:	Eurostretch B.V. Akkermansbeekweg 15A 7061 ZA Terborg Nederland info@eurostretch.com	
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This document is valid for three tent sizes: 7.5x10m, 7.5x15m and 6.5x10m. Although no separate calculation is made for 6.5x10m size, all elements, sizes and anchoring forces can be based on 7.5x10m model.

General information:

Main dimensions:		7.5x10m / 6.5x10m	7.5x15m
Width:		7.5/6.5m	7.5m
Length:		10m	15m
Side height of membrane:		3m	3m
Max. height:		4m	4m
Main poles: *		4.0m Ø 95 mm [Spruce, C18]	4.0m Ø 95 mm [Spruce, C18]
Side poles: *		3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]
Corner poles: *		2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
Stormbelts:		50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
Guy ropes:		Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **
Long side – combined:		3 sections	3 sections
Long side - V-ropes:		2 sections	2 sections
Short side – combined:		3 sections	3 sections
Short side - V-ropes:		2 sections	2 sections
Belts:		50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
Fasteners:		Carabine 180x14	

	Side & corner pole base plate	M16 & M12 Eye bolt Ratchet, BL = 3.5 kN 25x25cm	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN 25x25cm
	Main pole base plate	30x30cm	30x30cm
User load:	A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.		
Snow load:	A snow load of 0.1 kN/m ² (4 cm) has been considered in accordance with the French CTS standard.		
Wind load:	According to EN 13782 The following pressure values have been used: $h < 5m$: 0.50kN/m ² This value corresponds to wind force 8 on the Beaufort scale. Above this wind force (starting at wind force 9), the safety of the structure can no longer be guaranteed, and it must be evacuated.		

* The average diameter, as a minimum required at the middle of the pole.

** The required breaking load (BL_{tot}) may be achieved by using multiple rope sections (n): $BL_{rope} = BL_{tot} / n$, see page 47

Wind speeds for default European terrain categories (not Country specific)

Out of service ⁽¹⁾					
	Coast	Flattened, open area	Rural area	Village	City
10 min. average wind speed ⁽²⁾	> 17.94 m/s > 64.57 km/h	> 18.89 m/s > 67.99 km/h	> 21.08 m/s > 75.88 km/h	> 24.99 m/s > 89.97 km/h	> 26.08 m/s > 93.89 km/h
Beaufort ⁽³⁾	> 7 BFT	> 7 BFT	> 8 BFT	> 9 BFT	> 9 BFT
Peak wind speed ⁽⁴⁾	> 113 km/h	> 113 km/h	> 113 km/h	> 113 km/h	> 113 km/h

(1) 'Out of service' means: above the given wind speed the structure is no longer guaranteed regarding strength and/or stability.

(2) 10min average wind speed at 10m height measured at the nearest weather stations.

(3) wind data in Beaufort (BFT) are indicative values.

(4) 3 second peak wind speed measured on site at 10m height.

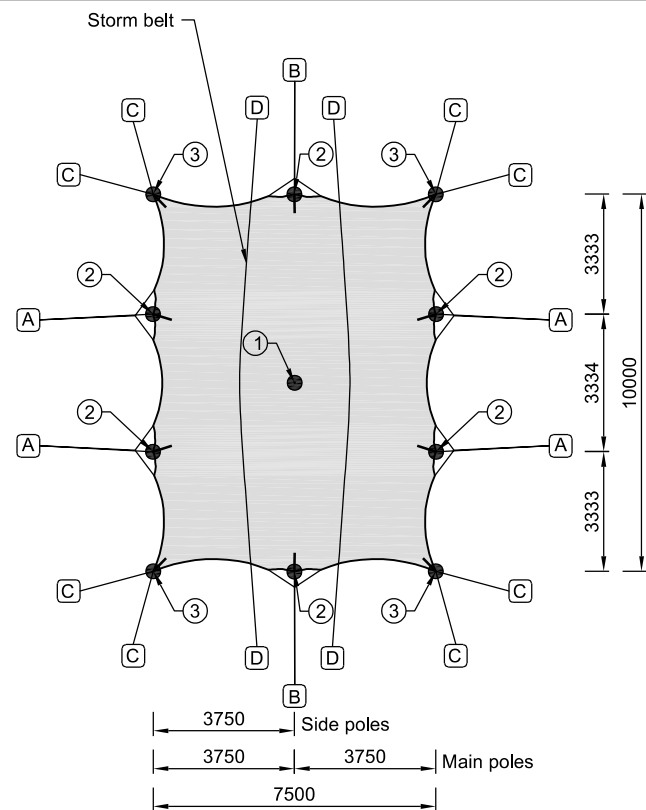
Safety against sliding, overturning and uplifting:

Anchor forces:	Assumptions: angle of 45 degrees The following design resistance of the anchor forces is required: (See H.8.3, page 56 for Anchor tests according to EN 13782)		
		7.5x10m / 6.5x10m	7.5x15m
	Long side guy ropes:	5.00 kN ($\gamma = 1.2$)	5.53 kN ($\gamma = 1.2$)
	Short side guy ropes:	4.09 kN ($\gamma = 1.2$)	4.93 kN ($\gamma = 1.2$)
	Corner ties:	4.32 kN ($\gamma = 1.2$)	4.86 kN ($\gamma = 1.2$)
Required anchors:	Stormbelt:	5.10 kN ($\gamma = 1.2$)	4.67 kN ($\gamma = 1.2$)
	Based on dense, non-cohesive soil (e.g. sandy soils). In the case anchors Ø30mm x 1000mm (effective length) are being used:		
		7.5x10m / 6.5x10m	7.5x15m
	Long side guy ropes:	1 anchor	2 anchors
	Short side guy ropes:	1 anchor	1 anchor
Required ballast:	Corner ties:	1 anchor	1 anchor
	Stormbelt:	1 anchor	1 anchor
	Ballast calculation is performed for 2 different situations:		
	- Concrete on concrete – concrete ballasts are placed on concrete - Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast		
	7.5x10m / 6.5x10m		
		Concrete on concrete	Concrete on wood or rubber
	Long side guy ropes:	1060 kg	940 kg
	Short side guy ropes:	870 kg	770 kg
	Corner ties:	920 kg	820 kg
	Stormbelt:	1080 kg	960 kg
	7.5x15m		
		Concrete on concrete	Concrete on wood or rubber
	Long side guy ropes:	1170 kg	1040 kg
	Short side guy ropes:	1050 kg	930 kg
	Corner ties:	1030 kg	920 kg
	Stormbelt:	990 kg	880 kg

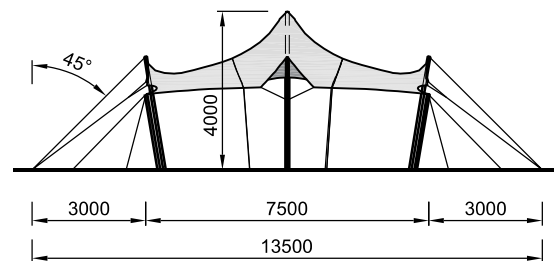
E. Drawings: main dimensions and anchoring

Drawings for the following sizes are presented in this chapter:

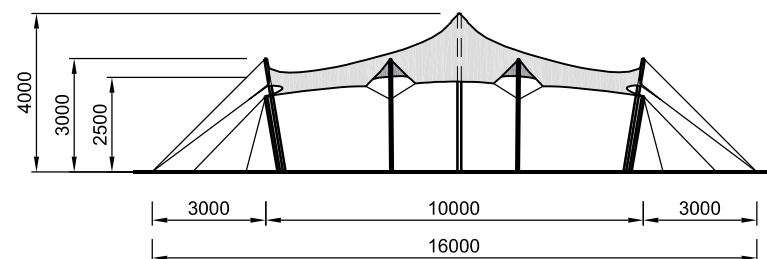
- 7.5x10m model
- 7.5x15m model
- 6.5x10m model (based on 7.5x10m model)



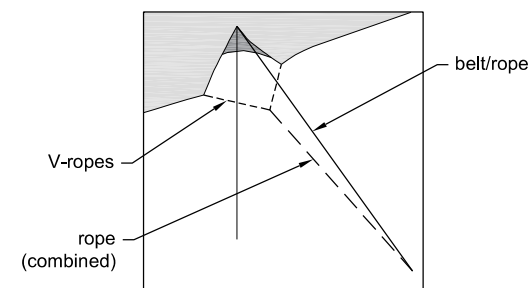
TOP VIEW
[1:200]



FRONT VIEW
[1:200]



SIDE VIEW
[1:200]



GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting

Based on dense, non-cohesive soil (e.g. sandy soils).

	Anchors (Ø30x1000mm)	Ballast (concrete on concrete)	Ballast (concrete on wood or rubber)
A Guying - long side	1x anchors	1060 kg	940 kg
B Guying - short side	1x anchors	870 kg	770 kg
C Guying - corner	1x anchors	920 kg	820 kg
D Storm belt (per end)	1x anchors	1080 kg	960 kg

All belts/ropes and storm belts of the membrane must be placed under a 45° angle of $\geq 45^\circ$ be attached to the anchors.

Poles

Main poles 4.0m:

① - Ø95mm [C18, Spruce]

Side poles 3.0m:

② - Ø75mm [C18, Spruce]

Corner poles 2.5m:

③ - Ø70mm [C18, Spruce]

Guying

Storm belt: [PES] BL 3000 daN = LC 1000 daN

Guying belt: [PES] BL 3000 daN = LC 1000 daN

Guying rope: rope: Ø 8mm polyester, breaking strength ≥ 900 daN

Combined & v-ropes: Rope

n - number of sections n

Guying - long side (combined) 3x

Guying - long side (v-ropes) 2x

Guying - short side (combined) 3x

Guying - short side (v-ropes) 2x

Belts (alternative rope):

n - number of sections (number) n

Guying - long side 1x 1x

Guying - short side 1x 1x

Guying - corner 1x 3x

Alternative configuration

7.5 x 10 m
7.5 x 15 m

* Basic
configurations

6.5 x 10 m

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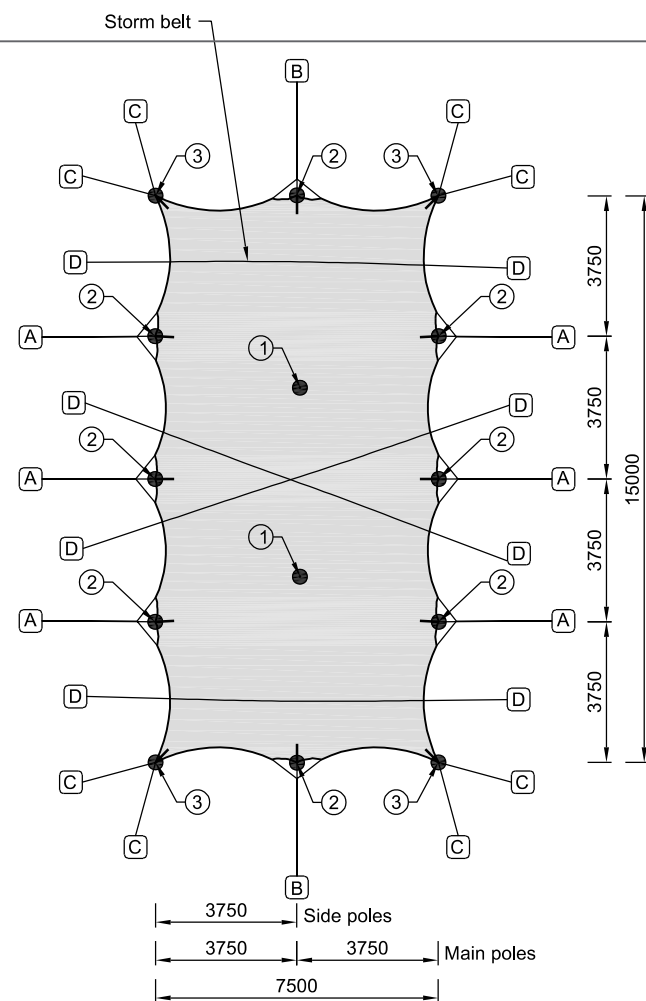
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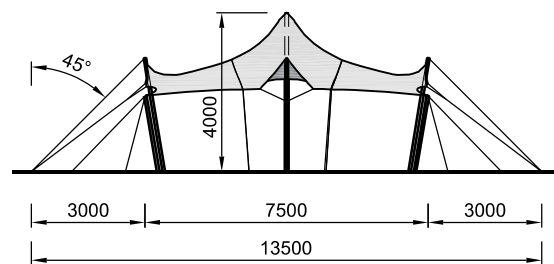
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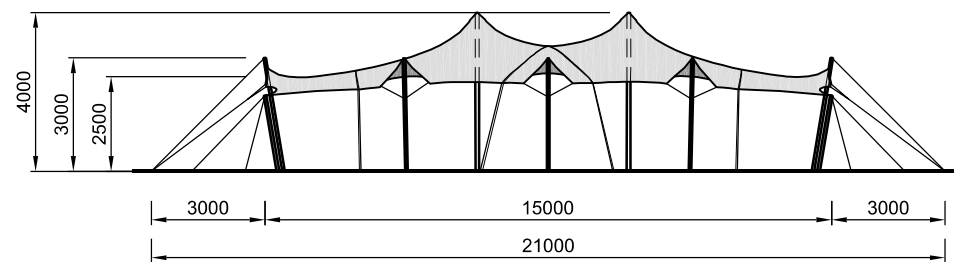
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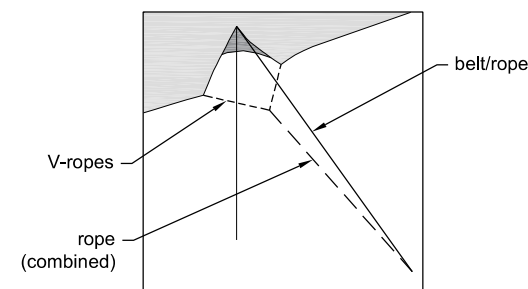
TOP VIEW
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FRONT VIEW
[1:200]



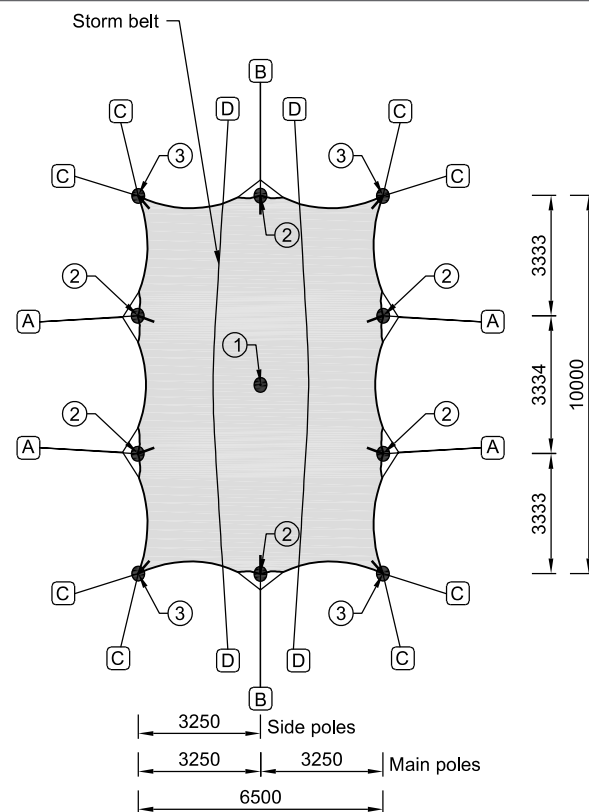
SIDE VIEW
[1:200]



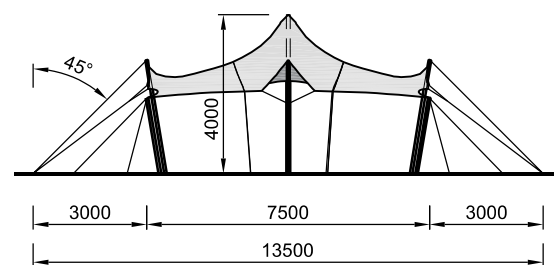
GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting				
Based on dense, non-cohesive soil (e.g. sandy soils).				
	<u>Anchors</u> (Ø30x1000mm)	<u>Ballast</u> (concrete on concrete)	<u>Ballast</u> (concrete on wood or rubber)	
A	Guying - long side	2x anchors	1170 kg	1040 kg
B	Guying - short side	1x anchors	1050 kg	930 kg
C	Guying - corner	1x anchors	1030 kg	920 kg
D	Storm belt (<i>per end</i>)	1x anchors	990 kg	880 kg
All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.				
Poles				
Main poles 4.0m:				
① - Ø95mm [C18, Spruce]				
Side poles 3.0m:				
② - Ø75mm [C18, Spruce]				
Corner poles 2.5m:				
③ - Ø70mm [C18, Spruce]				
Guying				
Storm belt:		[PES] BL 3000 daN = LC 1000 daN		
Guying belt:		[PES] BL 3000 daN = LC 1000 daN		
Guying rope:		rope: Ø 8mm polyester, breaking strength ≥ 900 daN		
<u>Combined & v-ropes:</u>		<u>Rope</u>		
n - number of sections		n		
Guying - long side (combined)		3x		
Guying - long side (v-ropes)		2x		
Guying - short side (combined)		3x		
Guying - short side (v-ropes)		2x		
<u>Belts (alternative rope):</u>		<u>Belt</u>	<u>Rope</u>	
n - number of sections		(number)	n	
Guying - long side		1x	1x	
Guying - short side		1x	1x	
Guying - corner		1x	3x	
Alternative configuration				
7.5 x 10 m		6.5 x 10 m		
7.5 x 15 m				
* Basic configurations				

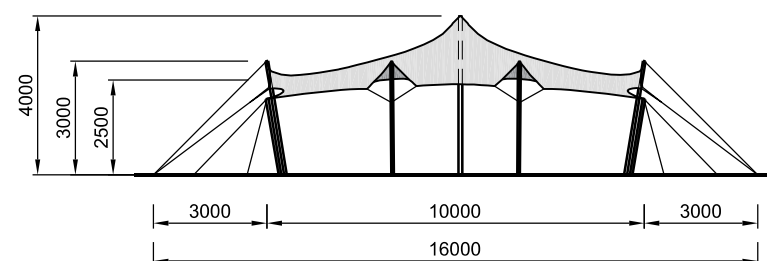
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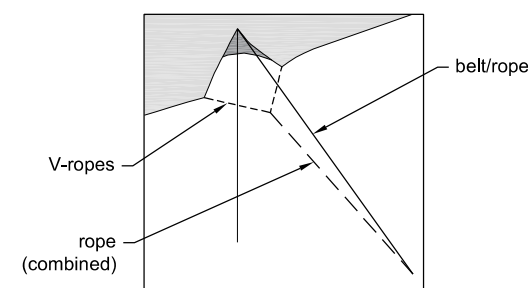
TOP VIEW
[1:200]



FRONT VIEW
[1:200]



SIDE VIEW
[1:200]



GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting

Based on dense, non-cohesive soil (e.g. sandy soils).

	Anchors (Ø30x1000mm)	Ballast (concrete on concrete)	Ballast (concrete on wood or rubber)
A Guying - long side	1x anchors	1060 kg	940 kg
B Guying - short side	1x anchors	870 kg	770 kg
C Guying - corner	1x anchors	920 kg	820 kg
D Storm belt (per end)	1x anchors	1080 kg	960 kg

All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.

Poles

Main poles 4.0m:

① - Ø95mm [C18, Spruce]

Side poles 3.0m:

② - Ø75mm [C18, Spruce]

Corner poles 2.5m:

③ - Ø70mm [C18, Spruce]

Guying

Storm belt: [PES] BL 3000 daN = LC 1000 daN

Guying belt: [PES] BL 3000 daN = LC 1000 daN

Guying rope: rope: Ø 8mm polyester, breaking strength ≥ 900 daN

Combined & v-ropes:

	Rope
n - number of sections	n
Guying - long side (combined)	3x
Guying - long side (v-ropes)	2x
Guying - short side (combined)	3x
Guying - short side (v-ropes)	2x

Belts (alternative rope):

	Belt (number)	Rope n
n - number of sections		
Guying - long side	1x	1x
Guying - short side	1x	1x
Guying - corner	1x	3x

Alternative configuration

7.5 x 10 m	6.5 x 10 m
7.5 x 15 m	

* Basic
configurations

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F. Important terms and conditions

This document applies to the built structure if the following principles and conditions are met:

- The used materials, parts and sections (membrane, poles, ties, anchoring) are in accordance with this document.
- The dimensions of the built structure match the dimensions stated in this document.
- Parts (poles, ties, anchors) may not be removed.
- Obstacles should be placed at least 0.5m from the membrane (measured perpendicular to the fabric); The fabric needs a certain freedom to deform in all directions to prevent damages caused by collision with objects located closely to the fabric (see also EN 13782, article 8.7).
- Above the maximum allowable wind speeds (see D) the structure should be evacuated and access for the public must be denied.
- Only decorations, music- and light installations of less than 25 kg per pole, can be attached to the structure. They must be attached centrally.
- A conventional load of 0.1 kN/m^2 is considered according to EN 13782, which corresponds with the required snow load (4cm) according to the French CTS.
- Anchoring is based on dense, non-cohesive soil. When soil differs, additional anchoring might be necessary, or anchor tests need to be performed.
- Poles must be stored dry. If they become wet, there needs to be a possibility for them to dry.
- All water accumulation must be removed as soon as possible.

G. Allowed wind speeds

The maximum wind speed is converted into a basic wind speed for a coastal area, flattened/open area, rural area, village and city according to EN 1991-1-4. Terrain roughness is taken according to the recommended general values for the different terrain categories for Europe. Illustrations of these terrain categories are presented on page 16.

Full wind load		
Wind pressure p_w	500 N/m ²	At 5m height
Corresponding peak wind pressure $p_{w,peak}$	605 N/m ²	At 10m height

Peak wind speed at 10m height

Equation:

$$605 = \frac{1}{2} \times \rho \times v^2 = \frac{1}{2} \times 1.25 \times v^2 \rightarrow v = 31.1 \text{ m/s} \rightarrow \pm 113 \text{ km/h}$$

Eq. 4.10 EN 1991-1-4
Basic wind pressure

Wind speed coastal area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.003}{0.05}\right)^{0.07} = 0.156$$

Eq. 4.5 EN 1991-1-4
Terrain factor for coastal area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.156 \times \ln\left(\frac{4.0}{0.003}\right) = 1.123$$

Eq. 4.4 EN 1991-1-4
Roughness factor
 $Z = 4.0 > Z_{min} = 1$

$$V_m = C_r \times V_b = 1.123 \times V_b$$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.141 \times V_b$$

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.156 \times V_b}{1.158 \times V_b} = 0.139$$

Eq. 4.7 EN 1991-1-4
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.554 \times V_b^2$$

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Equation:

$$500 = 1.554 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 17.94 \text{ m/s}$$

Characteristic wind speed

Wind speed flattened, open area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.01}{0.05}\right)^{0.07} = 0.170$$

Eq. 4.5 EN 1991-1-4
Terrain factor for coastal area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.170 \times \ln\left(\frac{4.0}{0.01}\right) = 1.017$$

Eq. 4.4 EN 1991-1-4
Roughness factor at 3.5m height
 $Z = 4.0 > Z_{min} = 1$

$$V_m = C_r \times V_b = 1.017 \times V_b$$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.170 \times V_b$$

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.170 \times V_b}{1.017 \times V_b} = 0.167$$

Eq. 4.7 EN 1991-1-4
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.402 \times V_b^2$$

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Equation:

$$500 = 1.402 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 18.89 \text{ m/s}$$

Characteristic wind speed

Wind speed rural area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.05}{0.05}\right)^{0.07} = 0.190$$

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.190 \times \ln\left(\frac{4.0}{0.05}\right) = 0.833$$

$$V_m = C_r \times V_b = 0.833 \times V_b$$

$$\sigma_v = K_r \times V_b = 0.190 \times V_b$$

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.190 \times V_b}{0.833 \times V_b} = 0.228$$

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.125 \times V_b^2$$

Equation:

$$500 = 1.125 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 21.08 \text{ m/s}$$

Eq. 4.5 EN 1991-1-4
Terrain factor for unbuilt area

Eq. 4.4 EN 1991-1-4
Roughness factor at 4m height
 $Z = 4.0 > Z_{\min} = 2$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

Eq. 4.7 EN 1991-1-4
Turbulence intensity

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Characteristic wind speed

Wind speed village at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.3}{0.05}\right)^{0.07} = 0.215$$

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.215 \times \ln\left(\frac{5}{0.3}\right) = 0.606$$

$$V_m = C_r \times V_b = 0.606 \times V_b$$

$$\sigma_v = K_r \times V_b = 0.215 \times V_b$$

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.215 \times V_b}{0.606 \times V_b} = 0.355$$

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 0.801 \times V_b^2$$

Equation:

$$500 = 0.801 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 24.99 \text{ m/s}$$

Eq. 4.5 EN 1991-1-4
Terrain factor for unbuilt area

Eq. 4.4 EN 1991-1-4
Roughness factor at 7m height
 $Z = Z_{\min} = 5$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

Eq. 4.7 EN 1991-1-4
Turbulence intensity

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Characteristic wind speed

Wind speed city at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{1}{0.05}\right)^{0.07} = 0.234$$

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.234 \times \ln\left(\frac{10}{1}\right) = 0.540$$

$$V_m = C_r \times V_b = 0.540 \times V_b$$

$$\sigma_v = K_r \times V_b = 0.234 \times V_b$$

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.234 \times V_b}{0.540 \times V_b} = 0.434$$

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 0.735 \times V_b^2$$

Equation:

$$500 = 0.735 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 26.08 \text{ m/s}$$

Eq. 4.5 EN 1991-1-4
Terrain factor for unbuilt area

Eq. 4.4 EN 1991-1-4
Roughness factor at 7m height
 $Z = Z_{\min} = 10$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

Eq. 4.7 EN 1991-1-4
Turbulence intensity

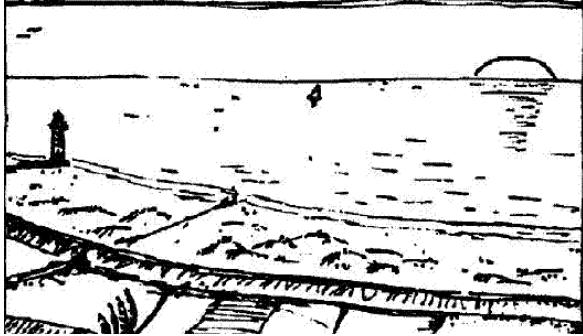
Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Characteristic wind speed

Terrain categories

Default European terrain categories (not Country specific)

0: Coastal area:



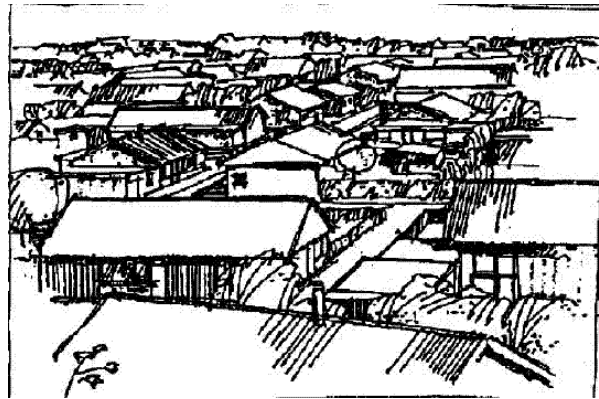
I: Flattened, open area:



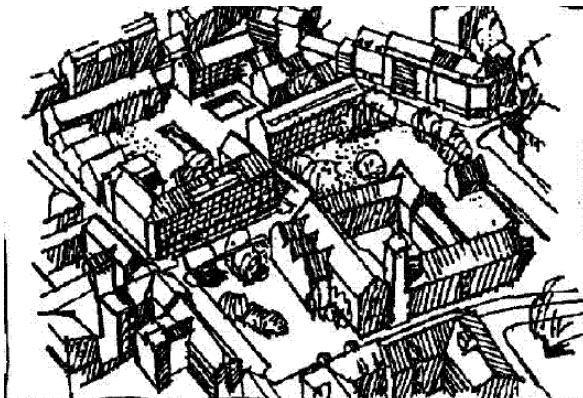
II: Rural area



III: Village



IV: City



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H.1. Project description

The principle of a stretchtent is based on a rectangular piece of stretchable membrane with a reinforcement along the boundary. The membrane is supported by poles, both at the edge and under the membrane. The side poles are stabilized by guy ropes and belts. The poles do not require a fixed position which ensures a freedom of shape.

Tentech B.V. has performed a static analysis of two tents – 7.5x10m and 7.5x15m always assumed to have stormbelts. Further chapters include material and cross-section description, calculation method, load cases and combinations, results, element checks, anchor and ballast calculations. These chapters present calculation results valid for Europe according to EN 13782.

As an exception, two additions to the static analysis are made (see I and J on pages 59 and 68 respectively):

1. For use only in Netherlands – element check for reduced wind speeds.
2. For use in France or countries following French CTS – element check for combined wind and snow loads

For use in other countries than Netherlands or France, the statical analysis must be considered without its additions. When relevant information is mentioned for these specific cases either [NL] or [FR] immediately follows this information. This information is not considered while making the base calculations. Addition J on page 68 must always be considered for use in France.

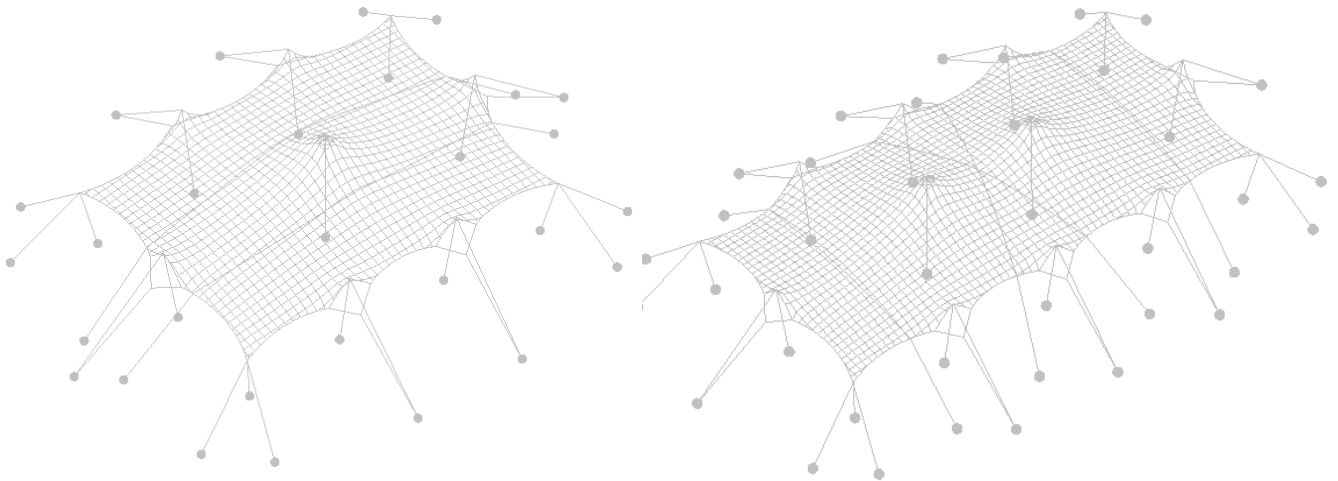


Figure 1. 7.5x10m and 7.5x15m models

H.2. Materials

H.2.1 Fabric

Design tensile strength	f_d	f_{tk} / γ_m	art 8.6 EN 13782
Characteristic tensile strength (warp)	$f_{tk, \text{warp}}$		
Characteristic tensile strength (weft)	$f_{tk, \text{weft}}$		
Material factor – global, permanent load	γ_m	2.5	tbl 4. EN13782
Material factor – global, short duration load	γ_m	2.0	tbl 4. EN13782

Table 1. Used symbols, codes and standard for fabric materials

Material	Type	Weight	$f_{rd; \text{warp}; \text{perm}}$	$f_{rd; \text{weft}; \text{perm}}$	$f_{rd; \text{warp}; \text{short}}$	$f_{rd; \text{weft}; \text{short}}$
Sioen RP5639	-	$\approx 730 \text{ gr/m}^2$	9.6 kN/m	7.2 kN/m	12.0 kN/m	9.0 kN/m

Table 2. Used fabrics

H.2.2 Guy ropes

Design resistance	F_{rd}	R_m / γ_{m1}	art 10.2. EN13782
Characteristic tensile strength	R_m		art 10.2. EN13782
Material factor	$\gamma_{m1} < 12\text{mm}$	4.0	art 10.3. EN13782
	$\gamma_{m1} > 12\text{mm}$	3.3	art 10.3. EN13782

Table 3. Used symbols, codes and standard for rope materials

Material	Cross section	Breaking strength	R_m	F_{rd}
8mm Polyester Braid rope	$\varnothing 8 \text{ mm}$	$\geq 900 \text{ daN}$	9.0 kN	2.25 kN

Table 4. Used guy ropes

H.2.3 Belts

Design resistance	F_{rd}	R_m / γ_{m1}	art 10.2. EN13782
Characteristic breaking strength	R_m	$LC \times \gamma_{m2}$	art 10.2. EN13782
Lashing capacity	LC		Conform EN 12195-2
Material factor	γ_{m1}	2.0	art 10.2. EN13782
Material factor	γ_{m2}	3.0	EN1492-1

Table 5. Used symbols, codes and standard for belt materials

Material	LC	R_m	F_{rd}
Belt [PES] EN 12195-2, 50mm	1000 daN 10 kN	3000 daN 30 kN	15 kN

Table 6. Used belts

H.2.4 Wood

Material	γ_{m1}	1.3	tbl. 2.3. EN 1995-1-1
----------	---------------	-----	-----------------------

Table 7. Used material factors

Material	Weight	$E_{0.05}$	F_{c0k}	F_{mk}
Wood, Spruce, C18	320 kg/m ³	6.0 kN/m ²	18 N/mm ²	18 N/mm ²

Table 8. Used wooden materials

H.3. Cross sections

Profile	Material	$\varnothing$ mm	G kg/m	A mm ²	I_y mm ⁴	$W_{el,y}$ mm ³
$\varnothing \approx 95$ mm *	Wood C18	95	2.26	7088	3998198	84173
$\varnothing \approx 75$ mm *	Wood C18	75	1.41	4418	1553156	41417
$\varnothing \approx 70$ mm *	Wood C18	70	1.23	3848	1178588	33674

Table 9. Used cross sections

* the average diameter, as a minimum required at the middle of the pole.

H.4. Calculation method

H.4.1 Modelling

The analysis of the structure is performed with the software package EASY FCS supplied by TECHNET GmbH, Berlin. This software is specially developed for structures with large deformability, such as membrane structures. The performed analysis is a full non-linear second order analysis.

The membrane structure is modelled in 3D. The membrane is modelled as a cable net structure and is supported by poles. These centre poles will be stabilized by the tensioned membrane. The side poles are stabilized and tied down by tension belts or ropes, which are attached to ground anchors or ballast.

The membrane edges are reinforced. They consist of multiple layers of fabric about 10cm wide with additional patches where the lugs are placed. Storm belts are required constantly and placed in the valleys between the centre poles.

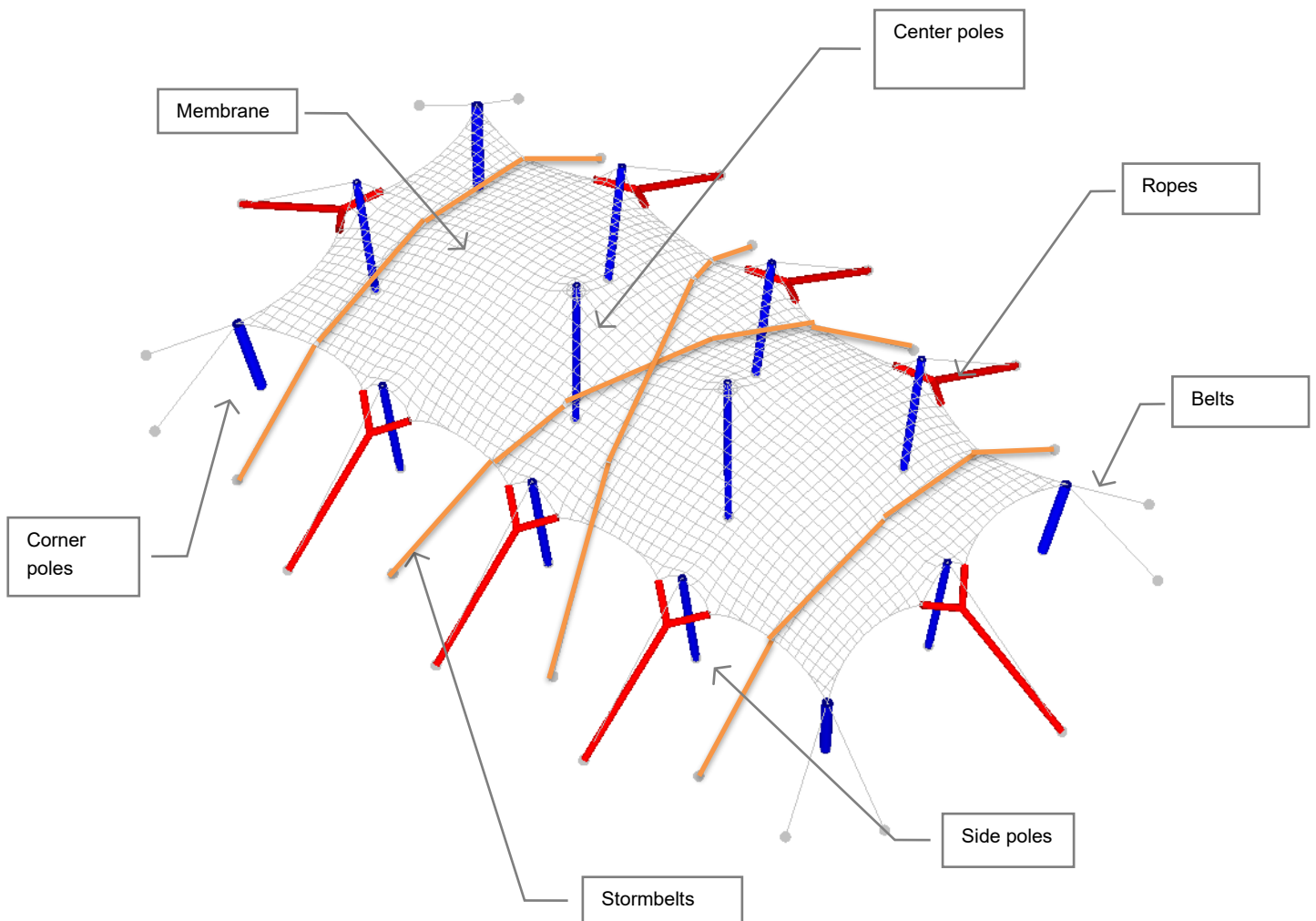


Figure 2. Calculation model

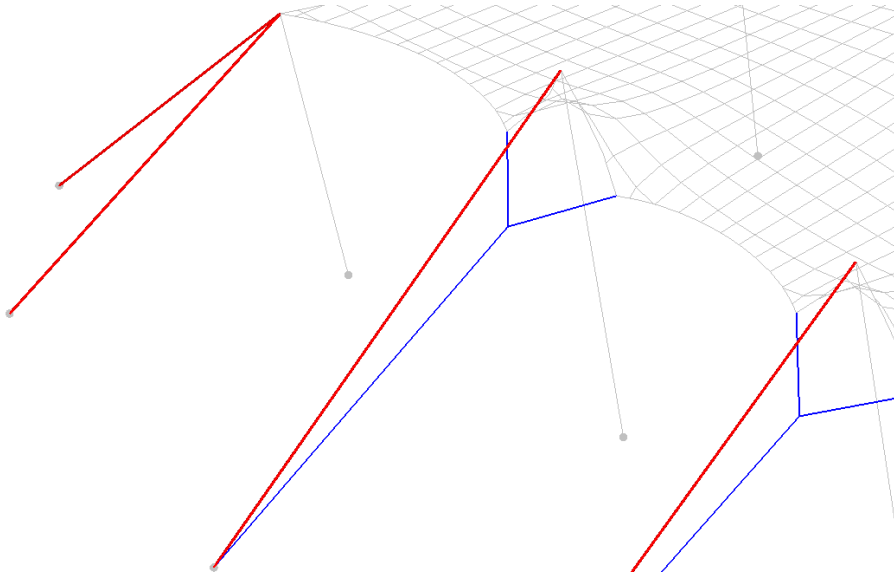


Figure 3. Principle of ties

Ties stabilize the tent and create the shape of the stretchable membrane. Guy ropes (see blue above) transfer the forces to the ground when wind suction is present. Belts (see red above) help transfer the pressure forces.

H.4.2 Structural behaviour of membrane structures

Since the fabric is a highly deformable material, it is only possible to calculate stresses and deformations with a non-linear method. FEM-software EASY is used to perform these calculations. Because of the non-linearity of the calculations the partial safety factors are not applied beforehand, since the deformations will be greater due to these safety factors, resulting in lower stresses in the fabric. See figure below.

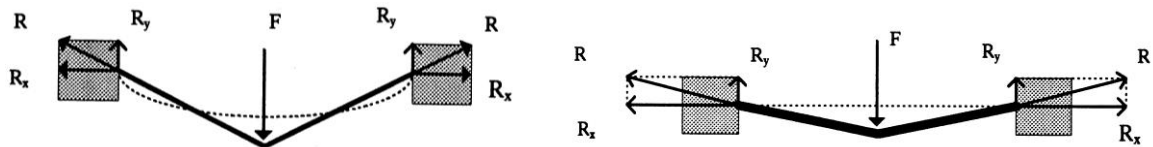


Figure 4. membrane behavior

As a membrane structure is a form-active structure, article 6.3 (4) b) of the EN-1990:2002 applies:

When the action effect increases less than the action, the partial factor γ_F should be applied to the action effect of the representative value of the action.

This means that no load factors are applied on the load beforehand, but afterwards.

H.4.3 Load combinations

H.4.3.1 Fundamental - Ultimate limit state

For determination of strength and to check elements and connections ultimate limit state is used.

	One variable load	Multiple variable loads
Unfavorable permanent load	$1.35 \times G + 1.5 \times Q$	$1.35 \times G + \sum 1.35 \times Q_i$
Favorable permanent load	$1.0 \times G + 1.5 \times Q$	$1.0 \times G + \sum 1.35 \times Q_i$

Table 10. Load combinations according to EN 13782

This means that the following load combinations will be checked/calculated:

1. 1.35 x Own weight + 1.5 x Wind load
2. 1.35 x Own weight + 1.5 x Snow load
3. 1.35 x Own weight + 1.35 x Wind load + 1.35 x Snow load [FR]

H.4.3.2 Safety against overturning, sliding and uplifting - Ultimate limit state

For determination and check of the needed ballast weight and/or anchors ultimate limit state is used

	One or multiple variable loads
Unfavorable permanent load	$1.1 \times G + 1.2 \times Q_{wind} + \sum 1.3 \times Q_i$
Favorable permanent load	$1.0 \times G + 1.2 \times Q_{wind} + \sum 1.3 \times Q_i$

Table 11. Load combinations according to EN 13782

This means the following load combinations will be checked/calculated:

1. 1.0 x Own weight + 1.2 x Wind load
2. 1.0 x Own weight + 1.2 x Wind load + 1.3 x Snow load [FR]

H.5. Load cases

H.5.1 Own weight

The own weight of the fabric is $730 \text{ g/m}^2 = 0.0073 \text{ kN/m}^2$ and is considered automatically within EASY software.

H.5.2 Pretension

The structure will be pretensioned with guy ropes and belts. With a ratchet it is possible to achieve 2-3 kN force in corner belts. Approximately 2kN force is achieved in side ropes. The tensile force in ropes and belts results in a membrane stress of approximately 0.50 kN/m^2 . Higher membrane forces can be present near connection points due to the stretchable nature of the fabric.

H.5.3 Conventional / snow load

Conventional load according to EN 13782: The stability shall be checked with a conventional vertical load of 0.1 kN/m^2 . This can be seen as a snow load of 0.1 kN/m^2 (4cm) according to the French CTS, where snow load must be combined with wind load. This condition is tackled in J. Addition 2 – element check for combined wind and snow load on page 68.

H.5.4 User load

A user defined load (for light, sound and/or decoration purposes) is 25 kg per pole and is added after the analysis while performing element checks.

H.5.5 Wind load

H.5.5.1 Wind pressure

Wind load according to EN 13782, 7.4.2.2:

$q_p(z_e)$ of **0.50 kN/m^2** if the height of the structure does not exceed 5m.

Wind load can be reduced for use in the Netherlands. A reduction factor $\alpha=0.5$ is applied, which leads to $q_p(z_e)$ of **0.25 kN/m^2 [NL]**, this is tackled in I. Addition 1 – element check for reduced wind speeds [NL]

H.5.5.2 Wind shape values (Cp-factors)

Two different wind situations are reviewed for the membrane:

1. The whole tent is subjected to wind suction (Cp values given in EN 13782)
2. The whole tent is subjected to wind pressure (Cp values given in EN 13782)

Wind suction

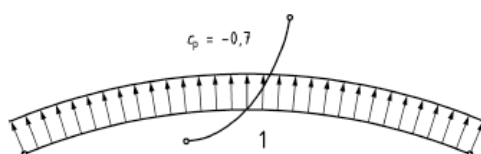


Figure 5. C_p values for rectangle structures EN 13782

P_w	$p_{w,rep}$	
500 N/m ²	$-0.7 \times 0.500 = -0.350$ kN/m ²	
250 N/m ²	$-0.7 \times 0.250 = -0.125$ kN/m ²	[NL]

Wind pressure

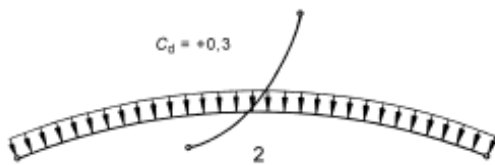


Figure 6. C_p values for rectangle structures EN 13782

P_w	$p_{w,rep}$
500 N/m ²	$0.3 \times 0.500 = 0.150$ kN/m ²

H.6. Calculation results

H.6.1 Listing of calculated load combinations

LC1 = Pretension

LC2 = Own weight

LC3 = Snow load

LC4 = Wind pressure

LC5 = Wind suction

LC6 = Wind suction reduced

The following load combinations are considered:

partial safety factors are added after the static analysis (see H.4.2).

	LC1	LC2	LC3	LC4	LC5	LC6 [NL]
CO1	1 x	1 x				
CO2	1 x	1 x	1 x			
CO3	1 x	1 x		1 x		
CO4	1 x	1 x			1 x	
CO5 [NL]	1 x	1 x				1 x
CO6 [FR]	1 x	1 x	1 x	1 x		

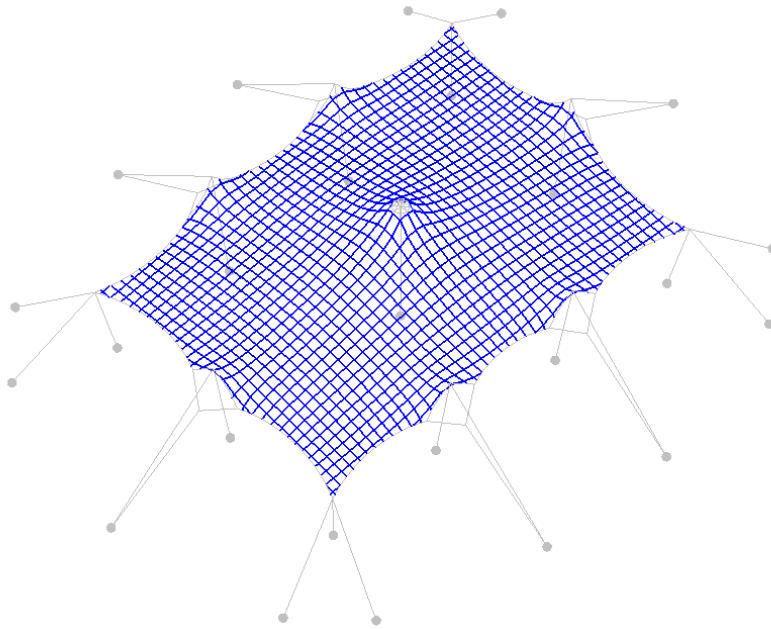
Table 12. Combinations (CO)

The results for load combinations CO5 and CO6 are mentioned in the upcoming chapter but considered in calculations only in chapters I and J on pages 59 and 68 respectively.

H.6.2 Overview: Global results of static analysis

H.6.2.1 Membrane

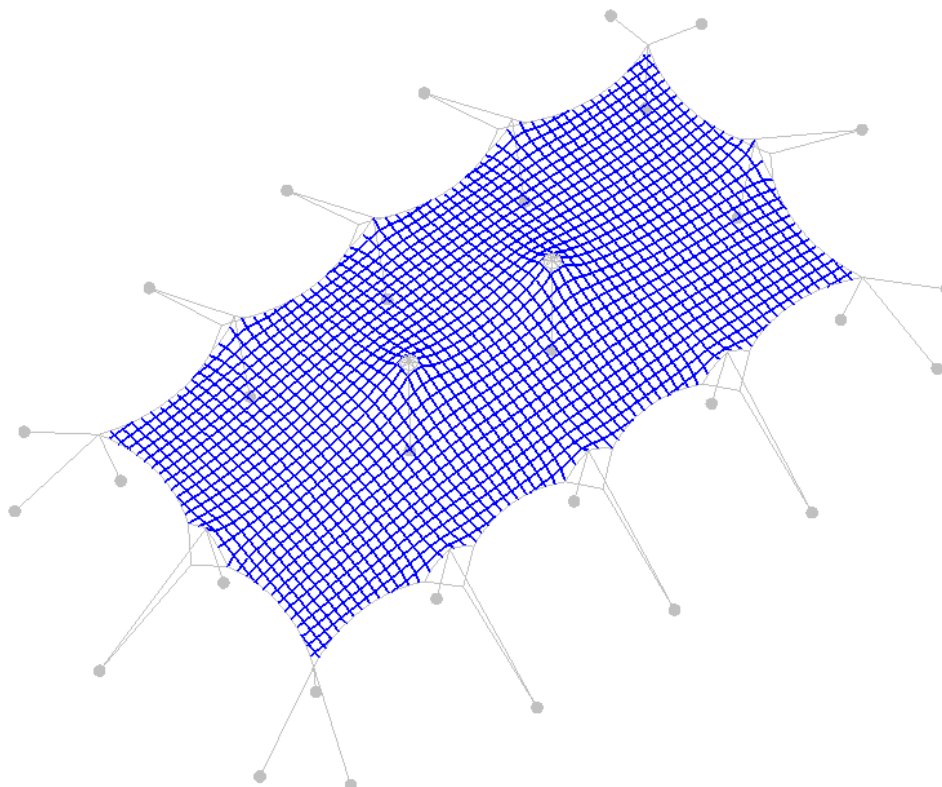
H.6.2.1.1. 7.5x10m model



	Load combination	F _{rep}	Pag.
Warp	CO1. Own weight + pretension	0.96 kN/m	101
	CO2. Own weight + pretension + snow	1.93 kN/m	105
	CO3. Own weight + pretension + wind pressure	2.37 kN/m	109
	Max CO4. Own weight + pretension + wind suction	2.98 kN/m	113
	CO5. Own weight + pretension + wind suction reduced [NL]	2.06 kN/m	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.46 kN/m	121
Weft	CO1. Own weight + pretension	1.02 kN/m	101
	CO2. Own weight + pretension + conventional / snow	2.55 kN/m	105
	Max CO3. Own weight + pretension + wind pressure	3.33 kN/m	109
	CO4. Own weight + pretension + wind suction	0.98 kN/m	113
	CO5. Own weight + pretension + wind suction reduced [NL]	1.11 kN/m	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.54 kN/m	121
Edge	CO1. Own weight + pretension	2.04 kN	101
	CO2. Own weight + pretension + conventional / snow	2.30 kN	105
	CO3. Own weight + pretension + wind pressure	2.55 kN	109
	Max CO4. Own weight + pretension + wind suction	3.26 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	2.45 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.92 kN	121

Table 13. Leading forces in membrane – 7.5x10m

H.6.2.1.2. 7.5x15m model

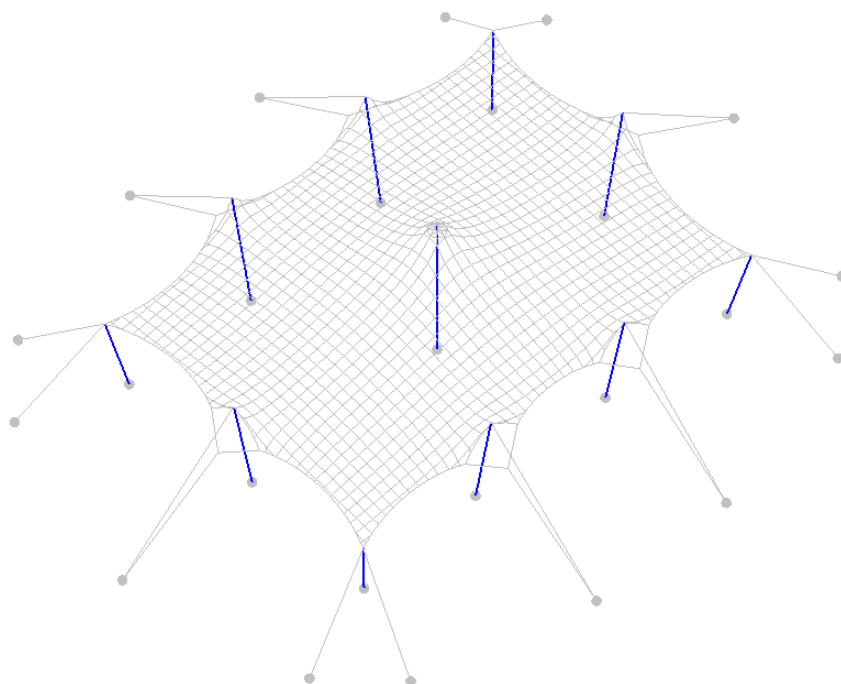


	Load combination	F _{rep}	Pag.
Warp	CO1. Own weight + pretension	1.22 kN/m	125
	CO2. Own weight + pretension + snow	2.56 kN/m	129
	Max CO3. Own weight + pretension + wind pressure	3.38 kN/m	133
	CO4. Own weight + pretension + wind suction	2.23 kN/m	137
	CO5. Own weight + pretension + wind suction reduced [NL]	1.24 kN/m	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.72 kN/m	145
Weft	CO1. Own weight + pretension	1.05 kN/m	125
	CO2. Own weight + pretension + conventional / snow	1.84 kN/m	129
	CO3. Own weight + pretension + wind pressure	2.43 kN/m	133
	Max CO4. Own weight + pretension + wind suction	2.52 kN/m	137
	CO5. Own weight + pretension + wind suction reduced [NL]	2.14 kN/m	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.39 kN/m	145
Edge	CO1. Own weight + pretension	1.96 kN	125
	CO2. Own weight + pretension + conventional / snow	2.29 kN	129
	CO3. Own weight + pretension + wind pressure	2.62 kN	133
	Max CO4. Own weight + pretension + wind suction	3.36 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	2.60 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.04 kN	145

Table 14. Leading forces in membrane – 7.5x15m

H.6.2.2 Poles

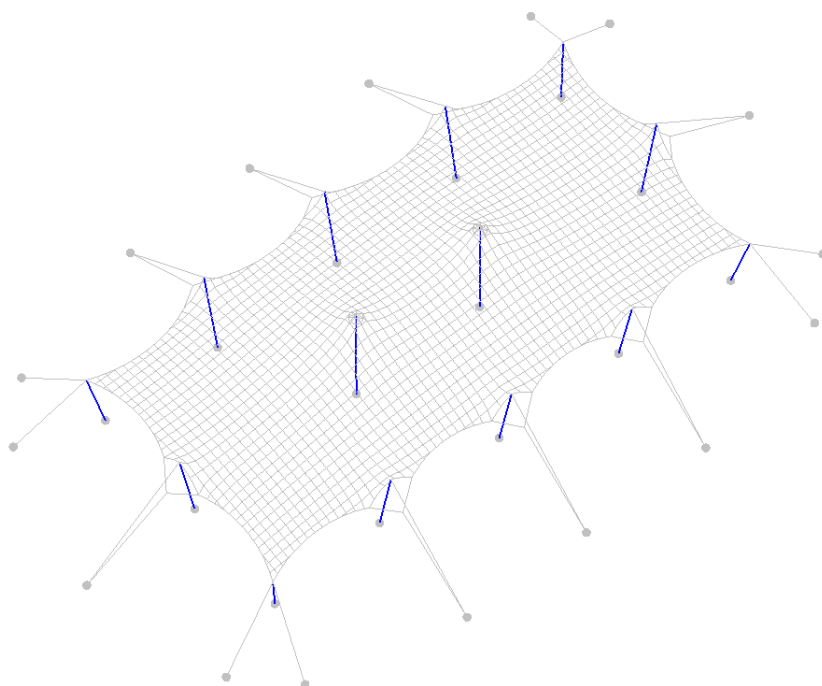
H.6.2.2.1. 7.5x10m model



	Load combination	F _{rep}	Pag.
Main poles	CO1. Own weight + pretension	-2.19 kN	101
	CO2. Own weight + pretension + snow	-4.72 kN	105
	Max CO3. Own weight + pretension + wind pressure	-5.91 kN	109
	CO4. Own weight + pretension + wind suction	-0.06 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	-0.22 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	-8.57 kN	121
Side poles	CO1. Own weight + pretension	-1.50 kN	101
	CO2. Own weight + pretension + conventional / snow	-2.61 kN	105
	Max CO3. Own weight + pretension + wind pressure	-3.32 kN	109
	CO4. Own weight + pretension + wind suction	-2.29 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	-0.82 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	-4.49 kN	121
Corner poles	CO1. Own weight + pretension	-2.69 kN	101
	CO2. Own weight + pretension + conventional / snow	-3.26 kN	105
	Max CO3. Own weight + pretension + wind pressure	-3.79 kN	109
	CO4. Own weight + pretension + wind suction	-3.92 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	-3.15 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	-4.52 kN	121

Table 15. Leading forces in poles – 7.5x10m model

H.6.2.2.2. 7.5x15m model

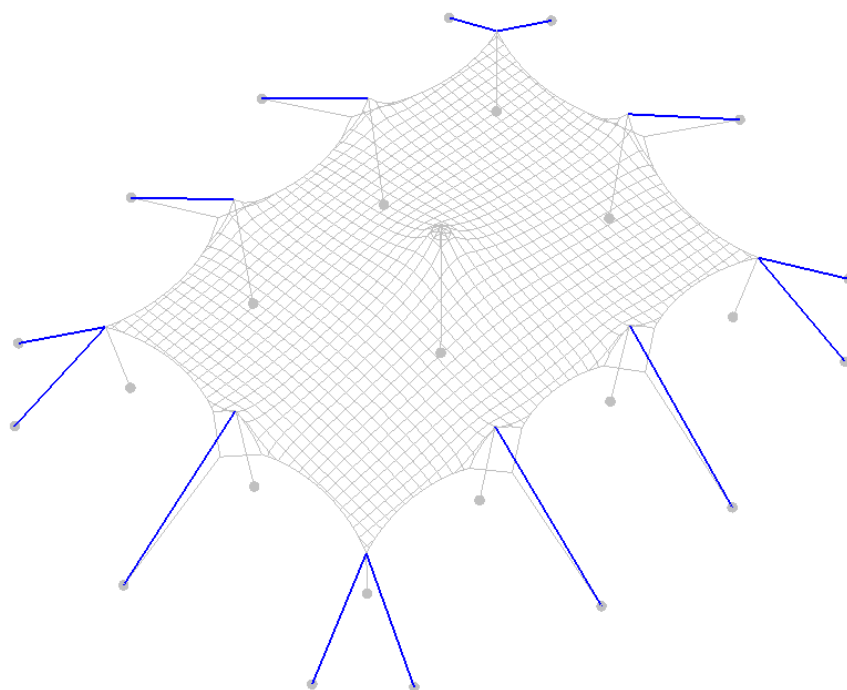


	Load combination	F _{rep}	Pag.
Main poles	CO1. Own weight + pretension	-1.94 kN	125
	CO2. Own weight + pretension + snow	-4.40 kN	129
	Max CO3. Own weight + pretension + wind pressure	-5.60 kN	133
	CO4. Own weight + pretension + wind suction	-0.01 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	-0.10 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	-8.20 kN	145
Side poles	CO1. Own weight + pretension	-1.55 kN	125
	CO2. Own weight + pretension + conventional / snow	-2.83 kN	129
	Max CO3. Own weight + pretension + wind pressure	-3.69 kN	133
	CO4. Own weight + pretension + wind suction	-1.89 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	-0.78 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	-5.04 kN	145
Corner poles	CO1. Own weight + pretension	-2.80 kN	125
	CO2. Own weight + pretension + conventional / snow	-3.45 kN	129
	CO3. Own weight + pretension + wind pressure	-4.08 kN	133
	Max CO4. Own weight + pretension + wind suction	-4.44 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	-3.27 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	-4.90 kN	145

Table 16. Leading forces in poles – 7.5x15m model

H.6.2.3 Belts

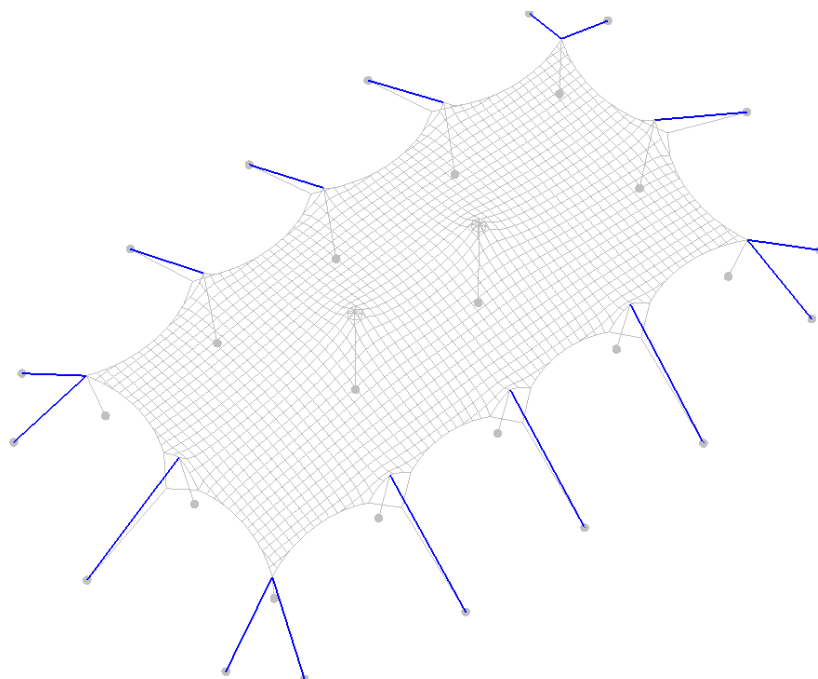
H.6.2.3.1. 7.5x10m model



	Load combination	F _{rep}	Pag.
Corner belts	CO1. Own weight + pretension	2.21 kN	101
	CO2. Own weight + pretension + snow	2.48 kN	105
	CO3. Own weight + pretension + wind pressure	2.82 kN	109
	Max CO4. Own weight + pretension + wind suction	3.60 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	3.13 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.26 kN	121
Belts – long side	CO1. Own weight + pretension	0.12 kN	101
	CO2. Own weight + pretension + conventional / snow	0.35 kN	105
	Max CO3. Own weight + pretension + wind pressure	0.51 kN	109
	CO4. Own weight + pretension + wind suction	0.12 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	0.15 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	0.82 kN	121
Belts – short side	CO1. Own weight + pretension	0.24 kN	101
	CO2. Own weight + pretension + conventional / snow	0.72 kN	105
	Max CO3. Own weight + pretension + wind pressure	1.01 kN	109
	CO4. Own weight + pretension + wind suction	0.07 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	0.01 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	1.47 kN	121

Table 17. Leading forces in belts – 7.5x10m model

H.6.2.3.2. 7.5x15m model

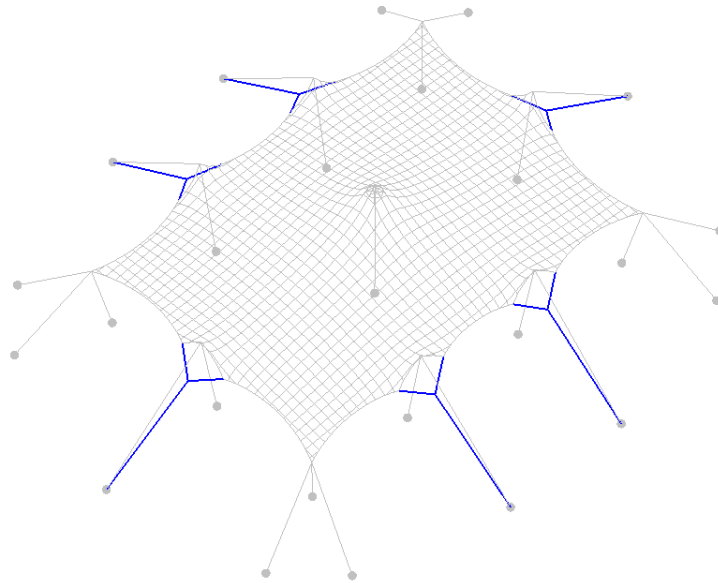


	Load combination	F _{rep}	Pag.
Corner belts	CO1. Own weight + pretension	2.24 kN	125
	CO2. Own weight + pretension + snow	2.55 kN	129
	CO3. Own weight + pretension + wind pressure	2.95 kN	133
	Max CO4. Own weight + pretension + wind suction	4.05 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	3.22 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.47 kN	145
Belts – long side	CO1. Own weight + pretension	0.17 kN	125
	CO2. Own weight + pretension + conventional / snow	0.42 kN	129
	Max CO3. Own weight + pretension + wind pressure	0.61 kN	133
	CO4. Own weight + pretension + wind suction	0.34 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	0.18 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	0.93 kN	145
Belts – short side	CO1. Own weight + pretension	0.25 kN	125
	CO2. Own weight + pretension + conventional / snow	0.83 kN	129
	Max CO3. Own weight + pretension + wind pressure	1.23 kN	133
	CO4. Own weight + pretension + wind suction	0.14 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	0.01 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	1.85 kN	145

Table 18. Leading forces in belts – 7.5x15m model

H.6.2.4 Guy ropes

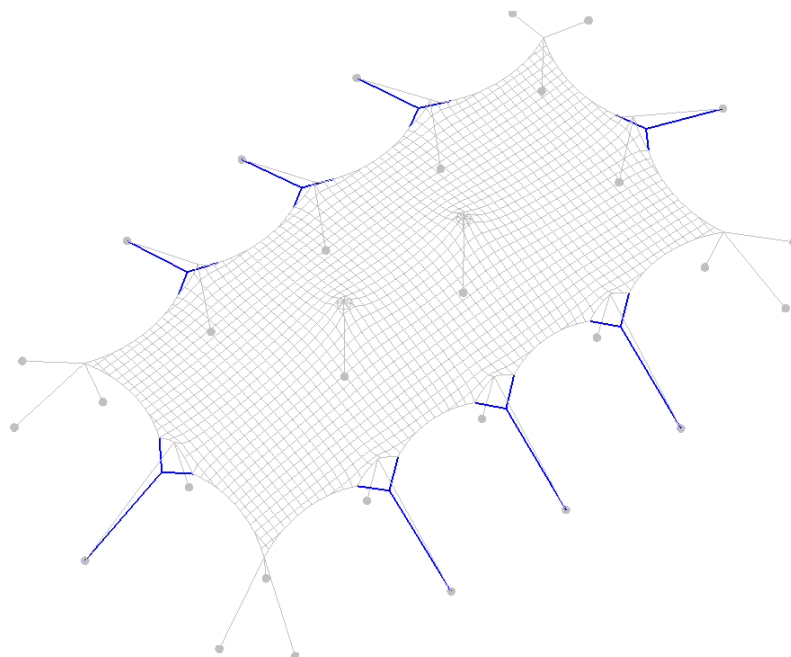
H.6.2.4.1. 7.5x10m model



	Load combination	F _{rep}	Pag.
Combined rope – long side	CO1. Own weight + pretension	1.83 kN	101
	CO2. Own weight + pretension + snow	1.94 kN	105
	CO3. Own weight + pretension + wind pressure	2.16 kN	109
	Max CO4. Own weight + pretension + wind suction	4.05 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	3.28 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.43 kN	121
V-ropes – long side	CO1. Own weight + pretension	1.38 kN	101
	CO2. Own weight + pretension + conventional / snow	1.48 kN	105
	CO3. Own weight + pretension + wind pressure	1.64 kN	109
	Max CO4. Own weight + pretension + wind suction	2.97 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	2.43 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	1.83 kN	121
Combined rope – short side	CO1. Own weight + pretension	1.96 kN	101
	CO2. Own weight + pretension + conventional / snow	2.08 kN	105
	CO3. Own weight + pretension + wind pressure	2.30 kN	109
	Max CO4. Own weight + pretension + wind suction	3.36 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	2.92 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.56 kN	121
V-ropes – short side	CO1. Own weight + pretension	1.43 kN	101
	CO2. Own weight + pretension + conventional / snow	1.47 kN	105
	CO3. Own weight + pretension + wind pressure	1.60 kN	109
	Max CO4. Own weight + pretension + wind suction	2.56 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	2.19 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	1.75 kN	121

Table 19. Leading forces in ropes – 7.5x10m model

H.6.2.4.2. 7.5x15m model

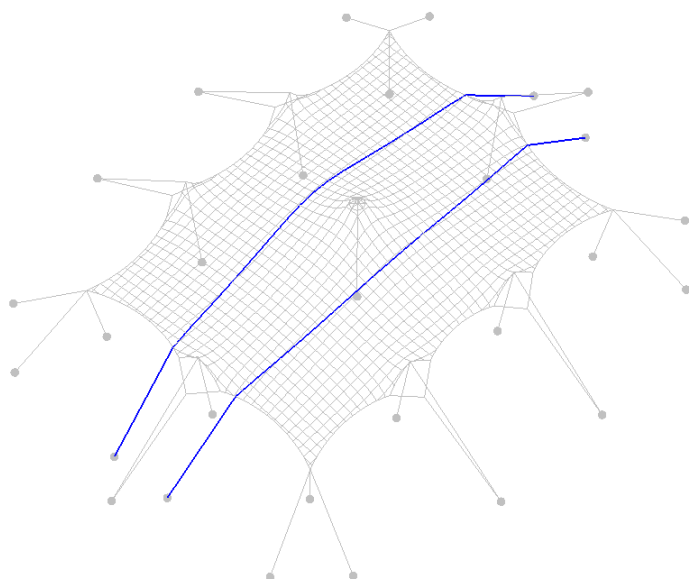


	Load combination	F _{rep}	Pag.
Combined rope – long side	CO1. Own weight + pretension	2.11 kN	125
	CO2. Own weight + pretension + snow	2.18 kN	129
	CO3. Own weight + pretension + wind pressure	2.47 kN	133
	Max CO4. Own weight + pretension + wind suction	4.29 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	4.01 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.75 kN	145
V-ropes – long side	CO1. Own weight + pretension	1.47 kN	125
	CO2. Own weight + pretension + conventional / snow	1.59 kN	129
	CO3. Own weight + pretension + wind pressure	1.80 kN	133
	Max CO4. Own weight + pretension + wind suction	2.98 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	2.74 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.00 kN	145
Combined rope – short side	CO1. Own weight + pretension	1.96 kN	125
	CO2. Own weight + pretension + conventional / snow	2.28 kN	129
	CO3. Own weight + pretension + wind pressure	2.64 kN	133
	Max CO4. Own weight + pretension + wind suction	3.98 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	2.99 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.04 kN	145
V-ropes – short side	CO1. Own weight + pretension	1.42 kN	125
	CO2. Own weight + pretension + conventional / snow	1.59 kN	129
	CO3. Own weight + pretension + wind pressure	1.81 kN	133
	Max CO4. Own weight + pretension + wind suction	2.87 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	2.26 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.05 kN	145

Table 20. Leading forces in ropes – 7.5x15m model

H.6.2.5 Stormbelts

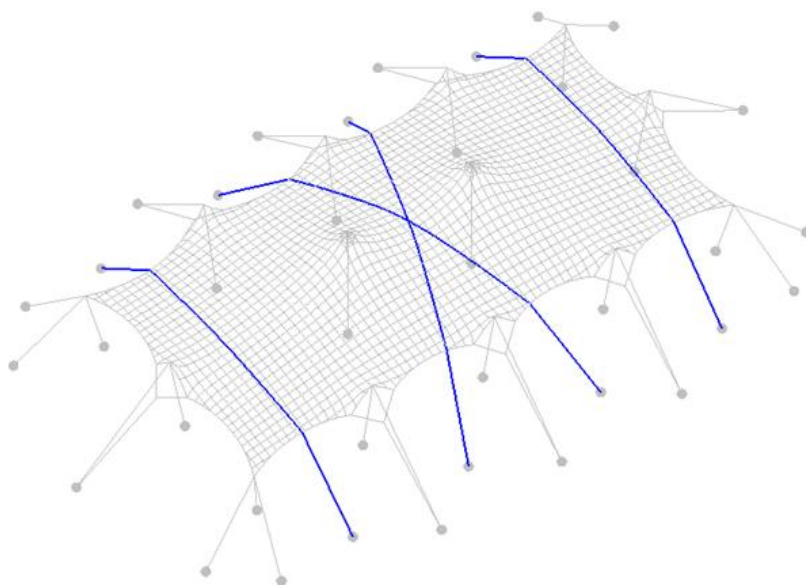
H.6.2.5.1. 7.5x10m model



	Load combination	F_{rep}	Pag.
Stormbelts	CO1. Own weight + pretension	0.00 kN	101
	CO2. Own weight + pretension + snow	0.00 kN	105
	CO3. Own weight + pretension + wind pressure	0.00 kN	109
	Max CO4. Own weight + pretension + wind suction	4.25 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	0.00 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	0.00 kN	121

Table 21. Leading forces in stormbelts – 7.5x10m model

H.6.2.5.2. 7.5x15m model

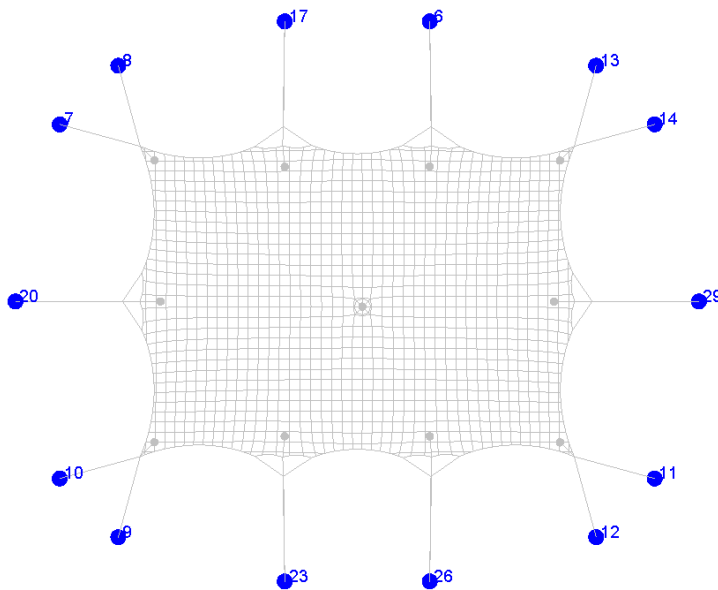


	Load combination	F _{rep}	Pag.
Stormbelts	CO1. Own weight + pretension	0.00 kN	125
	CO2. Own weight + pretension + snow	0.00 kN	129
	CO3. Own weight + pretension + wind pressure	0.00 kN	133
	Max CO4. Own weight + pretension + wind suction	3.89 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	0.00 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	0.00 kN	145

Table 22. Leading forces in stormbelts – 7.5x15m model

H.6.2.6 Reaction forces

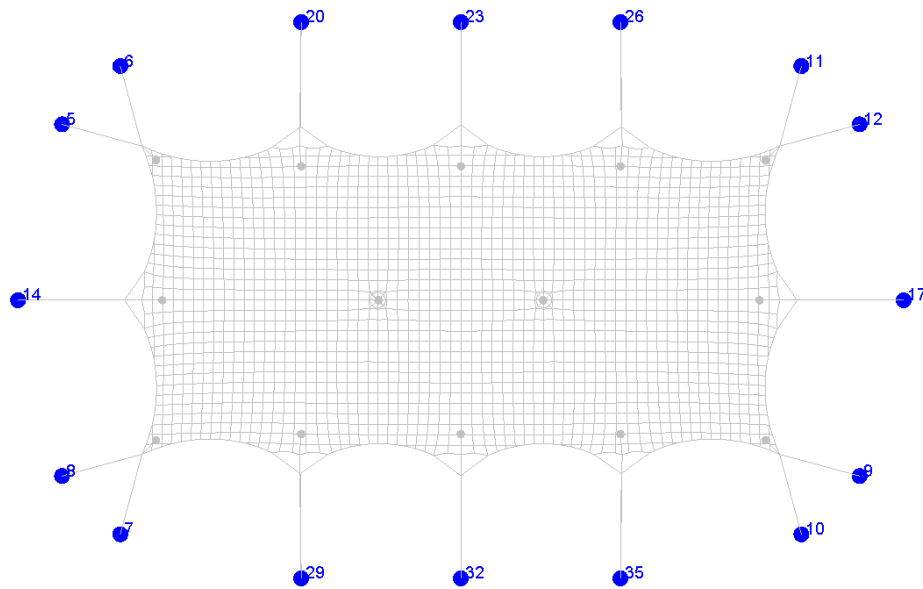
H.6.2.6.1. 7.5x10m model



	Load combination	F _{rep}	Pag.
Long side	CO1. Own weight + pretension	1.95 kN	101
	CO2. Own weight + pretension + snow	2.29 kN	105
	CO3. Own weight + pretension + wind pressure	2.67 kN	109
	Max CO4. Own weight + pretension + wind suction	4.17 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	3.43 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.29 kN	121
Short side	CO1. Own weight + pretension	2.19 kN	101
	CO2. Own weight + pretension + conventional / snow	2.79 kN	105
	CO3. Own weight + pretension + wind pressure	3.31 kN	109
	Max CO4. Own weight + pretension + wind suction	3.41 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	2.92 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.07 kN	121
Corner	CO1. Own weight + pretension	2.21 kN	101
	CO2. Own weight + pretension + conventional / snow	2.48 kN	105
	CO3. Own weight + pretension + wind pressure	2.82 kN	109
	Max CO4. Own weight + pretension + wind suction	3.80 kN	113
	CO5. Own weight + pretension + wind suction reduced [NL]	3.13 kN	117
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.30 kN	121

Table 23. Leading reaction forces – 7.5x10m model

H.6.2.6.2. 7.5x15m model



	Load combination	F _{rep}	Pag.
Long side	CO1. Own weight + pretension	2.21 kN	125
	CO2. Own weight + pretension + snow	2.59 kN	129
	CO3. Own weight + pretension + wind pressure	3.07 kN	133
	Max CO4. Own weight + pretension + wind suction	4.61 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	4.18 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.72 kN	145
Short side	CO1. Own weight + pretension	2.21 kN	125
	CO2. Own weight + pretension + conventional / snow	3.10 kN	129
	CO3. Own weight + pretension + wind pressure	3.85 kN	133
	Max CO4. Own weight + pretension + wind suction	4.11 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	2.99 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.93 kN	145
Corner	CO1. Own weight + pretension	2.24 kN	125
	CO2. Own weight + pretension + conventional / snow	2.55 kN	129
	CO3. Own weight + pretension + wind pressure	2.95 kN	133
	Max CO4. Own weight + pretension + wind suction	4.05 kN	137
	CO5. Own weight + pretension + wind suction reduced [NL]	3.22 kN	141
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.51 kN	145

Table 24. Leading reaction forces – 7.5x15m model

H.6.3 Summary of forces

H.6.3.1 7.5x10m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.96	1.93	2.37	2.98	2.06	3.46
Weft	1.02	2.55	3.33	0.98	1.11	4.54
Edge	2.04	2.30	2.55	3.26	2.45	2.92
Main pole	-2.19	-4.72	-5.91	-0.06	-0.22	-8.57
Side pole	-1.50	-2.61	-3.32	-2.29	-0.82	-4.49
Corner pole	-2.69	-3.26	-3.79	-3.92	-3.15	-4.52
Corner belt	2.21	2.48	2.82	3.60	3.13	3.26
Belts - long side	0.12	0.35	0.51	0.12	0.15	0.82
Belts - short side	0.24	0.72	1.01	0.07	0.00	1.47
Stormbelts	0.00	0.00	0.00	4.25	0.00	0.00
Combined rope - long side	1.83	1.94	2.16	4.05	3.28	2.43
V-ropes - long side	1.38	1.48	1.64	2.97	2.43	1.83
Combined rope - short side	1.96	2.08	2.30	3.36	2.92	2.56
V-ropes short side	1.43	1.47	1.60	2.56	2.19	1.75
Reaction - long side	1.95	2.29	2.67	4.17	3.43	3.29
Reaction - short side	2.19	2.79	3.31	3.41	2.92	4.07
Reaction - corner	2.21	2.48	2.82	3.60	3.13	3.30

Table 25. Summary of forces for 7.5x10m model

H.6.3.2 7.5x15m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	1.22	2.56	3.38	2.23	1.24	4.72
Weft	1.05	1.84	2.43	2.52	2.14	3.39
Edge	1.96	2.29	2.62	3.36	2.60	3.04
Main pole	-1.94	-4.40	-5.60	-0.01	-0.10	-8.20
Side pole	-1.55	-2.83	-3.69	-1.89	-0.78	-5.04
Corner pole	-2.80	-3.45	-4.08	-4.44	-3.27	-4.90
Corner belt	2.24	2.55	2.95	4.05	3.22	3.47
Belts - long side	0.17	0.42	0.61	0.34	0.18	0.93
Belts - short side	0.25	0.83	1.23	0.14	0.00	1.85
Stormbelts	0.00	0.00	0.00	3.89	0.00	0.00
Combined rope - long side	2.11	2.18	2.47	4.29	4.01	2.75
V-ropes - long side	1.47	1.59	1.80	2.98	2.74	2.00
Combined rope - short side	1.96	2.28	2.64	3.98	2.99	3.04
V-ropes short side	1.42	1.59	1.81	2.87	2.26	2.05
Reaction - long side	2.21	2.59	3.07	4.61	4.18	3.72
Reaction - short side	2.21	3.10	3.85	4.11	2.99	4.93
Reaction - corner	2.24	2.55	2.95	4.05	3.22	3.51

Table 26. Summary of forces for 7.5x15m model

H.7. Element check

H.7.1 Membrane

	Load combination	Element	Representative stress	Design value stress	Pag.
7.5x15m	CO1. Own weight + pretension	Long term warp	1.22 kN/m	1.65 kN/m ($\gamma = 1.35$)	41
7.5x15m	CO1. Own weight + pretension	Long term weft	1.05 kN/m	1.42 kN/m ($\gamma = 1.35$)	41
7.5x15m	CO3. Own weight + pretension + wind pressure	Short term warp	3.38 kN/m	5.07 kN/m ($\gamma = 1.5$)	41
7.5x15m	CO4. Own weight + pretension + wind suction	Short term weft	2.52 kN/m	3.78 kN/m ($\gamma = 1.5$)	41

Sioen RP5639 fabric is being used.

UC.1a	Long term warp	$S_{Ed} / S_{rd} < 1$	$1.65 / 9.6 = 0.17 < 1$	OK
UC.1b	Long term weft	$S_{Ed} / S_{rd} < 1$	$1.42 / 7.2 = 0.20 < 1$	OK
UC.1c	Short term warp	$S_{Ed} / S_{rd} < 1$	$5.07 / 12.0 = 0.42 < 1$	OK
UC.1d	Short term weft	$S_{Ed} / S_{rd} < 1$	$3.78 / 9.0 = 0.42 < 1$	OK

For capacity of membrane see H.2.1, page 21

H.7.2 Main poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x10m	CO3. Own weight + pretension + wind pressure	Main pole 4.0m	-5.91 kN	-8.87 kN ($\gamma = 1.5$)	40

User load of max. 25 kg is applied, loaded centrically.

4.0m pole $N_{ed} = -5.91 \times 1.50 + (-0.25 \times 1.35) = -9.20 \text{ kN}$

H.7.2.1 4.0m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 95 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 4.00m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.2	4.0m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.30 / (12.46 \times 0.109) = 0.96 \leq 1$	OK
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See Annex E.1 Main pole 4.0m – Wood, on page 149 for the elaborate check

H.7.3 Side poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x15m	CO3. Own weight + pretension + wind pressure	Side pole 3.0m	-3.69 kN	-5.54 kN ($\gamma = 1.5$)	41

User load of max. 25 kg is applied, loaded centrally.

3m pole $N_{ed} = -3.69 \times 1.50 + (-0.25 \times 1.35) = -5.87 \text{ kN}$

H.7.3.1 3m pole - Wood, Spruce C18

Profile = Pole, $\varnothing \approx 75 \text{ mm}$
average diameter, as a minimum required at the middle of the pole
Length = max. 3.0m
Quality = C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.3	3m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.33 / (12.46 \times 0.12) = 0.89 < 1$	OK
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See Annex E.2 Side pole 3m – Wood, Spruce C18, on page 150 for the elaborate check

H.7.4 Corner poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x15m	CO4. Own weight + pretension + wind suction	Corner pole 2.5m	-4.44 kN	-6.66 kN ($\gamma = 1.5$)	41

User load of max. 25 kg is applied, loaded centrically.

2.5m corner pole $N_{ed} = -4.44 \times 1.50 + (-0.25 \times 1.35) = -7.00 \text{ kN}$

H.7.4.1 2.5m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 70 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 2.5m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.4	2.5m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.82 / (12.46 \times 0.149) = 0.98 < 1$	OK
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See Annex E.3 Corner pole 2.5m - Wood, C18, on page 151 for the elaborate check

H.7.5 Guy ropes

	Load combination	Element	Representative force	Design value force	Pag.
<u>7.5x15m</u>	CO4. Own weight + pretension + wind suction	long side - combined	4.29 kN	6.44 kN ($\gamma = 1.5$)	41
7.5x10m	CO4. Own weight + pretension + wind suction	long side - combined	4.05 kN	6.08 kN ($\gamma = 1.5$)	40
<u>7.5x15m</u>	CO4. Own weight + pretension + wind suction	long side – V-ropes	2.98 kN	4.47 kN ($\gamma = 1.5$)	41
7.5x10m	CO4. Own weight + pretension + wind suction	long side – V-ropes	2.97 kN	4.46 kN ($\gamma = 1.5$)	40
<u>7.5x15m</u>	CO4. Own weight + pretension + wind suction	short side - combined	3.98 kN	5.97 kN ($\gamma = 1.5$)	41
7.5x10m	CO4. Own weight + pretension + wind suction	short side - combined	3.36 kN	5.04 kN ($\gamma = 1.5$)	40
<u>7.5x15m</u>	CO4. Own weight + pretension + wind suction	short side – V-ropes	2.87 kN	4.31 kN ($\gamma = 1.5$)	41
7.5x10m	CO4. Own weight + pretension + wind suction	short side – V-ropes	2.56 kN	3.84 kN ($\gamma = 1.5$)	40

Synthetic rope Ø8mm, BL 900 daN. Is being used $\rightarrow F_{rd,8mm} = 2.25$ kN.

In a case of using individual guy ropes made of multiple sections, the ropes need to be tied back, creating multiple rope sections that together will take the load.

7.5x10m model

UC.5a	long side - combined	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$6.08 / (1 \times 3 \times 2.25) = 0.92 < 1$	OK
UC.5b	long side – V-ropes	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$4.46 / (1 \times 2 \times 2.25) = 0.99 < 1$	OK
UC.5c	short side - combined	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$5.04 / (1 \times 3 \times 2.25) = 0.78 < 1$	OK
UC.5d	short side – V-ropes	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$3.84 / (1 \times 2 \times 2.25) = 0.87 < 1$	OK

For capacity of ropes see H.2.2, page 21

7.5x15m model

UC.5e	long side - combined	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$6.44 / (1 \times 3 \times 2.25) = 0.95 < 1$	OK
UC.5f	long side – V-ropes	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$4.47 / (1 \times 2 \times 2.25) = 0.99 < 1$	OK
UC.5g	short side - combined	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$5.97 / (1 \times 3 \times 2.25) = 0.88 < 1$	OK

UC.5h	short side – V-ropes	1 individual rope made of 2 sections Ø8mm	$F_d / F_{rd} < 1$	$4.31 / (1 \times 2 \times 2.25) = 0.96 < 1$	OK
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For capacity of ropes see H.2.2, page 21

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

7.5x10m model

Long side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2432 \text{ kg}$

Long side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1784 \text{ kg}$

Short side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2016 \text{ kg}$

Short side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1536 \text{ kg}$

7.5x15m model

Long side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2576 \text{ kg}$

Long side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1788 \text{ kg}$

Short side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2388 \text{ kg}$

Short side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1724 \text{ kg}$

H.7.6 Belts

	Load combination	Element	Representative force	Design value force	Pag.
7.5x15m	CO4. Own weight + pretension + wind suction	Corner belt	4.05 kN	6.08 kN ($\gamma = 1.5$)	41
7.5x15m	CO3. Own weight + pretension + wind pressure	Belt – long side	0.61 kN	0.92 kN ($\gamma = 1.5$)	41
7.5x15m	CO3. Own weight + pretension + wind pressure	Belt – short side	1.23 kN	1.85 kN ($\gamma = 1.5$)	41

H.7.6.1 Belt capacity check

PES belt with a breaking strength of 3000kg is being used.

UC.6a	Corner belt	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$6.08 / (1 \times 15) = 0.41 < 1$	OK
UC.6b	Belt – long side	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$0.92 / (1 \times 15) = 0.06 < 1$	OK
UC.6c	Belt – short side	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$1.85 / (1 \times 15) = 0.12 < 1$	OK

For capacity of belts see H.2.3, page 21

H.7.6.2 Alternative – rope capacity check

It is possible to use either a rope or a belt. Synthetic rope Ø8mm, BL 900 daN. Is being used $\rightarrow F_{rd,8mm} = 2.25$ kN.

In a case of using individual guy ropes, the ropes need to be tied back, creating multiple rope sections that together will take the load.

UC.6d	Corner rope	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$6.08 / (1 \times 3 \times 2.25) = 0.90 < 1$	OK
UC.6e	Rope – long side	1 individual rope made of <u>1 section</u> Ø8mm	$F_d / F_{rd} < 1$	$0.92 / (1 \times 1 \times 2.25) = 0.41 < 1$	OK
UC.6f	Rope – short side	1 individual rope made of <u>1 section</u> Ø8mm	$F_d / F_{rd} < 1$	$1.85 / (1 \times 1 \times 2.25) = 0.82 < 1$	OK

For capacity of ropes see H.2.2, page 21

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

Corner rope: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2432$ kg

Rope – long side: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 368$ kg

Rope – long side: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 740$ kg

H.7.7 Stormbelts

	Load combination	Element	Representative force	Design value force	Pag.
7.5x10m	CO4. Own weight + pretension + wind suction	Stormbelt	4.25 kN	6.38 kN ($\gamma = 1.5$)	41

PES belt with a breaking strength of 3000kg is being used.

UC.7	Storm belt	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$6.38 / (1 \times 15) = 0.43 < 1.0$	OK
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For capacity of belts see H.2.3, page 21

H.7.8 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapters H.7.6 and H.7.5.

H.7.8.1 Corner belt connection

H.7.8.1.1. M16 Eye bolt, DIN580

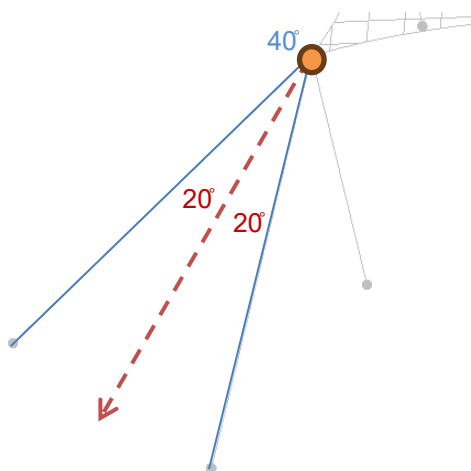
For $\leq 45^\circ$:

M16, WLL 700kg $\rightarrow$ BL = 6 x 7 kN = 42 kN $\rightarrow$ Frd = BL / γ_m = 42 / 2 = 21 kN

$F_d = 2 \times F_{\text{corner belt}} \times \cos 20^\circ = 2 \times 6.08 \times \cos 20^\circ = 11.56 \text{ kN}$

UC.8a	Corner eye bolt	$F_d / F_{rd} < 1$	$11.56 / 21 = 0.55 < 1.0$	OK
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For capacity of eye bolts see K.3 Capacity of Eye bolts on page 84.



H.7.8.1.2. Corner belt connection

Similar connection piece with a capacity of at least 11.5 kN can be used.

Orange Pin D-sluiting, WLL 1.0 t $\rightarrow$ BL = 6 x 10 kN = 60 kN $\rightarrow$ Frd = BL / γ_m = 60 kN / 2 = 30 kN

UC.8b	D-sluiting	$F_d / F_{rd} < 1$	$11.56 / 11.5 = 0.39 < 1.0$	OK
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For capacity of D-sluiting see K.4 Capacity of Orange Pin 1 Ton D-sluiting, DIN EN 13889 on page 84.

H.7.8.1.3. Rope ratchet – for alternative corner connection with ropes

Ratchet, BL = 3.5 kN $\rightarrow$ Frd = BL / γ_m = 3.5 kN / 1.5 = 2.33 kN

$F_d = 6.08 \text{ kN}$ (Instead of corner belts corner ropes are used.)

UC.8c	Corner ropes (alternative)	$F_d / F_{rd} < 1$	$6.08 / 3 \times 2.33 = 0.87 < 1$	OK
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For capacity of rope ratchet see K.6 Rope ratchet, page 85

* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

H.7.8.2 Side belt connection

H.7.8.2.1. M12 Eye bolt, DIN580

For $\leq 45^\circ$:

M12, WLL 340kg $\rightarrow$ BL = 6 x 3.4 kN = 20.4 kN $\rightarrow$ Frd = BL / γ_m = 20.4 / 2 = 10.2 kN

F_d=1.85 kN

UC.8d	Corner eye bolt	$F_d / F_{rd} < 1$	$1.85 / 10.2 = 0.18 < 1.0$	OK
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For capacity of eye bolts see K.3 Capacity of Eye bolts on page 84.

H.7.8.2.2. Carabine, DIN5299C

Carabine 180x14, WLL 580kg $\rightarrow$ BL = 2 x 5.8 kN = 11.6 kN $\rightarrow$ Frd = BL / γ_m = 11.6 kN / 2 = 5.8 kN

UC.8e	Corner eye bolt	$F_d / F_{rd} < 1$	$1.85 / 5.8 = 0.32 < 1.0$	OK
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For capacity of carabines see K.5 Carabine, DIN 5299C, page 85

H.7.8.3 Side rope connection

H.7.8.3.1. Carabine, DIN5299C

Carabine 180x14, WLL 580kg $\rightarrow$ BL = 2 x 5.8 kN = 11.6 kN $\rightarrow$ Frd = BL / γ_m = 11.6 kN / 2 = 5.8 kN

F_d=4.47 kN

UC.8f	$F_d / F_{rd} < 1$	$4.47 / 5.80 = 0.77 < 1$	OK
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For capacity of carabines see K.5 Carabine, DIN 5299C, page 85

H.7.8.3.2. Rope ratchet

Ratchet, BL = 3.5 kN $\rightarrow$ Frd = BL / γ_m = 3.5 kN / 1.5 = 2.33 kN

7.5x15m	UC.8g	Long side – combined rope	$F_d / F_{rd} < 1$	$6.44 / 3 \times 2.33 = 0.92 < 1$	OK
7.5x10m	UC.8h	Long side – combined rope	$F_d / F_{rd} < 1$	$6.08 / 3 \times 2.33 = 0.87 < 1$	OK
7.5x15m	UC.8i	Short side – combined rope	$F_d / F_{rd} < 1$	$5.97 / 3 \times 2.33 = 0.85 < 1$	OK
7.5x10m	UC.8j	Short side – combined rope	$F_d / F_{rd} < 1$	$5.04 / 3 \times 2.33 = 0.72 < 1$	OK

For capacity of rope ratchet see K.6 Rope ratchet, page 85

* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

H.7.8.4 Base plate

H.7.8.4.1. Side pole & corner pole

See H.6.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

1. Soil (base plate size 25x25 cm, $p_{\text{soil}} = 125 \text{ kN/m}^2$)

UC.8k	Soil	$\frac{F_d}{A} / p \leq 1$	$7.00 / (0.25 \cdot 0.25) / 125 = 0.90 < 1$	OK
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H.7.8.4.2. Main pole

See H.6.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

1. Soil (base plate size 30x30 cm, $p_{\text{soil}} = 150 \text{ kN/m}^2$)

UC.8l	Soil	$\frac{F_d}{A} / p \leq 1$	$8.74 / (0.30 \cdot 0.30) / 150 = 0.65 < 1$	OK
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H.8. Safety against overturning, sliding and uplifting

The calculations in this chapter provide a guideline for dense cohesionless soil, in case no anchor tests have been performed. Anchor tests on location can show that a different number of anchors or different size of anchors have sufficient capacity for the specific soil conditions. (see paragraph H.8.3)

H.8.1 Anchor capacity

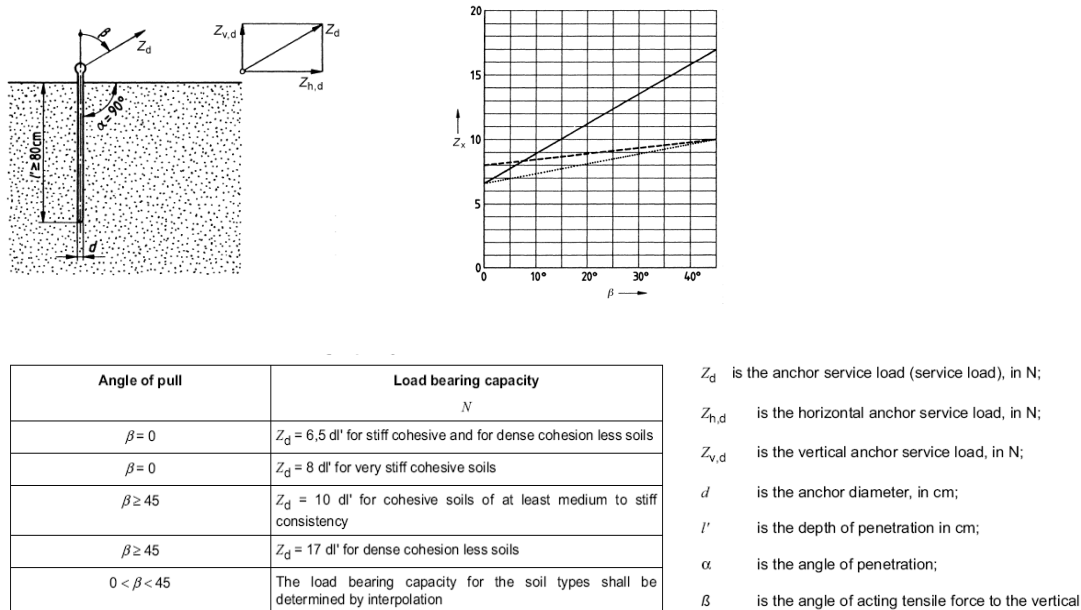


figure 7. Taken from EN 13782: Figures 4 & 5, table 5

Anchors of $\varnothing 30\text{mm}$ are used, considering an **effective length of at least 1000 mm**. If multiple anchors are placed at the same location, the anchors need to be at least $5 \times \varnothing$ apart to use the full capacity of the anchors.

Anchor capacity for anchors based in dense cohesionless soil (sandy soil)

Ø30 x 1000 mm		
Angle	β	≥ 45
Effective length anchor	l'	100 cm
Diameter anchor	d	3.0 cm
Anchor capacity*	Z_d	5.10 kN

H.8.2 Required anchor pins

	Load combination	Element	Representative force	Design value force	Pag.
7.5x10m	CO4. Own weight + pretension + wind suction	Long side anchor point	4.17 kN	5.00 kN ($\gamma = 1.2$)	40
<u>7.5x15m</u>	CO4. Own weight + pretension + wind suction	Long side anchor point	4.61 kN	5.53 kN ($\gamma = 1.2$)	41
7.5x10m	CO4. Own weight + pretension + wind suction	Short side anchor point	3.41 kN	4.09 kN ($\gamma = 1.2$)	40
<u>7.5x15m</u>	CO4. Own weight + pretension + wind suction	Short side anchor point	4.11 kN	4.93 kN ($\gamma = 1.2$)	41
7.5x10m	CO4. Own weight + pretension + wind suction	Corner anchor point	3.60 kN	4.32 kN ($\gamma = 1.2$)	40
<u>7.5x15m</u>	CO4. Own weight + pretension + wind suction	Corner anchor point	4.05 kN	4.86 kN ($\gamma = 1.2$)	41
7.5x10m	CO4. Own weight + pretension + wind suction	Stormbelt	4.25 kN	5.1 kN ($\gamma = 1.2$)	40
<u>7.5x15m</u>	CO4. Own weight + pretension + wind suction	Stormbelt	3.89 kN	4.67 kN ($\gamma = 1.2$)	41

Required anchor amount for dense cohesionless soil – 7.5x10m model

Long side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 5.00 / (1 \times 5.10) = 0.98 \leq 1$	OK
Short side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.09 / (1 \times 5.10) = 0.80 \leq 1$	OK
Corner ties	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.56 / (1 \times 5.10) = 0.85 \leq 1$	OK
Stormbelt	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 5.54 / (2 \times 5.10) = 1.00 \leq 1$	OK

Required anchor amount for dense cohesionless soil – 7.5x15m model

Long side guy ropes	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 5.53 / (2 \times 5.10) = 0.54 \leq 1$	OK
Short side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.93 / (1 \times 5.10) = 0.97 \leq 1$	OK
Corner ties	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.86 / (1 \times 5.10) = 0.95 \leq 1$	OK

Stormbelt	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.67 / (1 \times 5.10) = 0.92 \leq 1$	OK
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H.8.3 Anchor tests according to BS-EN 13782

It is advised to perform anchor test on location when there is a reason to doubt the “pull-out force” of the anchors, which could be when ground conditions differ from dense, non-cohesive soil.

Anchor tests should be carried out according to the following procedure:

Three anchors spread throughout the terrain should be put perpendicular into the ground. The anchors should be pulled out with the aid of a spring balance in the direction of the force acting on the anchor. The lowest of the three measured values should be used.

A safety factor of $\gamma = 1.6$ regarding ultimate limit load is to apply for the lowest test value to determine the load bearing capacity in subsequent calculation. The load bearing capacity determined in this manner shall not result in anchor movement which would result in stresses, deformations or instability inadmissible for the structure.

If the foundation conditions are comparable, test loadings carried out in another location may be adduced for substantiation purposes.

For example:

Force in belts: $F_{rep} = 3.60 \text{ kN}$

$F_{sd_{belt}} = 1.2 \times F_{rep} = 1.2 \times 3.60 = 4.32 \text{ kN}$

The partial safety factor $\gamma = 1.6$ is applied on the ultimate limit load:

$Z_{u,d,test} > 1.6 \times F_{sd} = 1.6 \times 4.32 = 6.91 \text{ kN}$

If for example the anchor test point has a minimal anchor capacity of 5 kN (500 kg), then 2 anchors are needed: $2 \times 5 = 10 \text{ kN} > Z_{u,d,test}$

H.8.4 Required ballast

The safety against sliding shall be calculated from:

$$\sum \mu * \gamma * N_k \geq \sum \gamma * H_k \quad (\text{eq. 5. NEN-EN 13782})$$

where

μ the coefficient of friction according to table 3

γ The safety factor according to Table 2, $\gamma = 1.2$ (unfavorable wind load)

N the vertical load components (characteristic value), corresponds to the required ballast

H the horizontal load components (characteristic value).

The safety against lifting shall be calculated from:

$$\sum \gamma * N_{Rk, stb} \geq \sum \gamma * N_{Ek, dst} \quad (\text{eq. 7. NEN-EN 13782})$$

where

$N_{Rk, stb}$ are the vertical stabilizing load components (characteristic value);

$N_{Ek, dst}$ are the vertical lifting destabilizing load components (characteristic value).

For load calculations against sliding and lifting, F_{rep} is split into the horizontal (F_h) and vertical (F_v) forces. If the separate reaction force factors are not known:

$$F_h = F_{rep} \times \cos \alpha \quad \text{and} \quad F_v = F_{rep} \times \sin \alpha$$

Where α is equal to the angle between F_{rep} and the ground plane.

It follows from the above equation:

$$\text{Required load (against sliding):} \quad B_h = (\gamma F_h) / \mu$$

$$\text{Required load (against lifting):} \quad B_v = \gamma F_v$$

$$B_{tot} = B_h + B_v$$

B_{tot} = total required load

B_h = required load due to horizontal forces

B_v = required load due to vertical forces

7.5x10m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Long side guy ropes	5.00	3.54	3.54	45	1060	940
Short side guy ropes	4.09	2.89	2.89	45	870	770
Corner ties	3.60	3.05	3.05	45	920	820
Stormbelts	5.10	3.61	3.61	45	1080	960

7.5x15m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Long side guy ropes	5.53	3.91	3.91	45	1170	1040
Short side guy ropes	4.93	3.49	3.49	45	1050	930
Corner ties	4.86	3.44	3.44	45	1030	920
Stormbelts	4.67	3.30	3.30	45	990	880

I. Addition 1 – element check for reduced wind speeds [NL]

For stretchtent use in the Netherlands, a reduction of maximal allowable wind speeds is specified. As a result, the stretchtent is calculated for two wind scenarios – full wind and reduced wind. Maximal wind speeds are specified and if the structure exceeds them, the tent must be evacuated and access must be denied. For these wind speeds no stormbelts are necessary.

A reduction factor $\alpha=0.5$ is applied, which leads to $q_p(z_e)$ of **0.25 kN/m² [NL]**. **No stormbelts are required.**

I.1. Summary of results – low wind in Netherlands

This document is valid for three tent sizes: 7.5x10m, 7.5x15m and 6.5x10m. Although no separate calculation is made for 6.5x10m size, all elements, sizes and anchoring forces can be based on 7.5x10m model.

Changes that come from lower wind and no stormbelts are highlighted in orange.

General information:			
Main dimensions:		7.5x10m / 6.5x10m	7.5x15m
	Width:	7.5m	7.5m
	Length:	10m	15m
	Side height of membrane:	3m	3m
	Max. height:	4m	4m
	Main poles: *	4.0m Ø 95 mm [Spruce, C18]	4.0m Ø 95 mm [Spruce, C18]
	Side poles: *	3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]
	Corner poles: *	2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
	Guy ropes:	Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **
	Long side – combined:	3 sections	3 sections
	Long side - V-ropes:	2 sections	2 sections
	Short side – combined:	2 sections	2 sections
	Short side - V-ropes:	2 sections	2 sections
	Belts:	50mm PES	50mm PES

		[EN 12195-2], LC=1000 daN	[EN 12195-2], LC=1000 daN
	Fasteners:	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN
	Side & corner pole base plate	25x25cm	25x25cm
	Main pole base plate	30x30cm	30x30cm
User load:	A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.		
Snow load:	A snow load of 0.1 kN/m ² (4 cm) has been considered in accordance with the French CTS standard.		
Wind load:	According to EN 13782 The following pressure values have been used: h < 5m: 0.35kN/m ²		

* The average diameter, as a minimum required at the middle of the pole.

** The required breaking load (BL_{tot}) may be achieved by using multiple rope sections (n): BL_{rope} = BL_{tot} / n, see page 63

Safety against sliding, overturning and uplifting:

Anchor forces:	Assumptions: angle of 45 degrees The following design resistance of the anchor forces is required: (See H.8.3, page 56 for Anchor tests according to EN 13782)		
		7.5x10m / 6.5x10m	7.5x15m
	Long side guy ropes:	4.12 kN (γ = 1.2)	5.02 kN (γ = 1.2)
	Short side guy ropes:	3.97 kN (γ = 1.2)	4.62 kN (γ = 1.2)
Required anchors:	Corner ties:	3.76 kN (γ = 1.2)	3.86 kN (γ = 1.2)
	Based on dense, non-cohesive soil (e.g. sandy soils). In the case anchors Ø30mm x 1000mm (effective length) are being used:		
		7.5x10m / 6.5x10m	7.5x15m
	Long side guy ropes:	1 anchor	1 anchor
Required ballast:	Short side guy ropes:	1 anchor	1 anchor
	Corner ties:	1 anchor	1 anchor
	Ballast calculation is performed for 2 different situations: - Concrete on concrete – concrete ballasts are placed on concrete - Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast		
	7.5x10m / 6.5x10m		

	Concrete on concrete	Concrete on wood or rubber
Long side guy ropes:	870 kg	780 kg
Short side guy ropes:	840 kg	750 kg
Corner ties:	800 kg	710 kg
7.5x15m		
	Concrete on concrete	Concrete on wood or rubber
Long side guy ropes:	1060 kg	950 kg
Short side guy ropes:	980 kg	870 kg
Corner ties:	820 kg	730 kg

I.2. Summary of forces

Further chapters include the calculation only for highlighted elements in the CO5 [NL] column if they show higher values than calculated in the previous chapters or if recalculating them show a clear difference in results. Other element checks can be looked up in H.7.

I.2.1 7.5x10m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.96	1.93	2.37	2.98	2.06	3.46
Weft	1.02	2.55	3.33	0.98	1.11	4.54
Edge	2.04	2.30	2.55	3.26	2.45	2.92
Main pole	-2.19	-4.72	-5.91	-0.06	-0.22	-8.57
Side pole	-1.50	-2.61	-3.32	-2.29	-0.82	-4.49
Corner pole	-2.69	-3.26	-3.79	-3.92	-3.15	-4.52
Corner belt	2.21	2.48	2.82	3.60	3.13	3.26
Belts - long side	0.12	0.35	0.51	0.12	0.15	0.82
Belts - short side	0.24	0.72	1.01	0.07	0.00	1.47
Stormbelts	0.00	0.00	0.00	4.25	0.00	0.00
Combined rope - long side	1.83	1.94	2.16	4.05	3.28	2.43
V-ropes - long side	1.38	1.48	1.64	2.97	2.43	1.83
Combined rope - short side	1.96	2.08	2.30	3.36	2.92	2.56
V-ropes short side	1.43	1.47	1.60	2.56	2.19	1.75
Reaction - long side	1.95	2.29	2.67	4.17	3.43	3.29
Reaction - short side	2.19	2.79	3.31	3.41	2.92	4.07
Reaction - corner	2.21	2.48	2.82	3.60	3.13	3.30

Table 27. Summary of forces for 7.5x10m model

I.2.2 7.5x15m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	1.22	2.56	3.38	2.23	1.24	4.72
Weft	1.05	1.84	2.43	2.52	2.14	3.39
Edge	1.96	2.29	2.62	3.36	2.60	3.04
Main pole	-1.94	-4.40	-5.60	-0.01	-0.10	-8.20
Side pole	-1.55	-2.83	-3.69	-1.89	-0.78	-5.04
Corner pole	-2.80	-3.45	-4.08	-4.44	-3.27	-4.90
Corner belt	2.24	2.55	2.95	4.05	3.22	3.47
Belts - long side	0.17	0.42	0.61	0.34	0.18	0.93
Belts - short side	0.25	0.83	1.23	0.14	0.00	1.85
Stormbelts	0.00	0.00	0.00	3.89	0.00	0.00
Combined rope - long side	2.11	2.18	2.47	4.29	4.01	2.75
V-ropes - long side	1.47	1.59	1.80	2.98	2.74	2.00
Combined rope - short side	1.96	2.28	2.64	3.98	2.99	3.04
V-ropes short side	1.42	1.59	1.81	2.87	2.26	2.05
Reaction - long side	2.21	2.59	3.07	4.61	4.18	3.72
Reaction - short side	2.21	3.10	3.85	4.11	2.99	4.93
Reaction - corner	2.24	2.55	2.95	4.05	3.22	3.51

Table 28. Summary of forces for 7.5x15m model

I.3. Element check - low wind in Netherlands

I.3.1 Guy ropes

	Load combination	Element	Representative force	Design value force	Pag.
<u>7.5x15m</u>	CO5. Own weight + pretension + wind suction reduced	long side - combined	4.01 kN	6.02 kN ($\gamma = 1.5$)	62
7.5x10m	CO5. Own weight + pretension + wind suction reduced	long side - combined	3.28 kN	4.92 kN ($\gamma = 1.5$)	61
<u>7.5x15m</u>	CO5. Own weight + pretension + wind suction reduced	long side – V-ropes	2.74 kN	4.11 kN ($\gamma = 1.5$)	62
7.5x10m	CO5. Own weight + pretension + wind suction reduced	long side – V-ropes	2.43 kN	3.65 kN ($\gamma = 1.5$)	61
<u>7.5x15m</u>	CO5. Own weight + pretension + wind suction reduced	short side - combined	2.99 kN	4.49 kN ($\gamma = 1.5$)	62
7.5x10m	CO5. Own weight + pretension + wind suction reduced	short side - combined	2.92 kN	4.38 kN ($\gamma = 1.5$)	61
<u>7.5x15m</u>	CO5. Own weight + pretension + wind suction reduced	short side – V-ropes	2.26 kN	3.39 kN ($\gamma = 1.5$)	62
7.5x10m	CO5. Own weight + pretension + wind suction reduced	short side – V-ropes	2.19 kN	3.29 kN ($\gamma = 1.5$)	61

Synthetic rope Ø8mm, BL 900 daN. Is being used $\rightarrow F_{rd,8mm} = 2.25$ kN.

In a case of using individual guy ropes made of multiple sections, the ropes need to be tied back, creating multiple rope sections that together will take the load.

7.5x10m model

UC.1a	long side - combined	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$4.92 / (1 \times 3 \times 2.25) = 0.73 < 1$	OK
UC.1b	long side – V-ropes	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$3.65 / (1 \times 2 \times 2.25) = 0.81 \leq 1$	OK
UC.1c	short side - combined	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$4.38 / (1 \times 2 \times 2.25) = 0.97 < 1$	OK
UC.1d	short side – V-ropes	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$3.29 / (1 \times 2 \times 2.25) = 0.73 < 1$	OK

For capacity of ropes see H.2.2, page 21

7.5x15m model

UC.1e	long side - combined	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$6.02 / (1 \times 3 \times 2.25) = 0.89 < 1$	OK
UC.1f	long side – V-ropes	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$4.11 / (1 \times 2 \times 2.25) = 0.91 < 1$	OK
UC.1g	short side - combined	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$4.49 / (1 \times 2 \times 2.25) = 1.00 \leq 1$	OK
UC.1h	short side – V-ropes	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$3.39 / (1 \times 2 \times 2.25) = 0.75 < 1$	OK

For capacity of ropes see H.2.2, page 21

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

7.5x10m model

Long side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1968 \text{ kg}$

Long side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1458 \text{ kg}$

Short side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1752 \text{ kg}$

Short side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1314 \text{ kg}$

7.5x15m model

Long side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2406 \text{ kg}$

Long side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1644 \text{ kg}$

Short side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1794 \text{ kg}$

Short side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1356 \text{ kg}$

I.3.2 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapter I.3.1.

I.3.2.1 Side rope connection

I.3.2.1.1. Carabine, DIN5299C

Carabine 180x14, WLL 580kg $\rightarrow$ BL = 2 x 5.8 kN = 11.6 kN $\rightarrow$ Frd = BL / γ_m = 11.6 kN / 2 = 5.8 kN
F_d=5.21 kN

UC.2a	F_d / F_{rd} < 1	5.21 / 5.8 = 0.90 ≤ 1	OK
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For capacity of carabines see K.5 Carabine, DIN 5299C, page 85

I.3.2.1.2. Rope ratchet

Ratchet, BL = 3.5 kN $\rightarrow$ Frd = BL / γ_m = 3.5 kN / 1.5 = 2.33 kN

7.5x15m	UC.2b	Long side – combined rope	F_d / F_{rd} < 1	6.02 / 3*x2.33= 0.86 < 1	OK
7.5x10m	UC.2c	Long side – combined rope	F_d / F_{rd} < 1	4.92 / 3*x2.33= 0.70 < 1	OK
7.5x15m	UC.2d	Short side – combined rope	F_d / F_{rd} < 1	4.49 / 2*x2.33= 0.96 < 1	OK
7.5x10m	UC.2e	Short side – combined rope	F_d / F_{rd} < 1	4.38 / 2*x2.33= 0.94 < 1	OK

For capacity of rope ratchet see K.6 Rope ratchet, page 85

* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

I.4. Safety against overturning, sliding and uplifting - low wind in Netherlands

The calculation in this chapter provides a guideline for dense cohesion less soil, in case no anchor tests have been performed. Anchor tests on location can show that a different number of anchors or different size of anchors have sufficient capacity for the specific soil conditions. (see paragraph H.8.3)

Assumed anchor capacity $F_{rd}=5.1$ kN (see H.8.1)

I.4.1 Required anchor pins

	Load combination	Element	Representative force	Design value force	Pag.
7.5x10m	CO5. Own weight + pretension + wind suction reduced	Long side anchor point	3.43 kN	4.12 kN ($\gamma = 1.2$)	61
7.5x15m	CO5. Own weight + pretension + wind suction reduced	Long side anchor point	4.18 kN	5.02 kN ($\gamma = 1.2$)	62
7.5x10m	CO3. Own weight + pretension + wind pressure	Short side anchor point	3.31 kN	3.97 kN ($\gamma = 1.2$)	61
7.5x15m	CO3. Own weight + pretension + wind pressure	Short side anchor point	3.85 kN	4.62 kN ($\gamma = 1.2$)	62
7.5x10m	CO5. Own weight + pretension + wind suction reduced	Corner anchor point	3.13 kN	3.76 kN ($\gamma = 1.2$)	61
7.5x15m	CO5. Own weight + pretension + wind suction reduced	Corner anchor point	3.22 kN	3.86 kN ($\gamma = 1.2$)	62

Required anchor amount for dense cohesionless soil – **7.5x10m model**

Long side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.12 / (1 \times 5.10) = 0.81 \leq 1$	OK
Short side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 3.97 / (1 \times 5.10) = 0.78 < 1$	OK
Corner ties	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 3.76 / (1 \times 5.10) = 0.74 < 1$	OK

Required anchor amount for dense cohesionless soil – **7.5x15m model**

Long side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 5.02 / (1 \times 5.10) = 0.98 < 1$	OK
Short side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.62 / (1 \times 5.10) = 0.91 < 1$	OK

Corner ties	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 3.86 / (1 \times 5.10) = 0.76 < 1$	OK
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I.4.2 Required ballast

For calculation method see H.8.4.

7.5x10m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Long side guy ropes	4.12	2.91	2.91	45	870	780
Short side guy ropes	3.97	2.81	2.81	45	840	750
Corner ties	3.76	2.66	2.66	45	800	710

7.5x15m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Long side guy ropes	5.02	3.55	3.55	45	1060	950
Short side guy ropes	4.62	3.27	3.27	45	980	870
Corner ties	3.86	2.73	2.73	45	820	730

J. Addition 2 – element check for combined wind and snow load [FR]

For stretchtent use in France, the static analysis must include a load combination including snow and wind loads, corresponding to article 7 of CTS. This addition represents changes in element checks and anchor and ballast determination if this load combination is considered. Stormbelts are always required.

J.1. Summary of results – snow+wind in France

This document is valid for three tent sizes: 7.5x10m, 7.5x15m and 6.5x10m. Although no separate calculation is made for 6.5x10m size, all elements, sizes and anchoring forces can be based on 7.5x10m model.

Changes that come from snow and wind combination are highlighted in orange.

General information:			
Main dimensions:		7.5x10m / 6.5x10m	7.5x15m
Width:		7.5m	7.5m
Length:		10m	15m
Side height of membrane:		3m	3m
Max. height:		4m	4m
Main poles: *		4.0m Ø 100 mm [Spruce, C18]	4.0m Ø 100 mm [Spruce, C18]
Side poles: *		3.0m Ø 80 mm [Spruce, C18]	3.0m Ø 80 mm [Spruce, C18]
Corner poles: *		2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
Stormbelts:		50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
Guy ropes:		Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **
Long side – combined:		3 sections	3 sections
Long side - V-ropes:		2 sections	2 sections
Short side – combined:		3 sections	3 sections
Short side - V-ropes:		2 sections	2 sections
Belts:		50mm PES [EN 12195-2],	50mm PES [EN 12195-2],

		LC=1000 daN	LC=1000 daN
	Fasteners:	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN
	Side & corner pole base plate	25x25cm	25x25cm
	Main pole base plate	30x30cm	30x30cm
User load:	A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.		
Snow load:	A snow load of 0.1 kN/m ² (4 cm) has been considered in accordance with the French CTS standard.		
Wind load:	According to EN 13782 The following pressure values have been used: h < 5m: 0.50kN/m ² This value corresponds to wind force 8 on the Beaufort scale. Above this wind force (starting at wind force 9), the safety of the structure can no longer be guaranteed, and it must be evacuated.		

* The average diameter, as a minimum required at the middle of the pole.

** The required breaking load (BL_{tot}) may be achieved by using multiple rope sections (n): BL_{rope} = BL_{tot} / n, see page 47

Safety against sliding, overturning and uplifting:

Anchor forces:	Assumptions: angle of 45 degrees The following design resistance of the anchor forces is required: (See H.8.3, page 56 for Anchor tests according to EN 13782)		
		7.5x10m / 6.5x10m	7.5x15m
	Long side guy ropes:	5.08 kN (γ = 1.2)	6.14 kN (γ = 1.2)
	Short side guy ropes:	4.88 kN (γ = 1.2)	5.92 kN (γ = 1.2)
	Corner ties:	4.56 kN (γ = 1.2)	4.92 kN (γ = 1.2)
	Stormbelt:	5.54 kN (γ = 1.2)	4.51 kN (γ = 1.2)
Required anchors:	Based on dense, non-cohesive soil (e.g. sandy soils). In the case anchors Ø30mm x 1000mm (effective length) are being used:		
		7.5x10m / 6.5x10m	7.5x15m
	Long side guy ropes:	1 anchor	2 anchors
	Short side guy ropes:	1 anchor	2 anchors
	Corner ties:	1 anchor	1 anchor
	Stormbelt:	1 anchor	1 anchor
Required ballast:	Ballast calculation is performed for 2 different situations: - Concrete on concrete – concrete ballasts are placed on concrete - Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast		

7.5x10m / 6.5x10m		
	Concrete on concrete	Concrete on wood or rubber
Long side guy ropes:	1080 kg	960 kg
Short side guy ropes:	1040 kg	920 kg
Corner ties:	970 kg	860 kg
Stormbelt:	1180 kg	1050 kg
7.5x15m		
	Concrete on concrete	Concrete on wood or rubber
Long side guy ropes:	1300 kg	1160 kg
Short side guy ropes:	1260 kg	1120 kg
Corner ties:	1045 kg	930 kg
Stormbelt:	960 kg	850 kg

J.2. Cross sections

Profile	Material	Ø mm	G kg/m	A mm ²	I _y mm ⁴	W _{el,y} mm ³
Ø ≈ 100 mm *	Wood C18	100	2.51	7854	4908738	98175
Ø ≈ 80 mm *	Wood C18	80	1.61	5026	2010619	50265
Ø ≈ 70 mm *	Wood C18	70	1.23	3848	1178588	33674

Table 29. Used cross sections

* The average diameter, as a minimum required at the middle of the pole.

J.3. Summary of forces

Further chapters include the calculation only for highlighted elements in the CO6 [FR] column if they show higher values than calculated in the previous chapters or if recalculating them show a clear difference in results. Other element checks can be looked up in H.7.

J.3.1 7.5x10m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.96	1.93	2.37	2.98	2.06	3.46
Weft	1.02	2.55	3.33	0.98	1.11	4.54
Edge	2.04	2.30	2.55	3.26	2.45	2.92
Main pole	-2.19	-4.72	-5.91	-0.06	-0.22	-8.57
Side pole	-1.50	-2.61	-3.32	-2.29	-0.82	-4.49
Corner pole	-2.69	-3.26	-3.79	-3.92	-3.15	-4.52
Corner belt	2.21	2.48	2.82	3.60	3.13	3.26
Belts - long side	0.12	0.35	0.51	0.12	0.15	0.82
Belts - short side	0.24	0.72	1.01	0.07	0.00	1.47
Stormbelts	0.00	0.00	0.00	4.25	0.00	0.00
Combined rope - long side	1.83	1.94	2.16	4.05	3.28	2.43

V-ropes - long side	1.38	1.48	1.64	2.97	2.43	1.83
Combined rope - short side	1.96	2.08	2.30	3.36	2.92	2.56
V-ropes short side	1.43	1.47	1.60	2.56	2.19	1.75
Reaction - long side	1.95	2.29	2.67	4.17	3.43	3.29
Reaction - short side	2.19	2.79	3.31	3.41	2.92	4.07
Reaction - corner	2.21	2.48	2.82	3.60	3.13	3.30

Table 30. Summary of forces for 7.5x10m model

J.3.2 7.5x15m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	1.22	2.56	3.38	2.23	1.24	4.72
Weft	1.05	1.84	2.43	2.52	2.14	3.39
Edge	1.96	2.29	2.62	3.36	2.60	3.04
Main pole	-1.94	-4.40	-5.60	-0.01	-0.10	-8.20
Side pole	-1.55	-2.83	-3.69	-1.89	-0.78	-5.04
Corner pole	-2.80	-3.45	-4.08	-4.44	-3.27	-4.90
Corner belt	2.24	2.55	2.95	4.05	3.22	3.47
Belts - long side	0.17	0.42	0.61	0.34	0.18	0.93
Belts - short side	0.25	0.83	1.23	0.14	0.00	1.85
Stormbelts	0.00	0.00	0.00	3.89	0.00	0.00
Combined rope - long side	2.11	2.18	2.47	4.29	4.01	2.75
V-ropes - long side	1.47	1.59	1.80	2.98	2.74	2.00
Combined rope - short side	1.96	2.28	2.64	3.98	2.99	3.04
V-ropes short side	1.42	1.59	1.81	2.87	2.26	2.05
Reaction - long side	2.21	2.59	3.07	4.61	4.18	3.72
Reaction - short side	2.21	3.10	3.85	4.11	2.99	4.93
Reaction - corner	2.24	2.55	2.95	4.05	3.22	3.51

Table 31. Summary of forces for 7.5x15m model

J.4. Element check – snow + wind in France

J.4.1 Membrane

	Load combination	Element	Representative stress	Design value stress	Pag.
7.5x15m	CO1. Own weight + pretension	Long term warp	1.22 kN/m	1.65 kN/m ($\gamma = 1.35$)	71
7.5x15m	CO1. Own weight + pretension	Long term weft	1.05 kN/m	1.42 kN/m ($\gamma = 1.35$)	70
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Short term warp	4.72 kN/m	6.37 kN/m ($\gamma = 1.35$)	71
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Short term weft	3.39 kN/m	4.58 kN/m ($\gamma = 1.35$)	71

Sioen RP5639 fabric is being used.

UC.1a	Long term warp	$S_{Ed} / S_{rd} < 1$	$1.65 / 9.6 = 0.17 < 1$	OK
UC.1b	Long term weft	$S_{Ed} / S_{rd} < 1$	$1.42 / 7.2 = 0.20 < 1$	OK
UC.1c	Short term warp	$S_{Ed} / S_{rd} < 1$	$6.37 / 12.0 = 0.53 < 1$	OK
UC.1d	Short term weft	$S_{Ed} / S_{rd} < 1$	$4.58 / 9.0 = 0.51 < 1$	OK

For capacity of membrane see H.2.1, page 21

J.4.2 Main poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x10m	CO6. Own weight + pretension + wind pressure + snow	Main pole 4.0m	-8.57 kN	-11.57 kN ($\gamma = 1.35$)	70

User load of max. 25 kg is applied, loaded centrically.

4.0m pole $N_{ed} = -8.57 \times 1.35 + (-0.25 \times 1.35) = -11.91 \text{ kN}$

J.4.2.1 4.0m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 100 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 4.00m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.2	4.0m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.52 / (12.46 \times 0.120) = 1.02 \leq 1$	OK
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See Annex E.4 Main pole 4.0m – Wood, Spruce C18 – for snow and wind combination [FR], on page 152 for the elaborate check

J.4.3 Side poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Side pole 3.0m	-5.04 kN	-6.80 kN ($\gamma = 1.35$)	71

User load of max. 25 kg is applied, loaded centrically.

3m pole $N_{ed} = -5.04 \times 1.35 + (-0.25 \times 1.35) = -7.14 \text{ kN}$

J.4.3.1 3m pole - Wood, Spruce C18

Profile = Pole, $\varnothing \approx 80 \text{ mm}$
average diameter, as a minimum required at the middle of the pole
Length = max. 3.0m
Quality = C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.3	3m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.42 / (12.46 \times 0.136) = 0.84 < 1$	OK
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See Annex E.5 Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR], on page 153 for the elaborate check

J.4.4 Corner poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Corner pole 2.5m	-4.90 kN	-6.62 kN ($\gamma = 1.35$)	71

User load of max. 25 kg is applied, loaded centrally.

2.5m corner pole $N_{ed} = -4.90 \times 1.35 + (-0.25 \times 1.35) = -6.95 \text{ kN}$

J.4.4.1 2.5m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 70 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 2.5m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.4	2.5m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.81 / (12.46 \times 0.149) = 0.97 < 1$	OK
------	------	---	----

See Annex E.6 Corner pole 2.5m – Wood, Spruce C18 – for snow and wind combination [FR], on page 154 for the elaborate check

J.4.5 Belts

	Load combination	Element	Representative force	Design value force	Pag.
7.5x15m	CO4. Own weight + pretension + wind suction	Corner belt	4.05 kN	6.08 kN ($\gamma = 1.5$)	71
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Belt – long side	0.93 kN	1.26 kN ($\gamma = 1.35$)	71
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Belt – short side	1.85 kN	2.50 kN ($\gamma = 1.35$)	71

J.4.5.1 Belt capacity check

PES belt with a breaking strength of 3000kg is being used.

UC.5a	Corner belt	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$6.08 / (1 \times 15) = 0.41 < 1$	OK
UC.5b	Belt – long side	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$1.26 / (1 \times 15) = 0.08 < 1$	OK
UC.5c	Belt – short side	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$2.50 / (1 \times 15) = 0.17 < 1$	OK

For capacity of belts see H.2.3, page 21

J.4.5.2 Alternative – rope capacity check

It is possible to use either a rope or a belt. Synthetic rope Ø8mm, BL 900 daN. Is being used → $F_{rd,8mm} = 2.25$ kN.

In a case of using individual guy ropes, the ropes need to be tied back, creating multiple rope sections that together will take the load.

UC.5d	Corner rope	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$6.08 / (1 \times 3 \times 2.25) = 0.90 < 1$	OK
UC.5e	Rope – long side	1 individual rope made of <u>1 section</u> Ø8mm	$F_d / F_{rd} < 1$	$1.26 / (1 \times 1 \times 2.25) = 0.56 < 1$	OK
UC.5f	Rope – short side	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$2.50 / (1 \times 2 \times 2.25) = 0.56 < 1$	OK

For capacity of ropes see H.2.2, page 21

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

Corner rope: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2432$ kg

Rope – long side: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 315$ kg

Rope – long side: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1000$ kg

J.4.6 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapters H.7.6 and H.7.5.

J.4.6.1 Side belt connection

J.4.6.1.1. M12 Eye bolt, DIN580

For $\leq 45^\circ$:

M12, WLL 340kg $\rightarrow$ BL = 6 x 3.4 kN = 20.4 kN $\rightarrow$ Frd = BL / γ_m = 20.4 / 2 = 10.2 kN

F_d=2.50 kN

UC.6a	Corner eye bolt	$F_d / F_{rd} < 1$	$2.50 / 10.2 = 0.25 < 1.0$	OK
-------	-----------------	--------------------	----------------------------	----

For capacity of eye bolts see K.3 Capacity of Eye bolts on page 84.

J.4.6.1.2. Carabine, DIN EN 5299C

Carabine 180x14, WLL 580kg $\rightarrow$ BL = 2 x 5.8 kN = 11.6 kN $\rightarrow$ Frd = BL / γ_m = 11.6 kN / 2 = 5.8 kN

UC.6b	Corner eye bolt	$F_d / F_{rd} < 1$	$2.50 / 5.8 = 0.43 < 1.0$	OK
-------	-----------------	--------------------	---------------------------	----

For capacity of carabines see K.5 Carabine, DIN 5299C, page 85

J.4.6.2 Base plate

J.4.6.2.1. Side pole & corner pole

See J.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 25x25 cm, $p_{\text{soil}} = 125 \text{ kN/m}^2$)

UC.6c	Soil	$\frac{F_d}{A} / p \leq 1$	$7.14 / (0.25 \cdot 0.25) / 125 = 0.91 < 1$	OK
-------	------	----------------------------	---	----

J.4.6.2.2. Main pole

See J.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 30x30 cm, $p_{\text{soil}} = 150 \text{ kN/m}^2$)

UC.6d	Soil	$\frac{F_d}{A} / p \leq 1$	$11.41 / (0.30 \cdot 0.30) / 150 = 0.85 < 1$	OK
-------	------	----------------------------	--	----

J.5. Safety against overturning, sliding and uplifting – snow + wind in France

The calculation in this chapter provides a guideline for dense cohesion less soil, in case no anchor tests have been performed. Anchor tests on location can show that a different number of anchors or different size of anchors have sufficient capacity for the specific soil conditions. (see paragraph H.8.3)

Assumed anchor capacity $F_{rd}=5.1$ kN (see H.8.1)

J.5.1 Required anchor pins

	Load combination	Element	Representative force	Design value force	Pag.
7.5x10m	CO5. Own weight + pretension + wind pressure + snow	Short side anchor point	4.07 kN	4.88 kN ($\gamma = 1.2$)	70
7.5x15m	CO5. Own weight + pretension + wind pressure + snow	Short side anchor point	4.93 kN	5.92 kN ($\gamma = 1.2$)	71

Required anchor amount for dense cohesionless soil – **7.5x10m model**

Short side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.88 / (1 \times 5.10) = 0.96 < 1$	OK
----------------------	----	---------------	--	----

Required anchor amount for dense cohesionless soil – **7.5x15m model**

Short side guy ropes	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 5.92 / (2 \times 5.10) = 0.58 < 1$	OK
----------------------	----	---------------	--	----

J.5.2 Required ballast

For calculation method see H.8.4.

7.5x10m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Short side guy ropes	4.88	3.45	3.45	45	1040	920

7.5x15m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Short side guy ropes	5.92	4.18	4.18	45	1260	1120

K. Material certificates

K.1. Fabric specifications

SIOEN FABRICS

Z.I. du Blanc Ballot, Avenue Urbino 6, B-7700 Mouscron - Tel +32(0)56 85 68 80 - Fax +32(0)56 34 61 31
sioenfabrics@sioen.be - <http://www.sioen.com>

Technical Data Sheet

RP5639

FLEXITENT SINGLE M2 (fiche provisoire) - Width 153 cm

Total weight / Poids total	730 [+/-10%] g/m ²	ISO 3801/5 1977
Weight of fabric [PES] / Poids tissu [PES]	280 [+/-10] g/m ²	ISO 3801/5 1977
Weight of coating [PU/PVC] / Poids enduction [PU/PVC]	450 [+/-10%] g/m ²	ISO 3801/5 1977
Breaking strength warp Force de rupture chaîne	1200 N/5cm	ISO 13934 1999
Breaking strength weft Force de rupture trame	900 N/5cm	ISO 13934 1999
Breaking extension warp Allongement à la rupture chaîne	110 %	ISO 13934 1999
Breaking extension weft Allongement à la rupture trame	110 %	ISO 13934 1999
Tear strenght warp Résistance à la déchirure chaîne	100 N	ISO 4674A1 1977
Tear strenght weft Résistance à la déchirure trame	180 N	ISO 4674A1 1977
Adhesion warp / Adhérence chaîne	23 N/5cm	D41 1015
Adhesion weft / Adhérence trame	23 N/5cm	D41 1015
Waterproofness before washing Imperméabilité avant lavage	OK:2000 G M D mm	ISO 811 1981
Waterproofness after washing Imperméabilité après lavage	OK:2000 G M D mm	ISO 811 1981
Flammability / Inflammabilité	OK:M2 -	NF P 92-507 (2004)



May be cleaned with mild, PH neutral, non-ignogenic detergents. Certainly avoid solvent containing cleaning agents (alcohols, polar solvents, etc.). Testing is always advised before using cleaning agents.
After surface cleaning, please let air dry.

All our technical characteristics are indicative
Rev. 10/2012

K.2. Fabric fire certificate



INSTITUT TECHNOLOGIQUE

INSTITUT TECHNOLOGIQUE

FCBA

Laboratoire d'essais feu

10 rue Galilée - 77420 Champs-sur-Marne

Laboratoire agréé par le Ministère de l'Intérieur et de l'Aménagement du Territoire

JO du 10 février 2007

PROCES-VERBAL DE CLASSEMENT DE REACTION AU FEU

D'UN MATERIAU PREVU A L'ARTICLE 5

DE L'ARRETE DU 21 NOVEMBRE 2002

N° CM - 21 - P - 096

et 1 annexe de 4 pages.

Valable 5 ans à compter du mardi 13 juillet 2021

Matériau présenté par : SIOEN FABRICS SA

Avenue Urbino 6

7700 MOUSCRON

Marque commerciale : F5639

Description sommaire:

Type de textile enduit tricot polyester enduit de PVC ignifugé

Composition support textile/ fabric 100% polyester (280 g/m²) + enduction/ coating PVC ignifugé (450 g/m²)

Masse surfacique 730 g/m²

Epaisseur 0,75 mm

Coloris tous coloris/ all colors

Nature de l'essai : Essai au brûleur électrique
Essai de persistance de flamme
Essai de gouttes

Classement : M2

Observations : /

Durabilité du classement (l'annexe 22) : Non limitée a priori

Usage : Revêtement de murs tendus, voilages, rideaux,... **Conditions d'entretien :** selon préconisations du client

Compte tenu des critères résultant des essais annexés décrits dans le(s) rapport(s) d'essais N° : FE 21-0216-1182

Ce procès-verbal atteste uniquement des caractéristiques de l'échantillon soumis aux essais et ne préjuge pas des caractéristiques de produits similaires. Il ne constitue donc pas une certification de produits au sens de l'article L.115-27 du code de la consommation et de la loi du 3 juin 1994.

NOTA : Seules sont autorisées les reproductions intégrales et par photocopies du présent procès-verbal de classement ou de l'ensemble du procès-verbal de classement et rapport(s) d'essais annexé(s). Ces conclusions ne portent que sur les performances de résistance / réaction au feu de l'élément objet du présent procès verbal de classement. Elles ne préjugent, en aucun cas, des autres performances liées à son incorporation à un ouvrage"

"Non valable pour toute application couverte par l'article AM 18 (Arrêté du 6 mars 2006) concernant les sièges rembourrés"

Champs-sur-Marne, le vendredi 23 juillet 2021

Technicien d'essais du Laboratoire Feu



Bilel HAMILA

Responsable Technique du laboratoire Feu
du Pôle Ameublement



Clotilde ETIEMVRE

Siège social
10 rue Galilée
77420 Champs-sur-Marne
Tél + 33 (0)1 72 84 97 84
www.fcba.fr
Siret 775 680 903 00132
APE 7219 Z
Code TVA CEE : FR 14 775 680 903



Institut technologique FCBA : Forêt, Cellulose, Bois - construction, Ameublement

Laboratory approved by the Ministry for the Interior
JO of February 10, 2007
OFFICIAL REPORT OF CLASSIFICATION OF FIRE PERFORMANCE
Of a MATERIAL ENVISAGED IN ARTICLE 5
DECREE OF November 21, 2002

N° **CM - 21 - P - 096**

Page 1

et 1 annexe de 4 pages.

Valid 5 years as from mardi 13 juillet 2021

Material presents by : **SIOEN FABRICS SA**

Avenue Urbino 6
7700 MOUSCRON

Trade mark: **F5639**

Summary description

Kind of coating fabric **tricot polyester enduit de PVC ignifugé**
Composition **support textile/ fabric 100% polyester (280 g/m²) + enduction/ coating PVC ignifugé (450 g/m²)**
Mass per unit area **730 g/m²**
Thickness **0,75 mm**
Color **tous coloris/ all colors**

Nature of the test : **Electrical burner test**
Flame persistence
Dripping test

Classification : **M2**

Observations :

Durability of the classification (Article 5 de l'annexe 2) Not limited

Usual term **Walls hung covering, sheers, curtains,...**

Cleaning conditions **Recommendation of Customer**

Taking into account the criteria resulting from the tests annexed described in the test report N°FE 21-0216-1182

This official report attests only characteristics of the sample submitted for testing and does not prejudice similar characteristics of products. It thus does not constitute a certification of products within the meaning of the L.115-27 article of the code of the consumption and the law of June 3rd, 1994.
NOTE: The integral reproductions and by photocopy of this official report of classification are only authorized or the official report unit of classification and report of annexed test. These conclusions relate only to the resistance / reaction to fire performance of the element that is the subject of the present classification report. They do not prejudice, in any case, other performances related to its incorporation to a work *

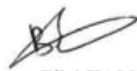
"Not legally acceptable for any use under the pretence of the article AM 18 (Decree of March 6th, 2006) for upholstered seat"

Champs-sur-Marne, le vendredi 23 juillet 2021

Technicien d'essais du Laboratoire Feu

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Bilal HAMILA



Only the french version is valid
Institut technologique FCBA : Forêt, Closely, Bois - construction, Ameublement



Clotilde ETIEMVRE

K.5. Carabine, DIN 5299C

Karta Techniczna Nr 10GM/KAR

1. Nazwa dostawcy: „GÓRALMET” M. i J. Góral Sp. J.

Adres dostawcy: ul. Krakowska 68, 32-860 Czchów, tel/fax 014 6635260

2. Nazwa wyrobu: Karabińczyk do lin 5299C

3. Opis i przeznaczenie: Zatrzaśniki stalowe, służące jako elementy łączące w zakończeniach lin, łańcuchów gospodarczych, oraz do innych zastosowań ogólnych. Do stosowania w zakresie parametrów technicznych podanych w punkcie 5.

4. Karabińczyki do lin nie mogą być użyte w zastosowaniach posiadających inne - uregulowane prawem czy normami wymagania bezpieczeństwa, np. do podnoszenia ładunków, jako środek do ochrony indywidualnej przed upadkiem z wysokości, lub osprzęt alpinistyczny. Nie przeciążać ponad wartość WLL (kG) podaną w kolumnie 13.

5. Parametry techniczne wyrobu:

LP.	Rozmiar [mm]	Wymiary nominalne [mm]										Masa [kg/szt]	WLL Obciążenie robocze [kG]	BF Siła niszcząca [min. kG]	Materiał	Powłoki
		A [mm]	C min [mm]	D [mm]	E min [mm]	Ø [mm]	L [mm]	---	---	---	---					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
1	4	19,5	4,5	6	4,5	3,9	40	---	---	---	0,008	70	140	C10 ~ C15	Ocynk elektrolityczny	
2	5	24	5,5	7,5	5,5	4,8	50	---	---	---	0,015	100	200			
3	6	29	8	8,5	6	5,5	60	---	---	---	0,023	120	240			
4	7	34	9	10,5	8,5	6,7	70	---	---	---	0,040	180	360			
5	8	39	10	11,5	9	7,4	80	---	---	---	0,056	230	460			
6	9	44	11	13,5	10	8,4	90	---	---	---	0,083	250	500			
7	10	48	11,5	14,5	11	9,4	100	---	---	---	0,115	350	700			
8	11	57	17	17,5	13	10,7	120	---	---	---	0,179	400	800			
9	12	67	22	20	15	11,5	140	---	---	---	0,246	450	900			
10	13	80	26	21	17	12,6	160	---	---	---	0,343	530	1060			
11	14	87	28	24	20	13,6	180	---	---	---	0,443	580	1160			
12	15	98	32	24	20	14,7	200	---	---	---	0,563	700	1400			

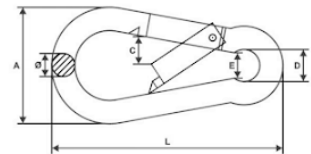
6. Informacje dodatkowe:

➤ Współczynnik bezpieczeństwa - 2

➤ BF = 2,0 x WLL

➤ Cechowanie: symbol producenta – GM lub GMG; kod rozmiaru

➤ Informacje dodatkowe – etykieta: nazwa wyrobu; rozmiar; ilość szt.; numer dostawy; dane producenta i adres strony internetowej.



K.6. Rope ratchet



3/8" Rope Ratchet

Carolina North Manufacturing
10022

\$10.89 ~~\$12.89~~

Maximum load limit 250 lbs. Outer casings are made from #6-33% glass filled nylon with interior mechanisms of die cast zinc. Perfect for large jobs:

- heavy loads for transport
- hoisting items
- straightening items
- securing objects under tension
- product weight 0.16 lbs

Product is compatible with our 3/8" S-hooks and 3/8" solid braided rope.

Bulk purchasing options available.

ADD TO CART

BUY IT NOW

Share:    

Collections: Bulk Rope Ratchets

Type: Bulk Ratchet

Annex

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Annex A. Software input (load combination) – 7.5x10m model

Annex A.1. CO1. Own weight + pretension

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)							
ORDERED BY LOADGROUPS							
LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.6020	0.007669
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0252	0.000320
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3575	71.495529
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3555	71.094423
SUM				0.0000	0.0000	-1.3401	

EXTERNAL LOADS (AREA-DEPENDENT)							
ORDERED BY LOADGROUPS							
LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
SUM				0.0000	0.0000	0.0000	

EXTERNAL LOADS (AREA-DEPENDENT)					
ORDERED BY LOADMODES					
	LOADMODE		SUM_X	SUM_Y	SUM_Z
SUM	AREA-LOADS		0.0000	0.0000	0.0000

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS			
	SUM_X	SUM_Y	SUM_Z
	0.0000	0.0000	-1.3401

Inner Energy= 1.342832

StatikD: Sum of external loads: 0.000000 0.000000 -1.340042
 StatikD: Sum of support forces: 0.000062 -0.000000 1.340657
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO1_OW+VO
 StatikD: Time of calculation: 0.25s
 StatikD: Version 2025.38 done.

Annex A.2. CO2. Own weight + pretension + conventional / snow

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT) ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.6020	0.007669
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0252	0.000320
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3575	71.495529
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3555	71.094423
SUM				0.0000	0.0000	-1.3401	

EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-6.8360	68.36
SUM				0.0000	0.0000	-6.8360	

EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-6.8360
SUM	AREA-LOADS	0.0000	0.0000	-6.8360

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	0.0000	-8.1761

Inner Energy= 1.919365

StatikD: Sum of external loads: 0.000000 0.000000 -8.176114
 StatikD: Sum of support forces: 0.000093 -0.000001 8.175749
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO2_OW+VO+SNOW
 StatikD: Time of calculation: 0.30s
 StatikD: Version 2025.38 done.

Annex A.3. CO3. Own weight + pretension + wind pressure

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT) ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.6020	0.007669
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0252	0.000320
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3575	71.495529
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3555	71.094423
SUM				0.0000	0.0000	-1.3401	

EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	-0.1500	1.00	-0.0000	0.0000	-10.2540	72.08
SUM				-0.0000	0.0000	-10.2540	

EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	-0.0000	0.0000	-10.2540
SUM	AREA-LOADS	-0.0000	0.0000	-10.2540

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0000	0.0000	-11.5941

Inner Energy= 2.514336

StatikD: Sum of external loads: -0.000008 -0.000005 -11.594129
 StatikD: Sum of support forces: 0.000107 0.000001 11.594147
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO3_OW+VO+WD
 StatikD: Time of calculation: 0.31s
 StatikD: Version 2025.38 done.

Annex A.4. CO4. Own weight + pretension + wind suction

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.6020	0.007669
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0345	0.000440
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3609	72.185733
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3588	71.761810
SUM				0.0000	0.0000	-1.3563	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	0.3500	1.00	-0.0002	0.0003	24.1239	72.73
SUM				-0.0002	0.0003	24.1239	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	-0.0002	0.0003	24.1239
SUM	AREA-LOADS	-0.0002	0.0003	24.1239

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0002	0.0003	22.7677

Inner Energy= 4.301083

StatikD: Sum of external loads: -0.000205 0.000300 22.767683
 StatikD: Sum of support forces: 0.000658 -0.000148 -22.767530
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO4_OW+VO+WU
 StatikD: Time of calculation: 1.11s
 StatikD: Version 2025.38 done.

Annex A.5. CO5. Own weight + pretension + wind suction reduced [NL]

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.6020	0.007669
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0252	0.000320
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3575	71.495529
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3555	71.094423
SUM				0.0000	0.0000	-1.3401	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	0.1750	1.00	0.0000	-0.0000	11.9630	72.08
SUM				0.0000	-0.0000	11.9630	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	0.0000	-0.0000	11.9630
SUM	AREA-LOADS	0.0000	-0.0000	11.9630

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	-0.0000	10.6229

Inner Energy= 2.730369

StatikD: Sum of external loads: 0.000003 0.000003 10.622948
 StatikD: Sum of support forces: 0.000016 -0.000042 -10.621775
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO5_OW+V0+WU_red
 StatikD: Time of calculation: 0.46s
 StatikD: Version 2025.38 done.

Annex A.6. CO6. Own weight + pretension + wind pressure + snow [FR]

Annex A.6.1. For element check

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT) ORDERED BY LOADGROUPS							
LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.6020	0.007669
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0252	0.000320
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3575	71.495529
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3555	71.094423
SUM				0.0000	0.0000	-1.3401	
EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADGROUPS							
LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-6.8360	68.36
1	WIND	-0.1500	1.00	-0.0000	0.0000	-10.2540	72.08
SUM				-0.0000	0.0000	-17.0900	
EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADMODES							
	LOADMODE			SUM_X	SUM_Y	SUM_Z	
SUM	SCHNEE			0.0000	0.0000	-6.8360	
SUM	WIND			-0.0000	0.0000	-10.2540	
SUM	AREA-LOADS			-0.0000	0.0000	-17.0900	
EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS							
				SUM_X	SUM_Y	SUM_Z	
				-0.0000	0.0000	-18.4301	
Inner Energy= 3.654297							
StatikD: Sum of external loads: -0.000008 -0.000005 -18.430164							
StatikD: Sum of support forces: 0.000129 0.000002 18.430528							
StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO6_OW+VO+WD+SNOW							
StatikD: Time of calculation: 0.30s							
StatikD: Version 2025.38 done.							

Annex A.6.2. For anchor and ballast calculation

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.6020	0.007669
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0252	0.000320
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3575	71.495529
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3555	71.094423
SUM				0.0000	0.0000	-1.3401	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1083	1.00	0.0000	0.0000	-7.4055	68.36
1	WIND	-0.1500	1.00	-0.0000	0.0000	-10.2540	72.08
SUM				-0.0000	0.0000	-17.6595	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-7.4055
SUM	WIND	-0.0000	0.0000	-10.2540
SUM	AREA-LOADS	-0.0000	0.0000	-17.6595

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0000	0.0000	-18.9996

Inner Energy= 3.756092

StatikD: Sum of external loads: -0.000008 -0.000005 -18.999588
 StatikD: Sum of support forces: 0.000130 0.000003 19.000350
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\C06a_OW+VO+WD+SNOW_anchors
 StatikD: Time of calculation: 0.27s
 StatikD: Version 2025.38 done.

Annex B. Software input (load combination) – 7.5x15m model

Annex B.1. CO1. Own weight + pretension

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.8365	0.010656
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0353	0.000449
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5402	108.042120
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5370	107.393517
SUM				0.0000	0.0000	-1.9490	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
SUM				0.0000	0.0000	0.0000	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM AREA-LOADS	0.0000	0.0000	0.0000

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	0.0000	-1.9490

Inner Energy= 1.864821

StatikD: Sum of external loads: 0.000000 0.000000 -1.948950
 StatikD: Sum of support forces: -0.000516 0.000859 1.948154
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\C01_OW+VO
 StatikD: Time of calculation: 0.55s
 StatikD: Version 2025.38 done.

Annex B.2. CO2. Own weight + pretension + conventional / snow

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.8365	0.010656
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0353	0.000449
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5402	108.042120
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5370	107.393517
SUM				0.0000	0.0000	-1.9490	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-10.3249	103.25
SUM				0.0000	0.0000	-10.3249	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-10.3249
SUM	AREA-LOADS	0.0000	0.0000	-10.3249

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	0.0000	-12.2739

Inner Energy= 2.729394

StatikD: Sum of external loads: 0.000000 0.000000 -12.273883
 StatikD: Sum of support forces: 0.000099 0.000429 12.274127
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO2_OW+VO+SNOW
 StatikD: Time of calculation: 1.03s
 StatikD: Version 2025.38 done.

Annex B.3. CO3. Own weight + pretension + wind pressure

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.8365	0.010656
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0353	0.000449
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5402	108.042120
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5370	107.393517
SUM				0.0000	0.0000	-1.9490	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	-0.1500	1.00	-0.0148	0.0059	-15.4874	108.86
SUM				-0.0148	0.0059	-15.4874	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	-0.0148	0.0059	-15.4874
SUM	AREA-LOADS	-0.0148	0.0059	-15.4874

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0148	0.0059	-17.4363

Inner Energy= 3.657800

StatikD: Sum of external loads: -0.014794 0.005925 -17.436344
 StatikD: Sum of support forces: 0.015695 -0.006346 17.435585
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO3_OW+VO+WD
 StatikD: Time of calculation: 0.98s
 StatikD: Version 2025.38 done.

Annex B.4. CO4. Own weight + pretension + wind suction

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.8365	0.010656
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0510	0.000650
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5402	108.042120
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5370	107.393517
SUM				0.0000	0.0000	-1.9647	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	0.3500	1.00	0.0392	-0.0134	36.0574	108.83
SUM				0.0392	-0.0134	36.0574	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	0.0392	-0.0134	36.0574
SUM	AREA-LOADS	0.0392	-0.0134	36.0574

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0392	-0.0134	34.0927

Inner Energy= 6.465364

StatikD: Sum of external loads: 0.039256 -0.013340 34.092686
 StatikD: Sum of support forces: -0.037114 0.012729 -34.123287
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO4_OW+VO+WU
 StatikD: Time of calculation: 1.51s
 StatikD: Version 2025.38 done.

Annex B.5. CO5. Own weight + pretension + wind suction reduced [NL]

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.8365	0.010656
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0353	0.000449
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5402	108.042120
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5370	107.393517
SUM				0.0000	0.0000	-1.9490	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	0.1750	1.00	0.0173	-0.0069	18.0686	108.86
SUM				0.0173	-0.0069	18.0686	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	0.0173	-0.0069	18.0686
SUM	AREA-LOADS	0.0173	-0.0069	18.0686

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0173	-0.0069	16.1196

Inner Energy= 4.271551

StatikD: Sum of external loads: 0.017233 -0.006919 16.119644
 StatikD: Sum of support forces: -0.016079 0.003593 -16.122554
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO5_OW+V0+WU_red
 StatikD: Time of calculation: 1.26s
 StatikD: Version 2025.38 done.

Annex B.6. CO6. Own weight + pretension + wind pressure + snow [FR]

Annex B.6.1. For element check

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.8365	0.010656
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0353	0.000449
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5402	108.042120
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5370	107.393517
SUM				0.0000	0.0000	-1.9490	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-10.3249	103.25
1	WIND	-0.1500	1.00	-0.0148	0.0059	-15.4874	108.86
SUM				-0.0148	0.0059	-25.8123	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-10.3249
SUM	WIND	-0.0148	0.0059	-15.4874
SUM	AREA-LOADS	-0.0148	0.0059	-25.8123

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0148	0.0059	-27.7612

Inner Energy= 5.357406

StatikD: Sum of external loads: -0.014794 0.005925 -27.761253
 StatikD: Sum of support forces: 0.015775 -0.006082 27.761015
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO6_OW+VO+WD+SNOW
 StatikD: Time of calculation: 1.06s
 StatikD: Version 2025.38 done.

Annex B.6.2. For anchor and ballast calculation

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.8365	0.010656
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0353	0.000449
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5402	108.042120
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5370	107.393517
SUM				0.0000	0.0000	-1.9490	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1083	1.00	0.0000	0.0000	-11.1850	103.25
1	WIND	-0.1500	1.00	-0.0148	0.0059	-15.4874	108.86
SUM				-0.0148	0.0059	-26.6724	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-11.1850
SUM	WIND	-0.0148	0.0059	-15.4874
SUM	AREA-LOADS	-0.0148	0.0059	-26.6724

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0148	0.0059	-28.6213

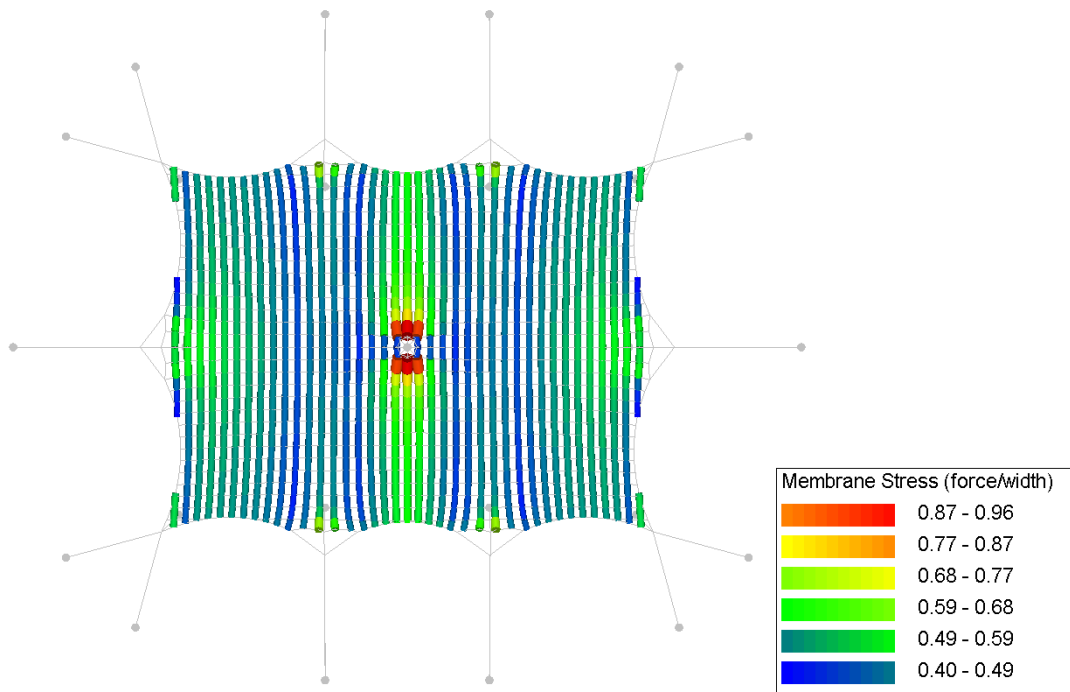
Inner Energy= 5.508935

StatikD: Sum of external loads: -0.014794 0.005925 -28.621299
 StatikD: Sum of support forces: 0.014705 -0.004778 28.622217
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\C06a_OW+VO+WD+SNOW_anchors
 StatikD: Time of calculation: 1.11s
 StatikD: Version 2025.38 done.

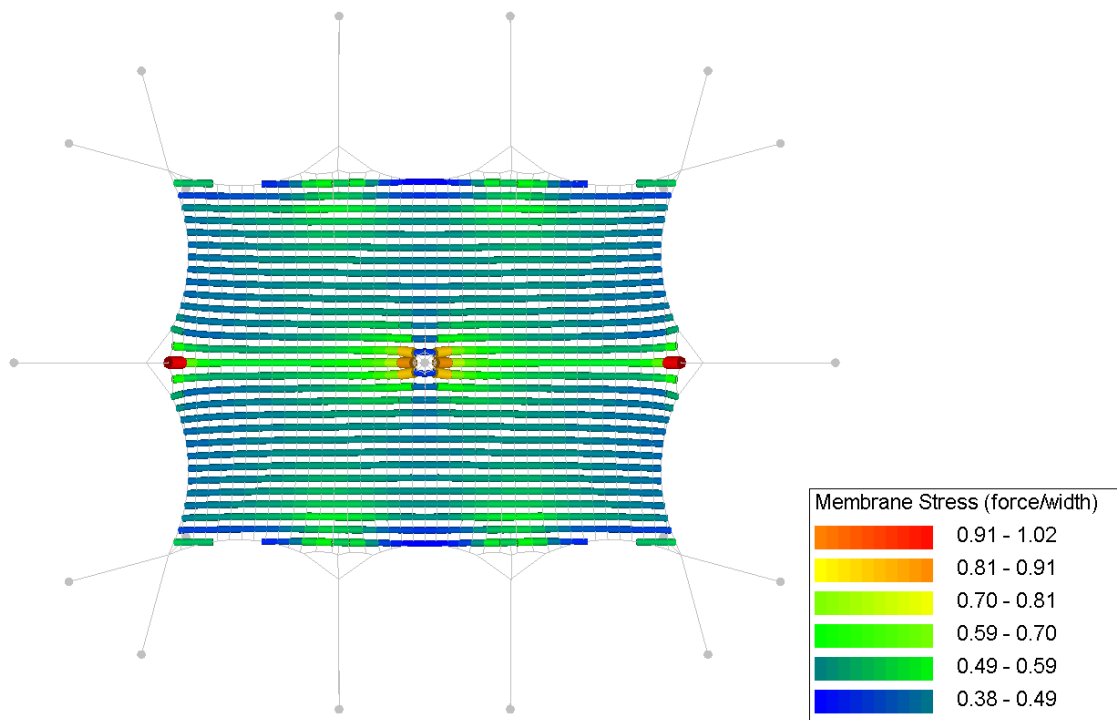
Annex C. Results per load combination – 7.5x10m model

Annex C.1. CO1. Own weight + pretension

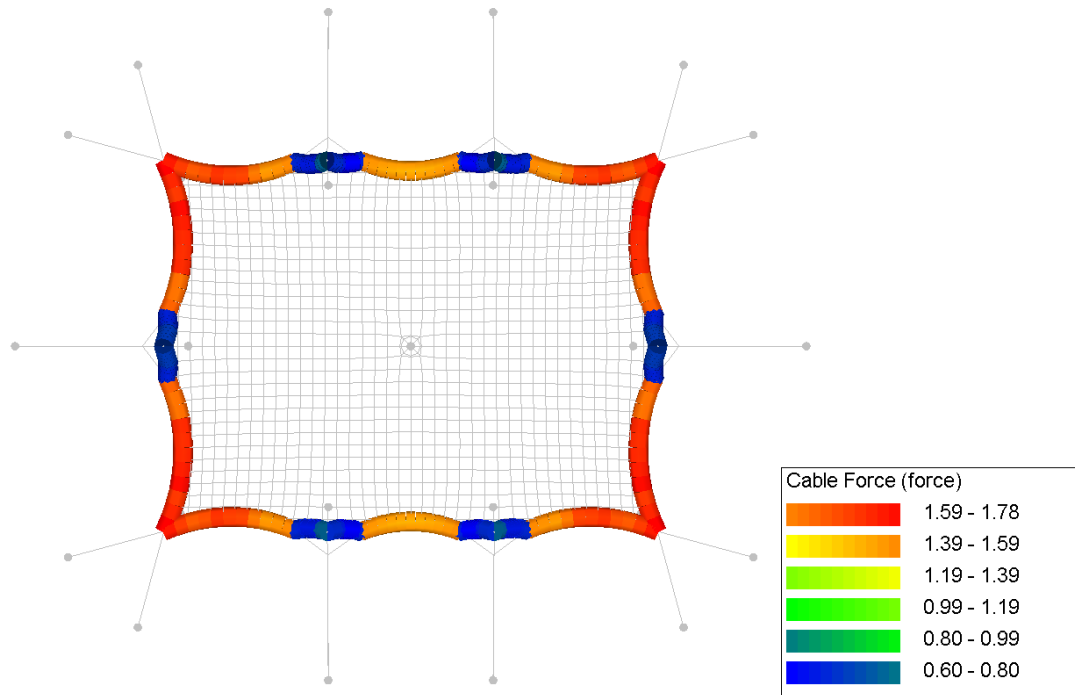
Annex C.1.1. Membrane stress (warp)



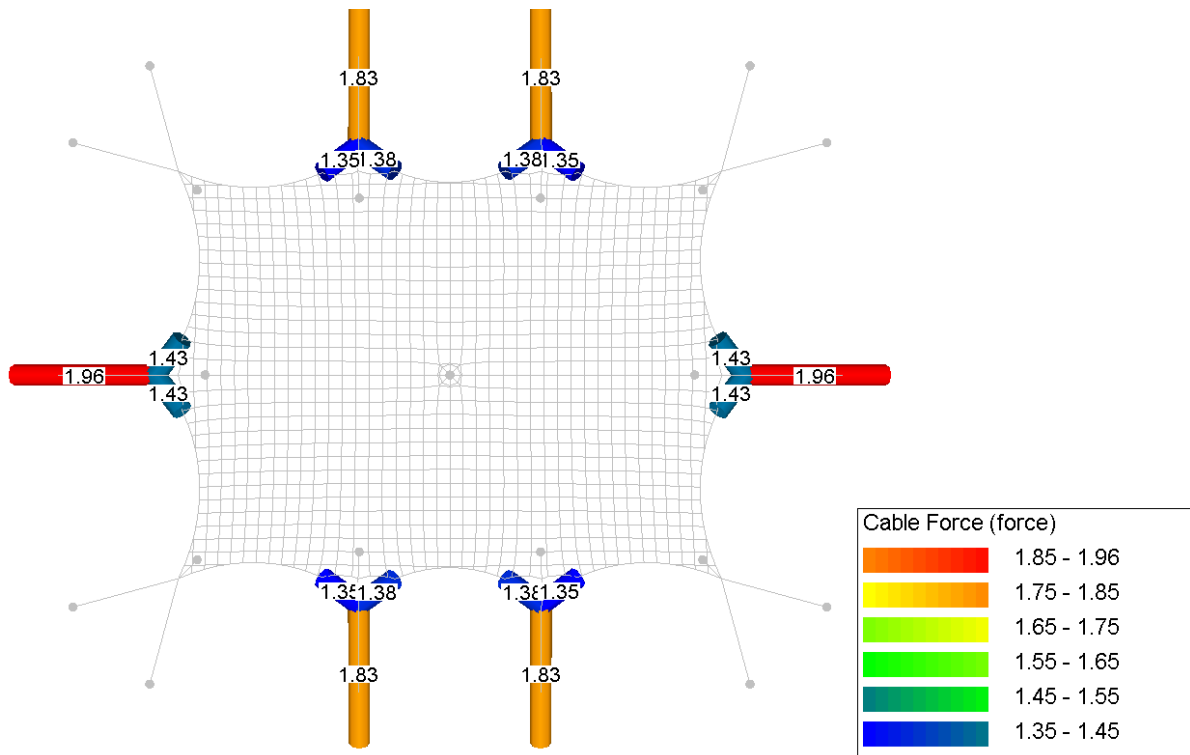
Annex C.1.2. Membrane stress (weft)



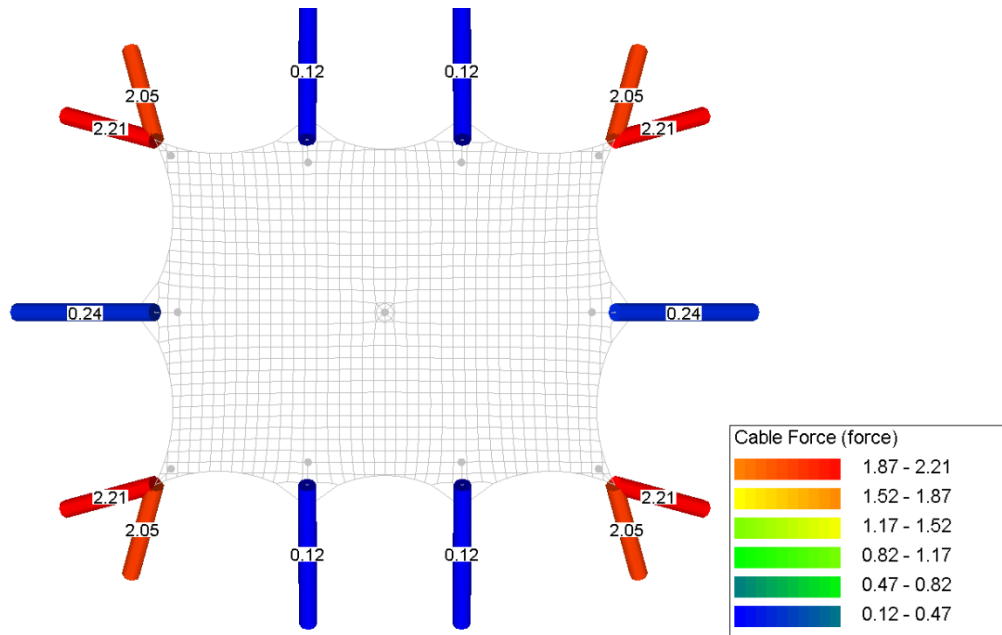
Annex C.1.3. Circumference rope forces



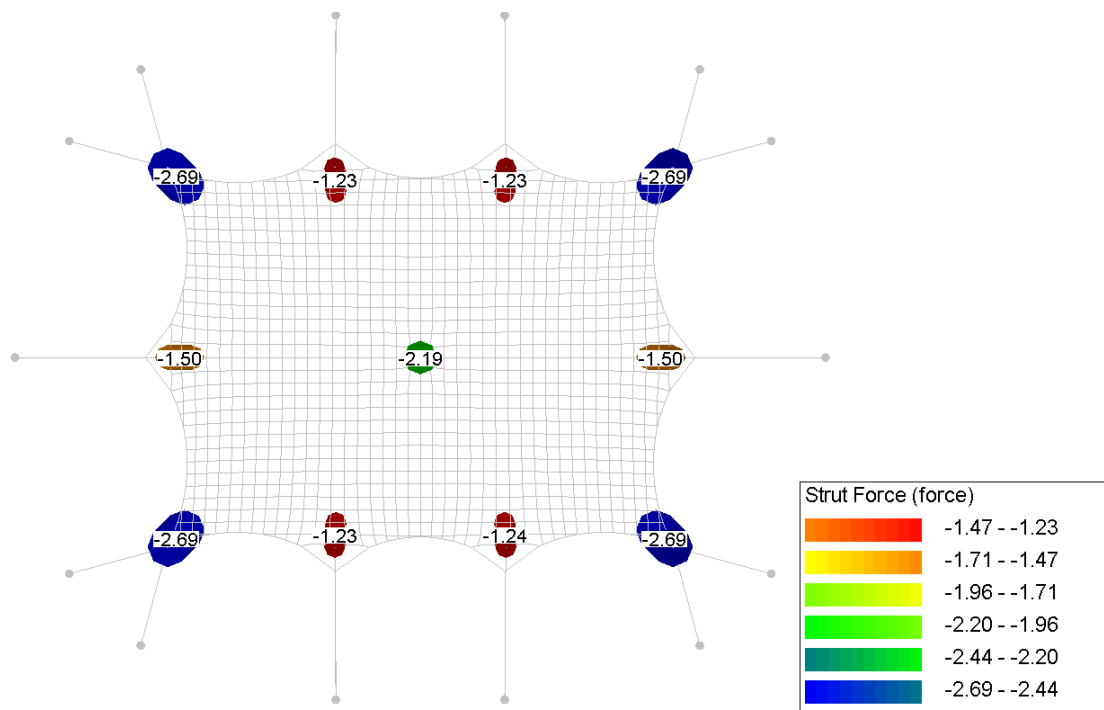
Annex C.1.4. Rope forces



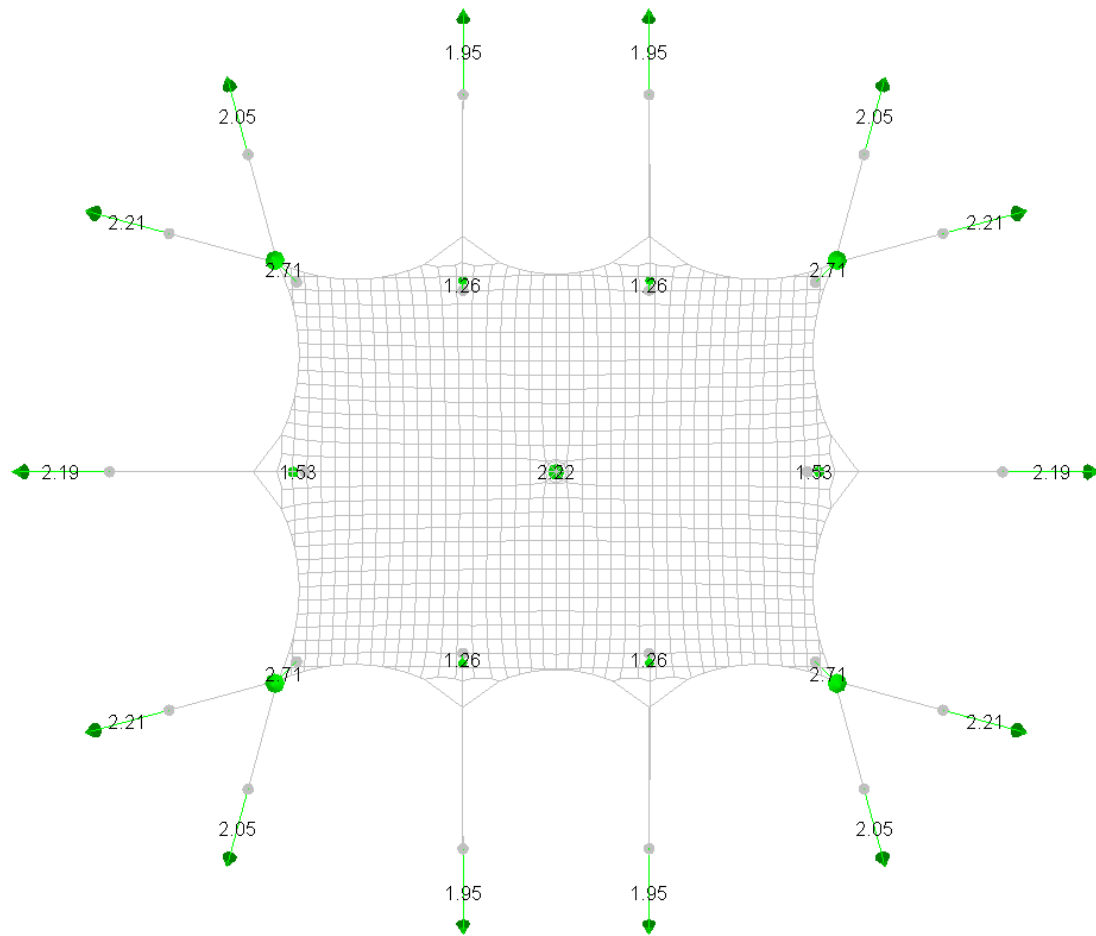
Annex C.1.5. Belt forces



Annex C.1.6. Pole forces

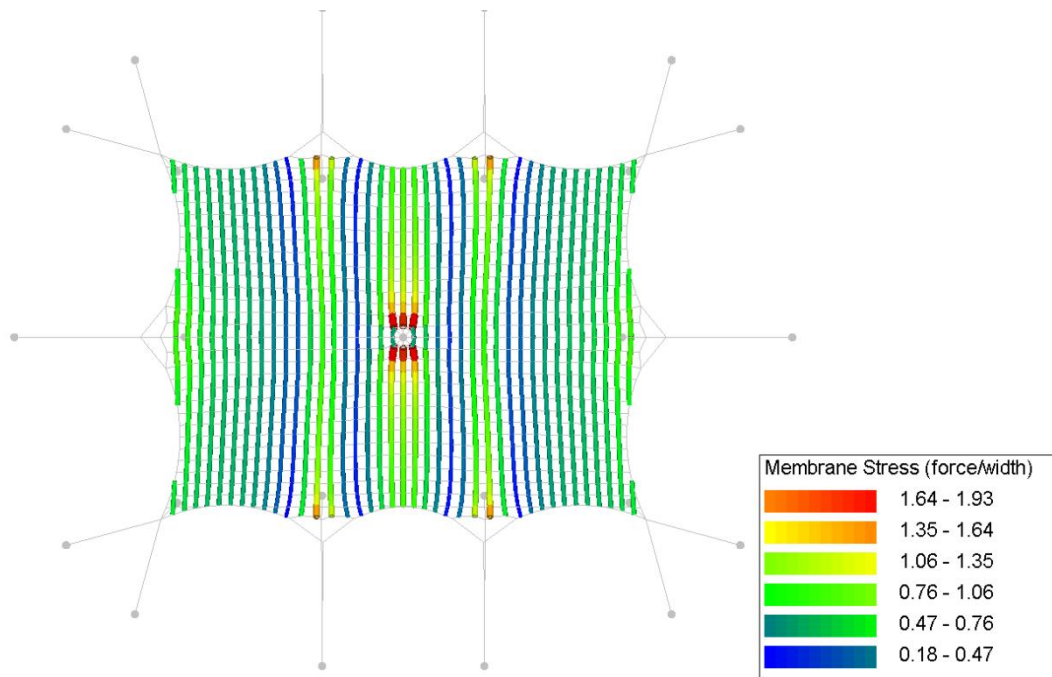


Annex C.1.7. Reaction forces

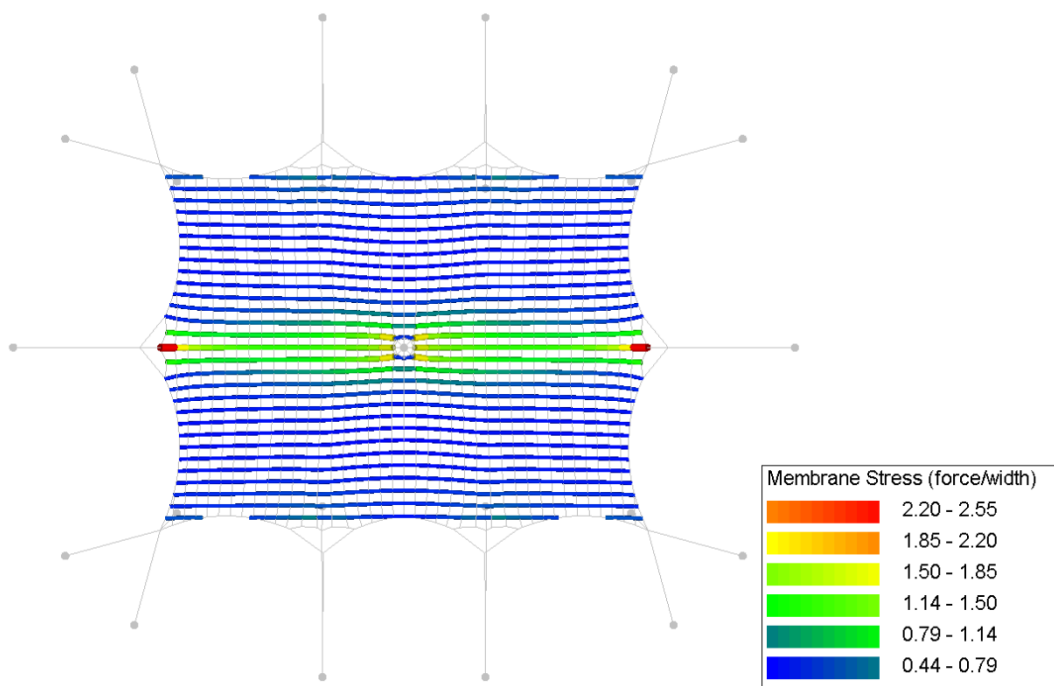


Annex C.2. CO2. Own weight + pretension + snow

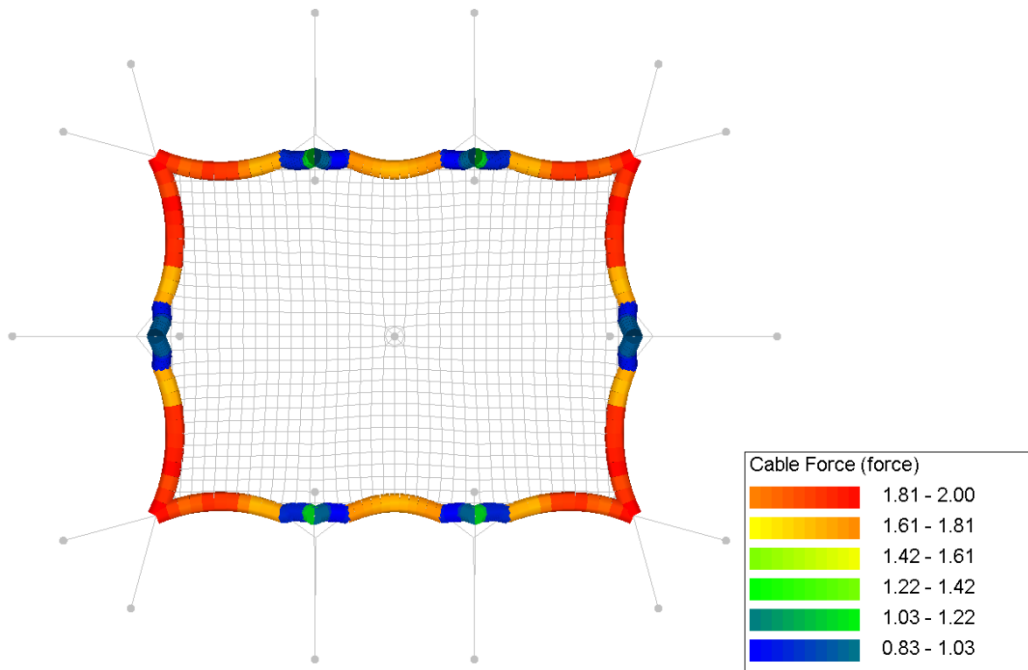
Annex C.2.1. Membrane stress (warp)



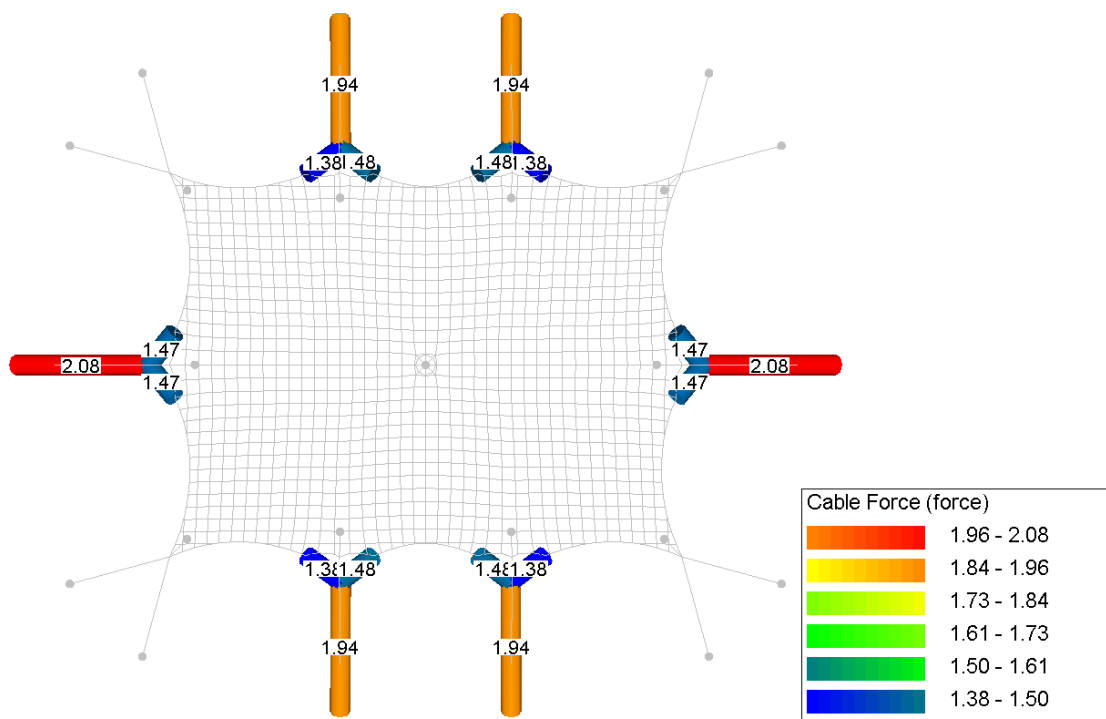
Annex C.2.2. Membrane stress (weft)



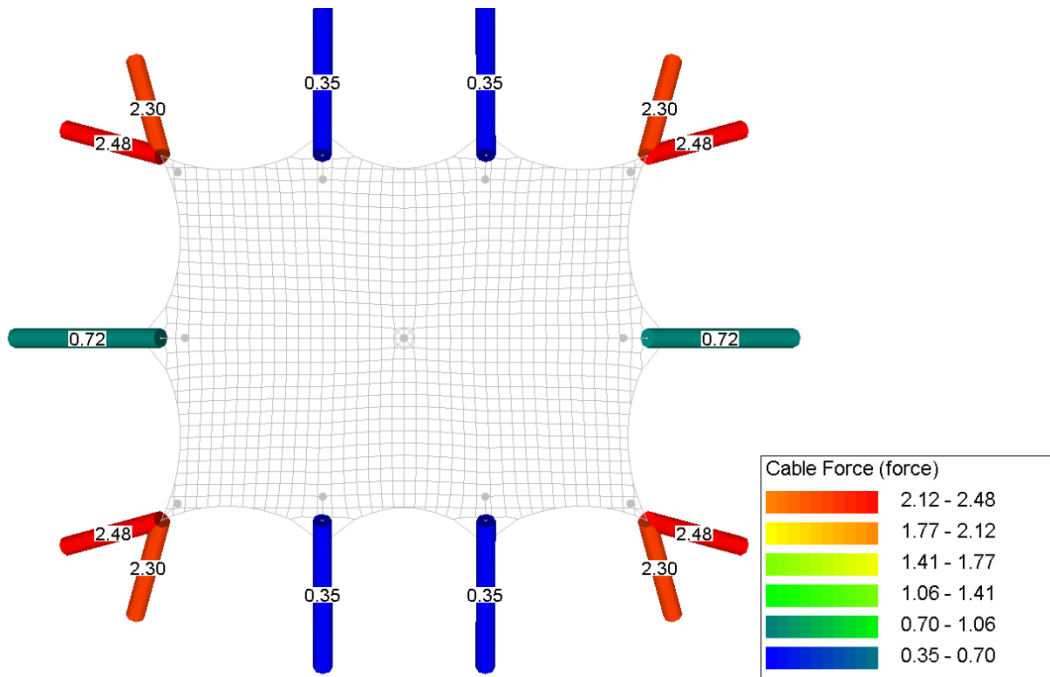
Annex C.2.3. Circumference rope forces



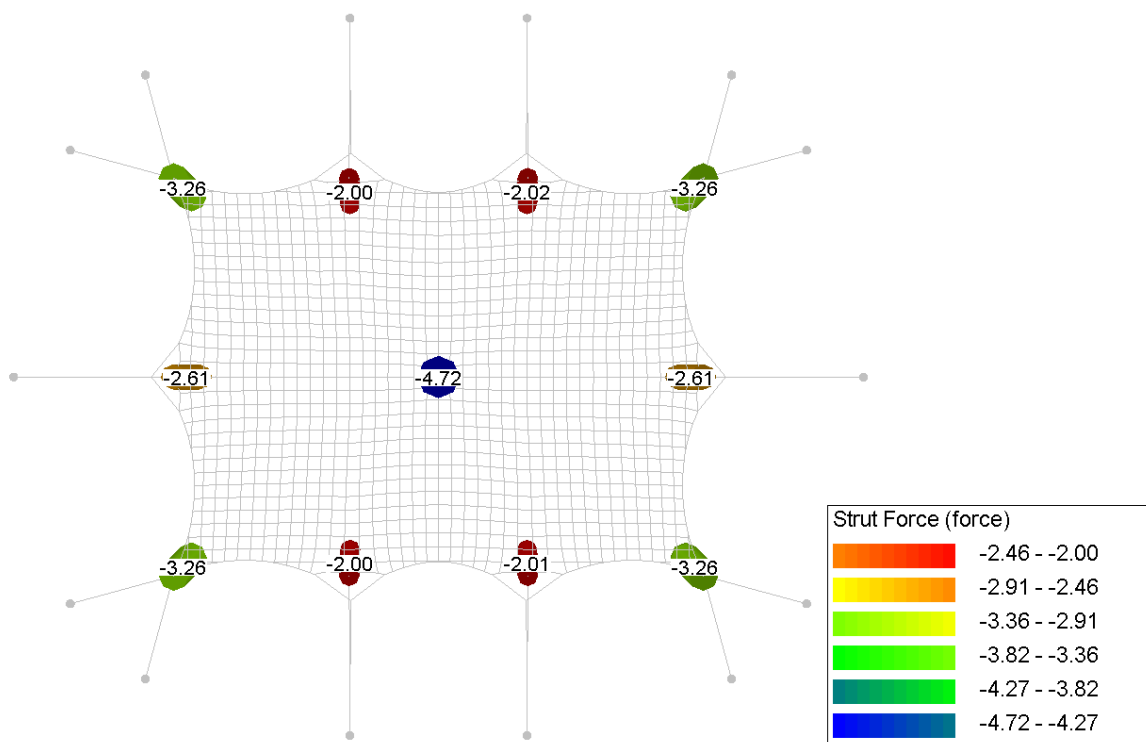
Annex C.2.4. Rope forces



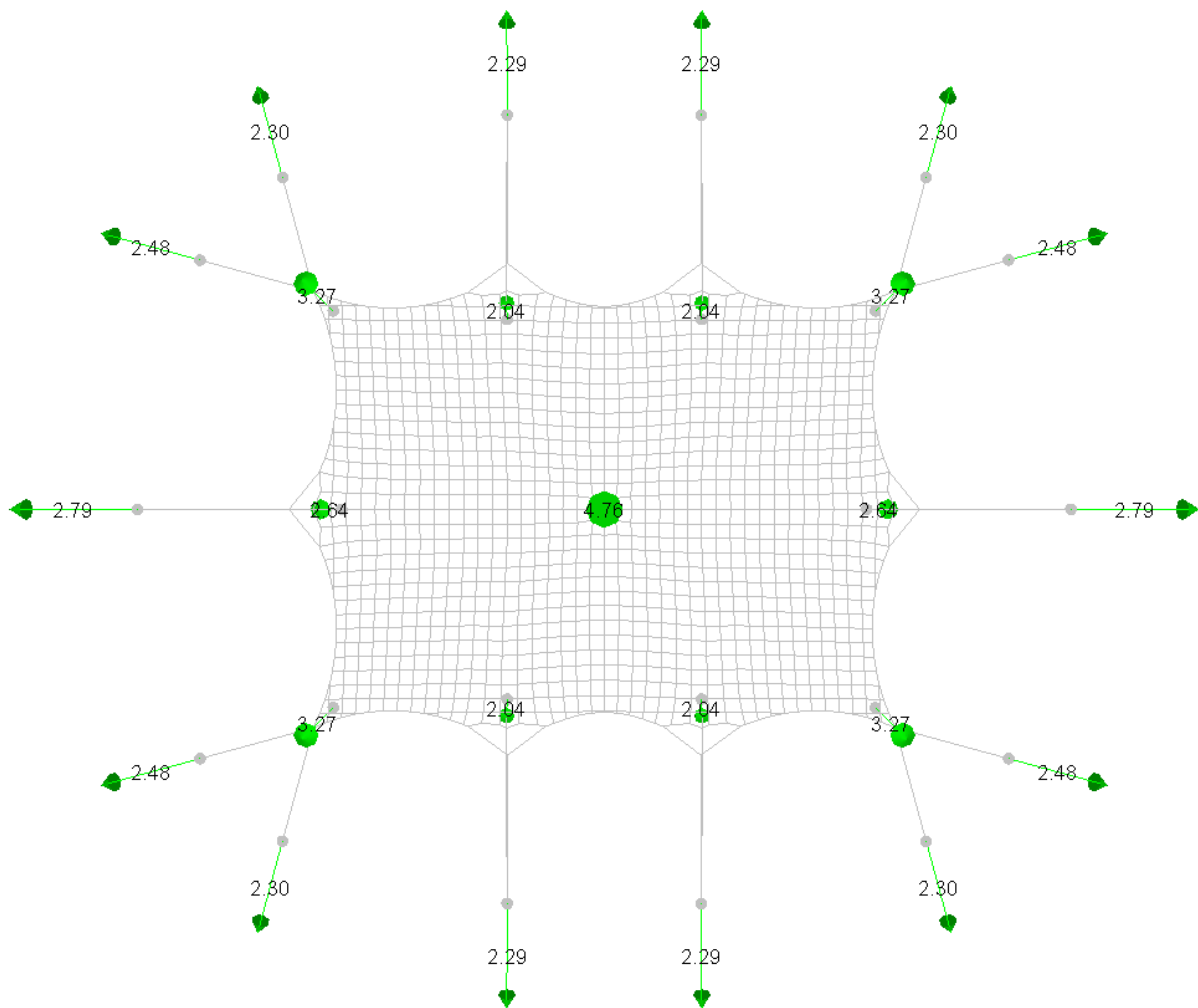
Annex C.2.5. Belt forces



Annex C.2.6. Pole forces

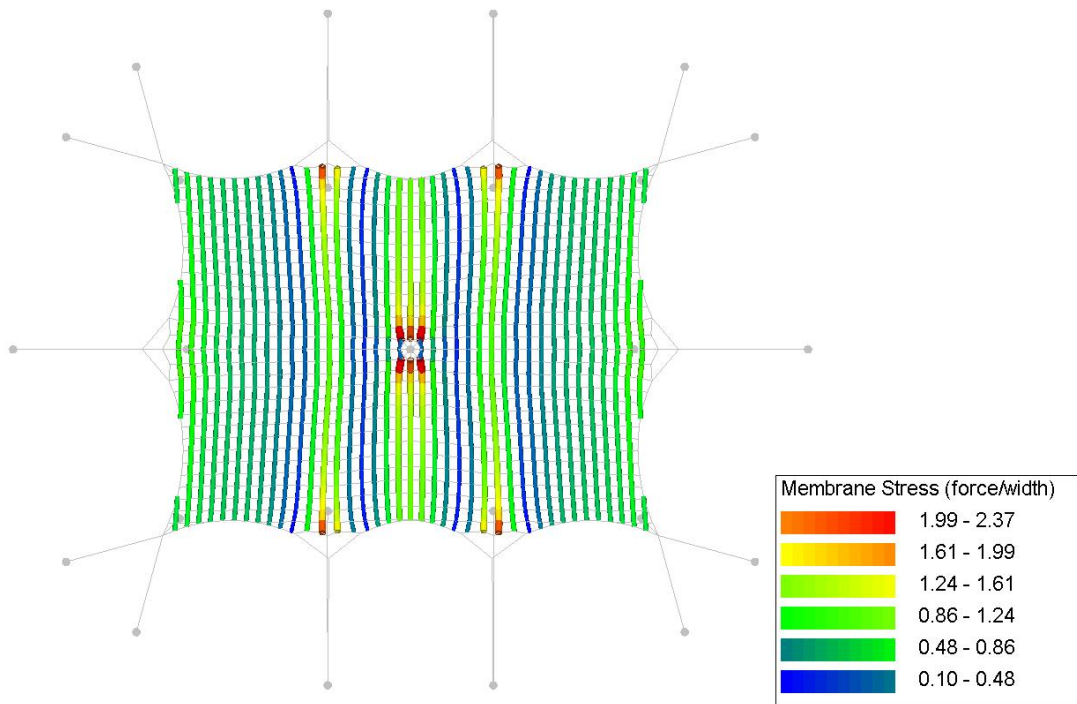


Annex C.2.7. Anchor forces

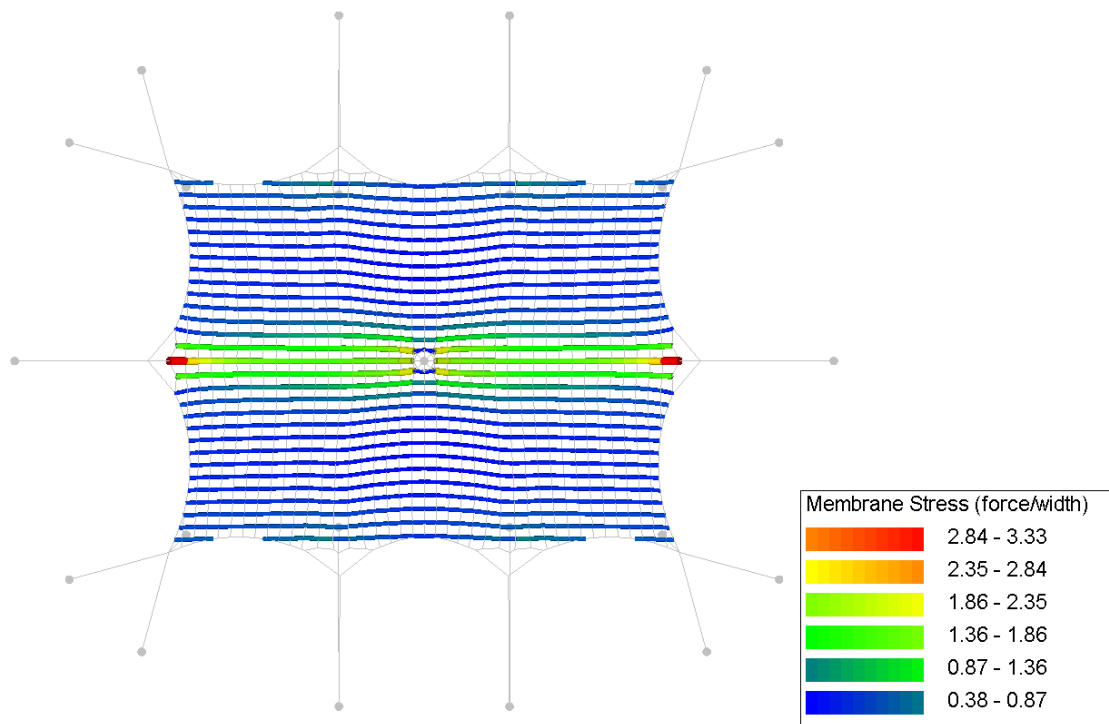


Annex C.3. CO3. Own weight + pretension + wind pressure

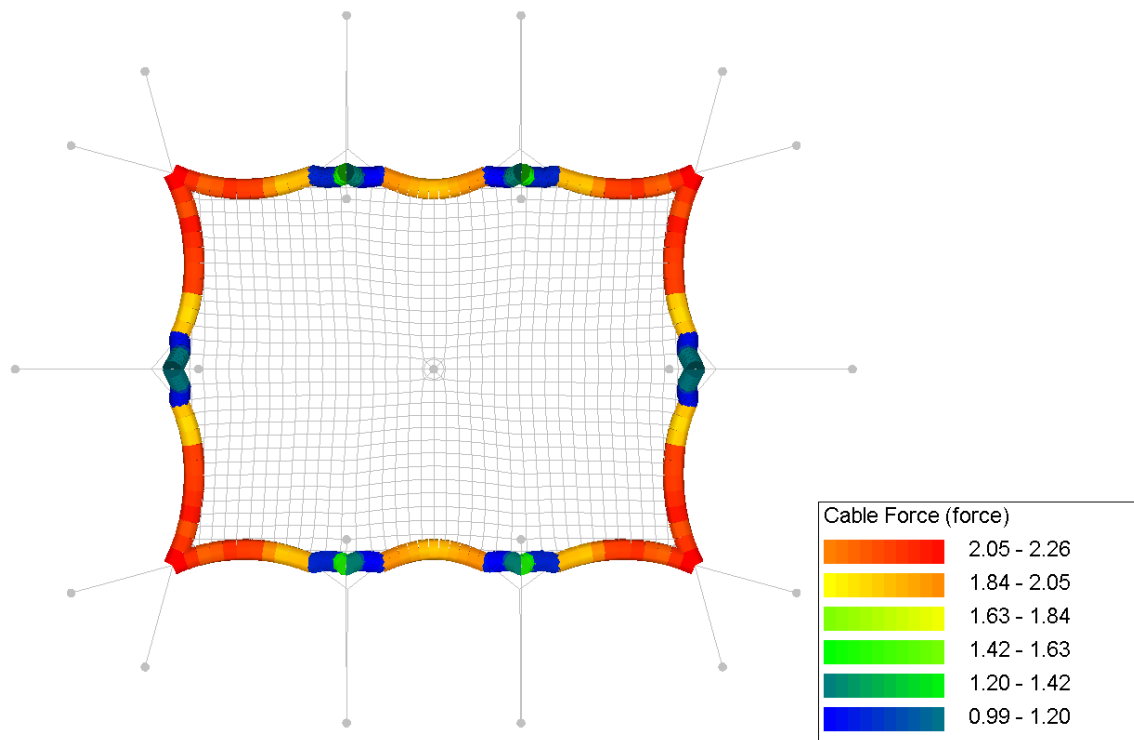
Annex C.3.1. Membrane stress (warp)



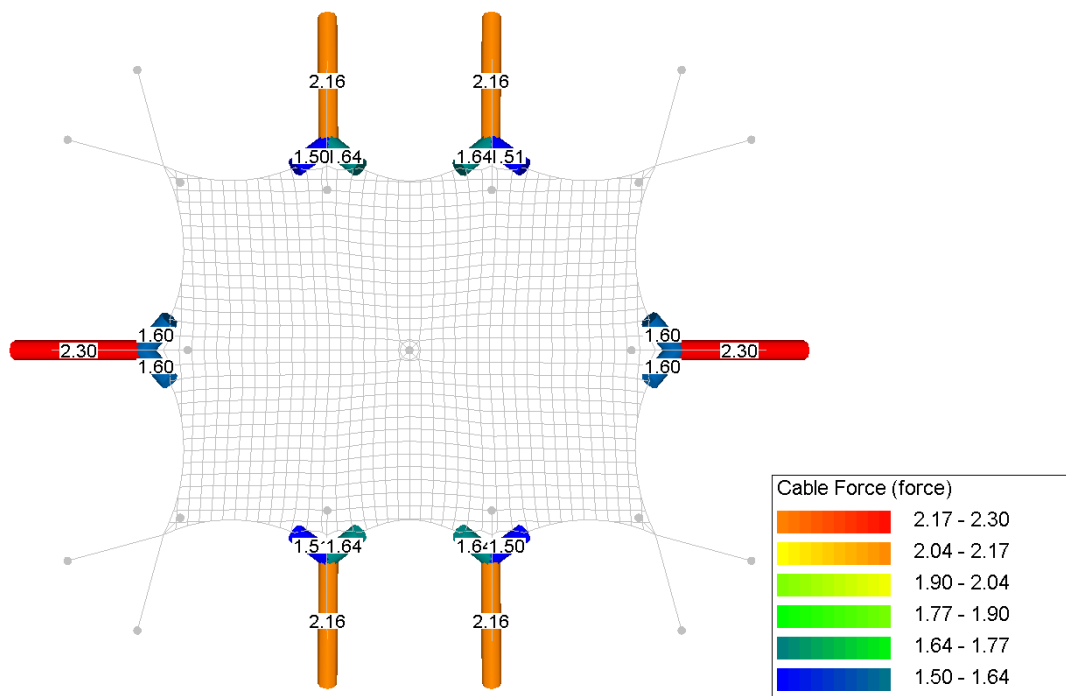
Annex C.3.2. Membrane stress (weft)



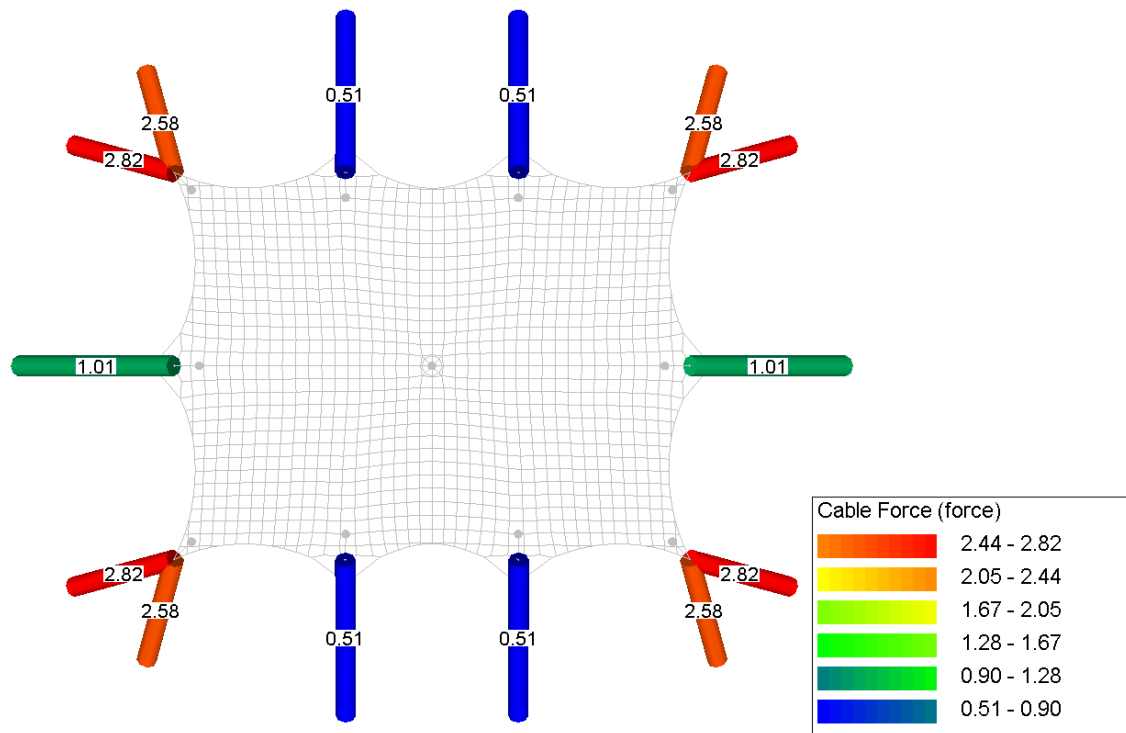
Annex C.3.3. Circumference rope forces



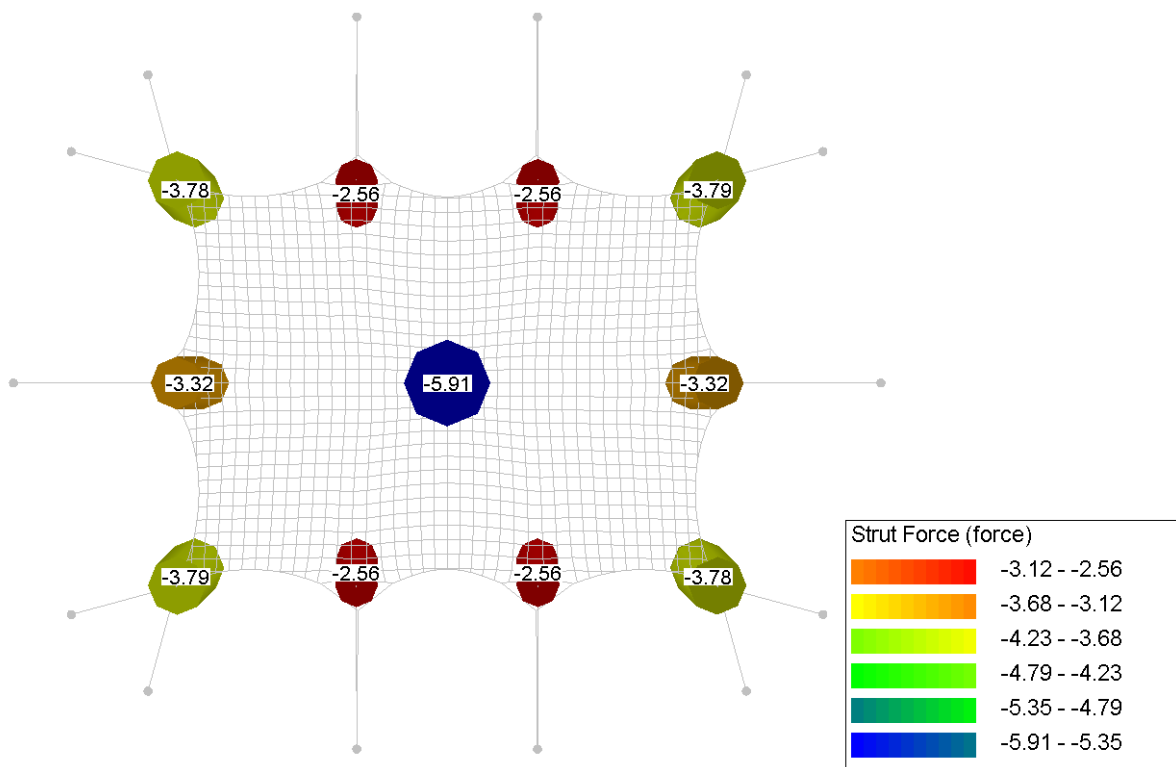
Annex C.3.4. Rope forces



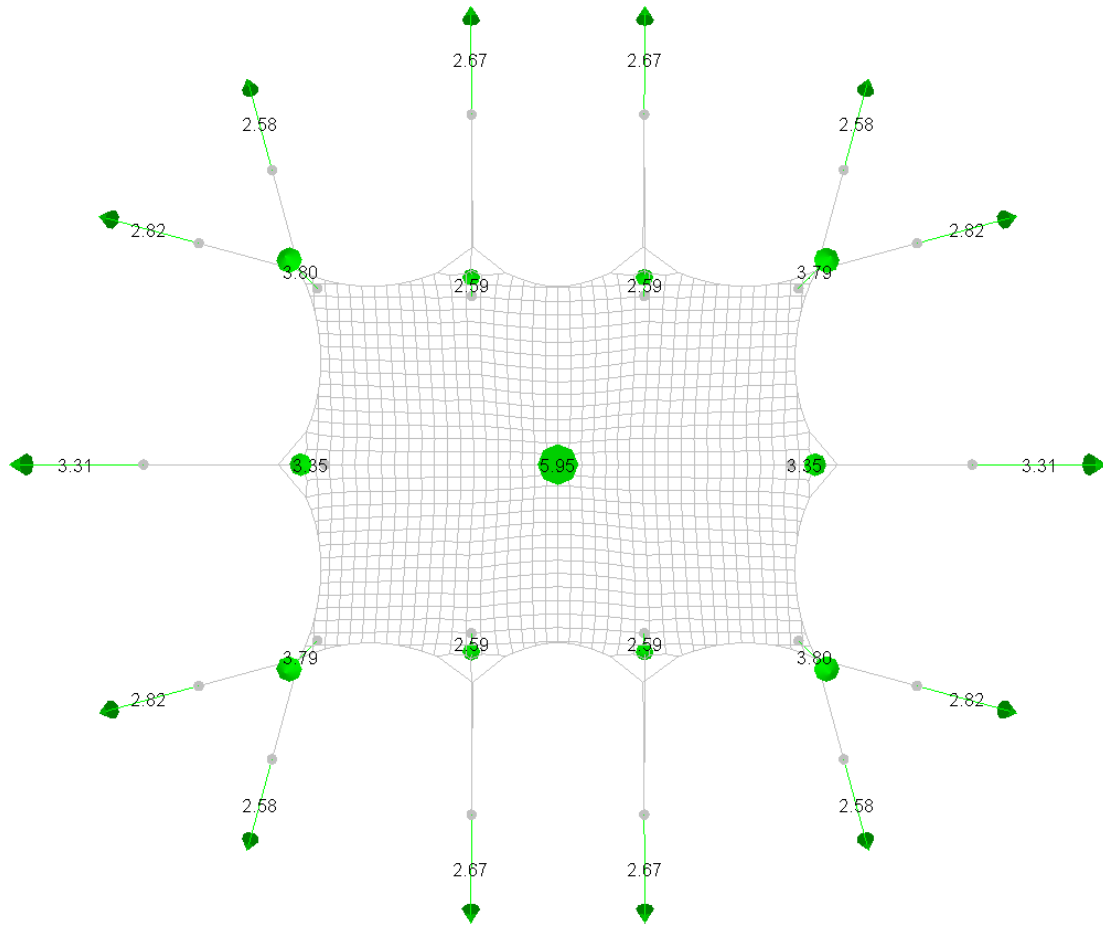
Annex C.3.5. Belt forces



Annex C.3.6. Poles forces

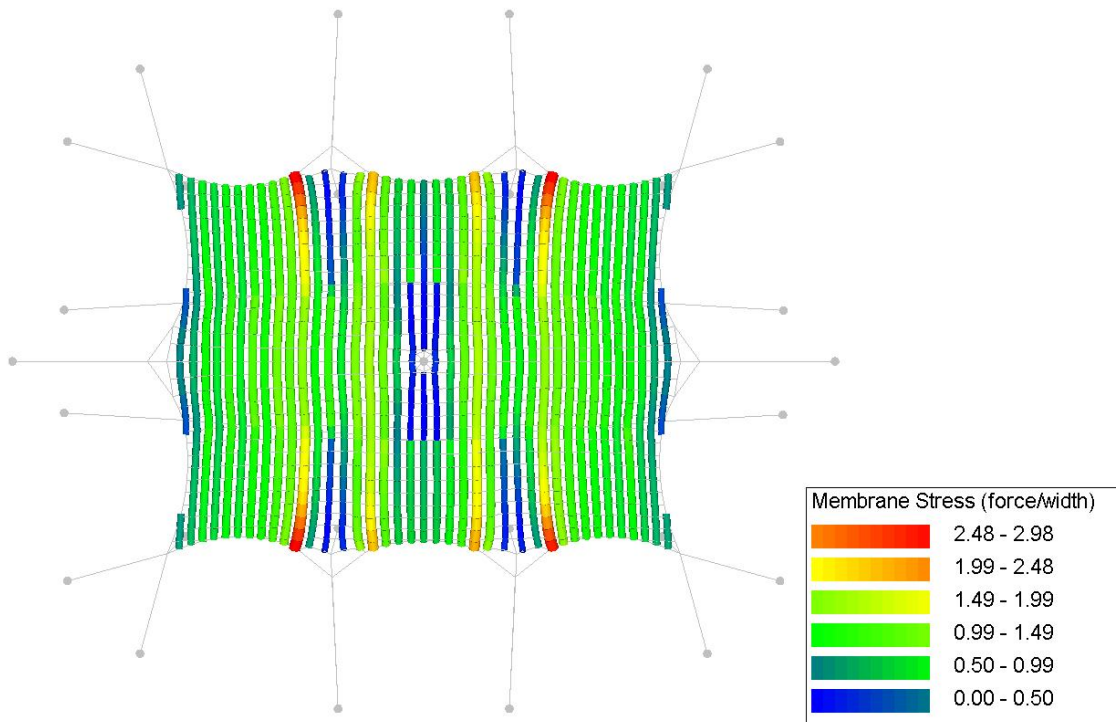


Annex C.3.7. Reaction forces

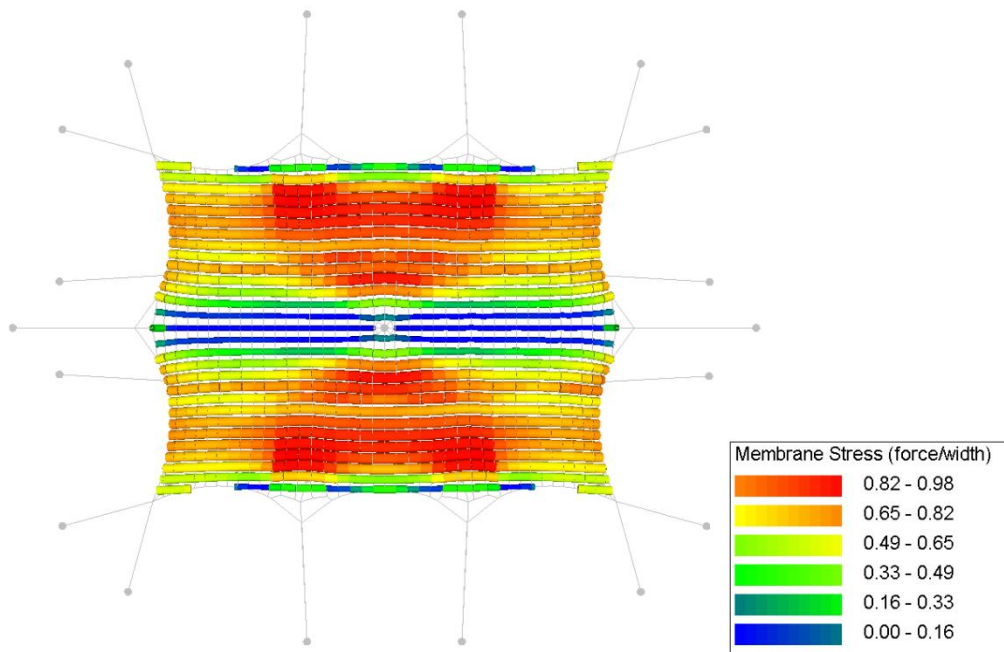


Annex C.4. CO4. Own weight + pretension + wind suction

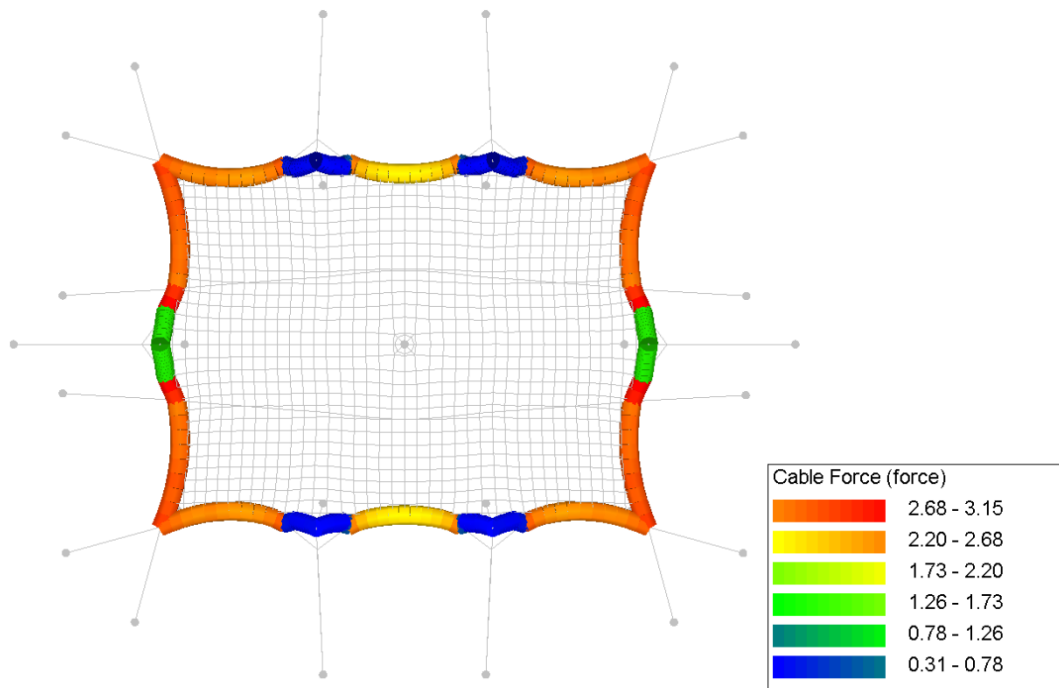
Annex C.4.1. Membrane stress (warp)



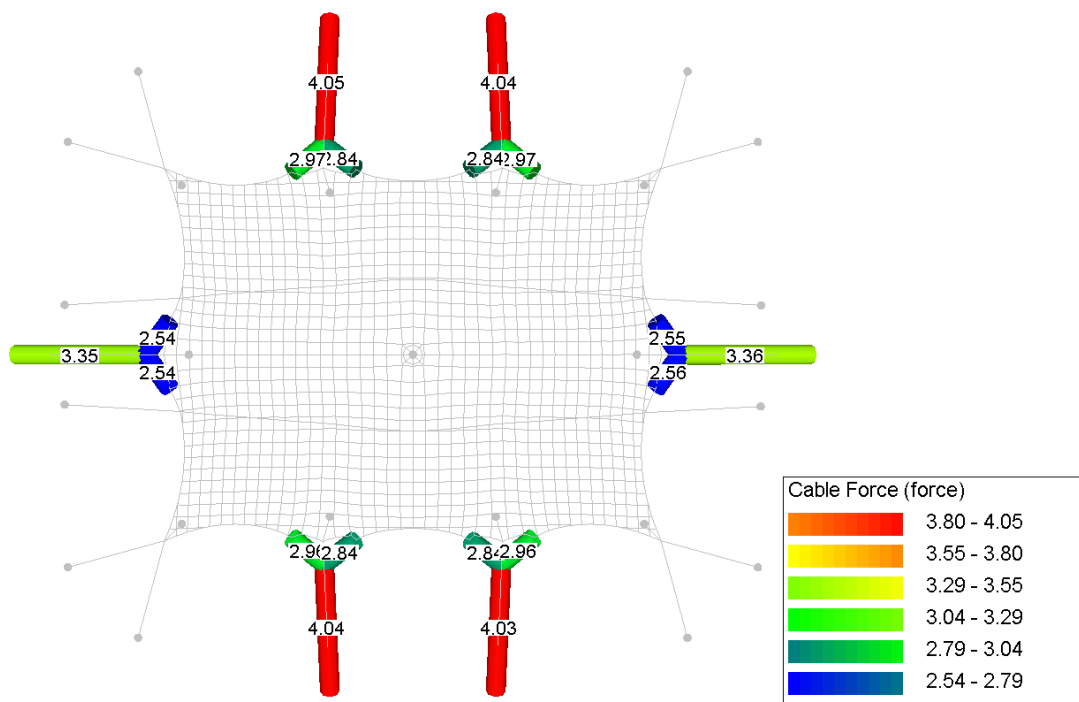
Annex C.4.2. Membrane stress (weft)



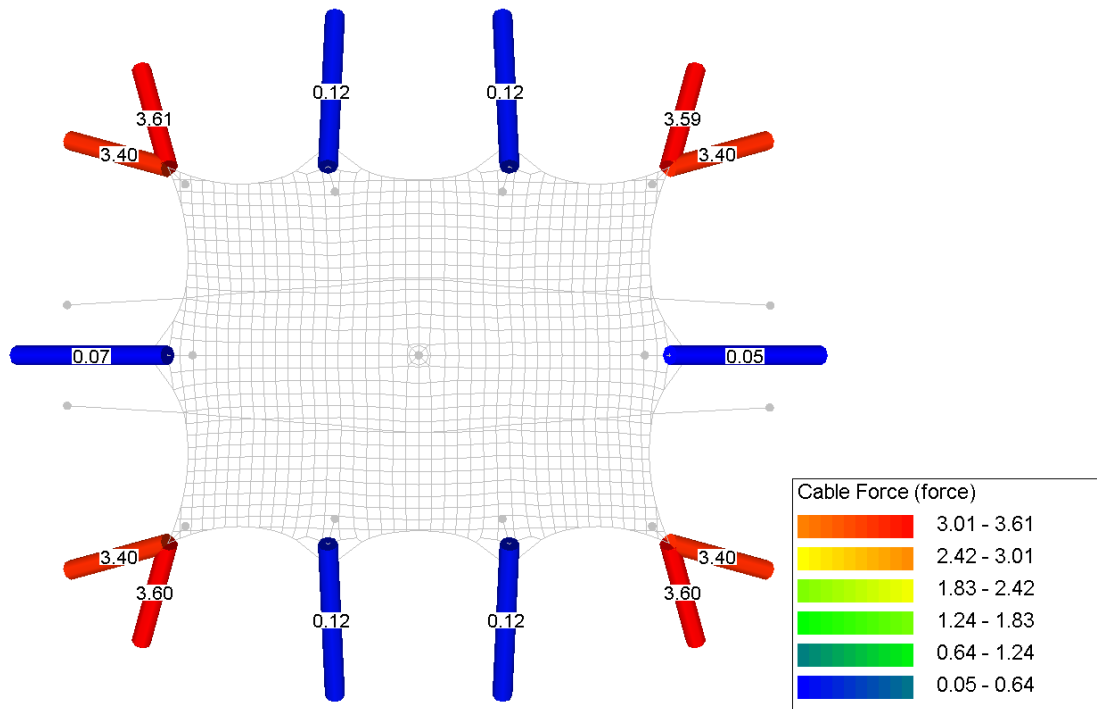
Annex C.4.3. Circumference rope forces



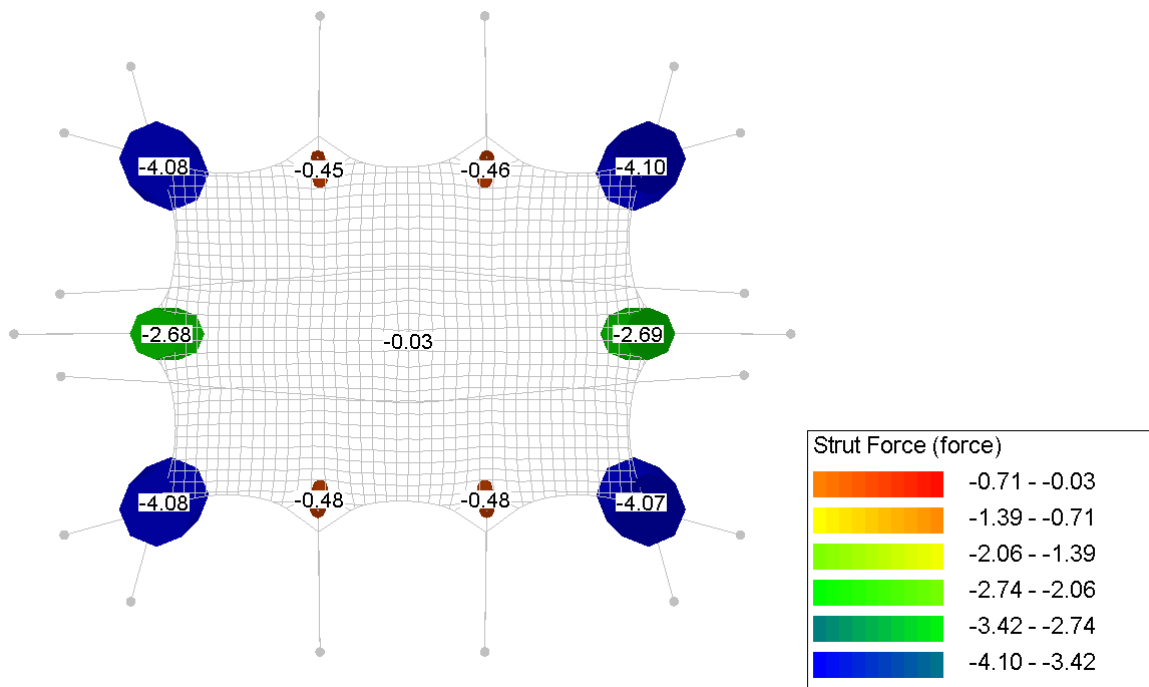
Annex C.4.4. Guy rope forces



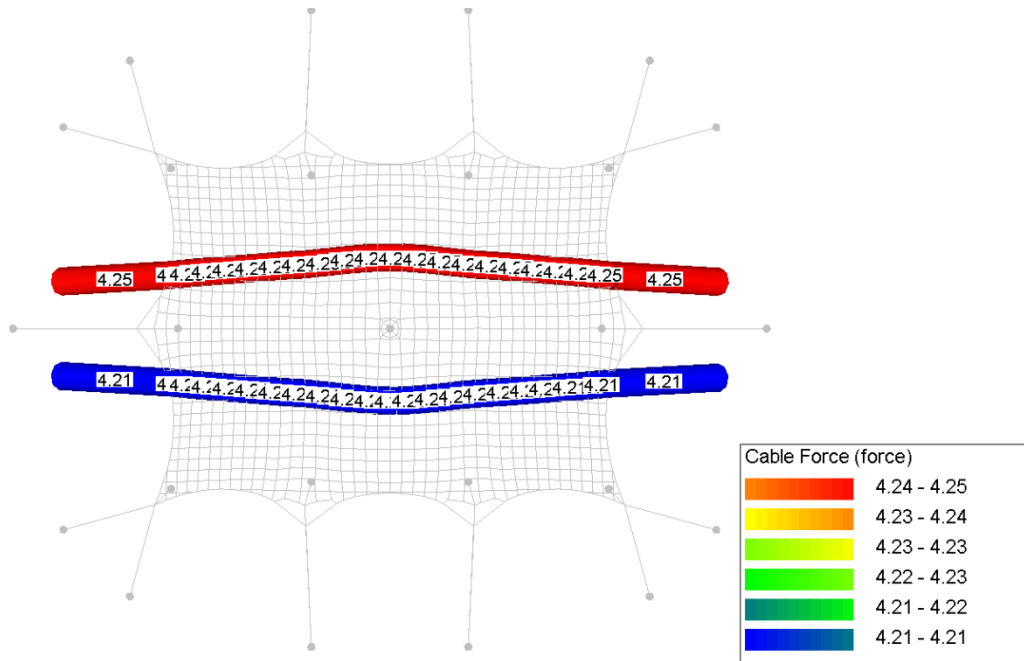
Annex C.4.5. Belt forces



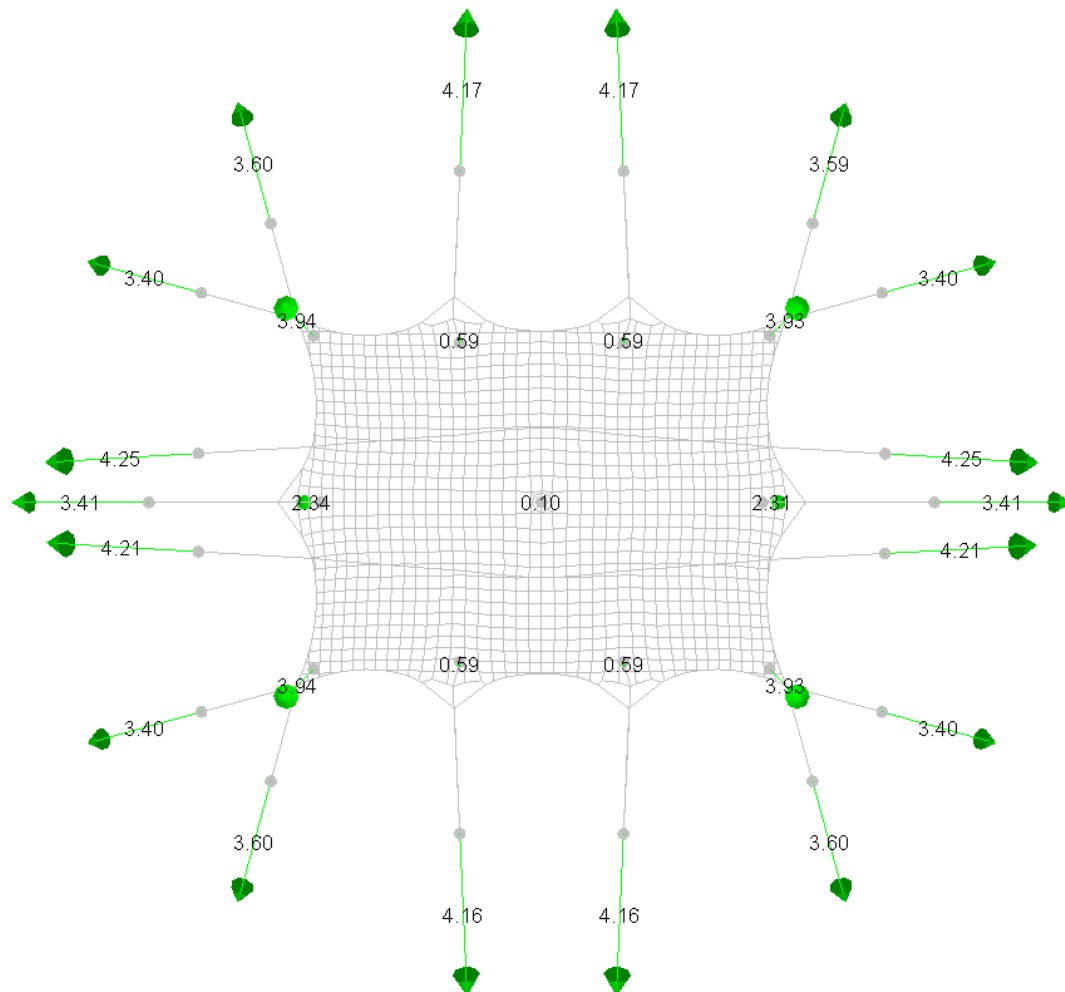
Annex C.4.6. Poles forces



Annex C.4.7. Storm belt forces

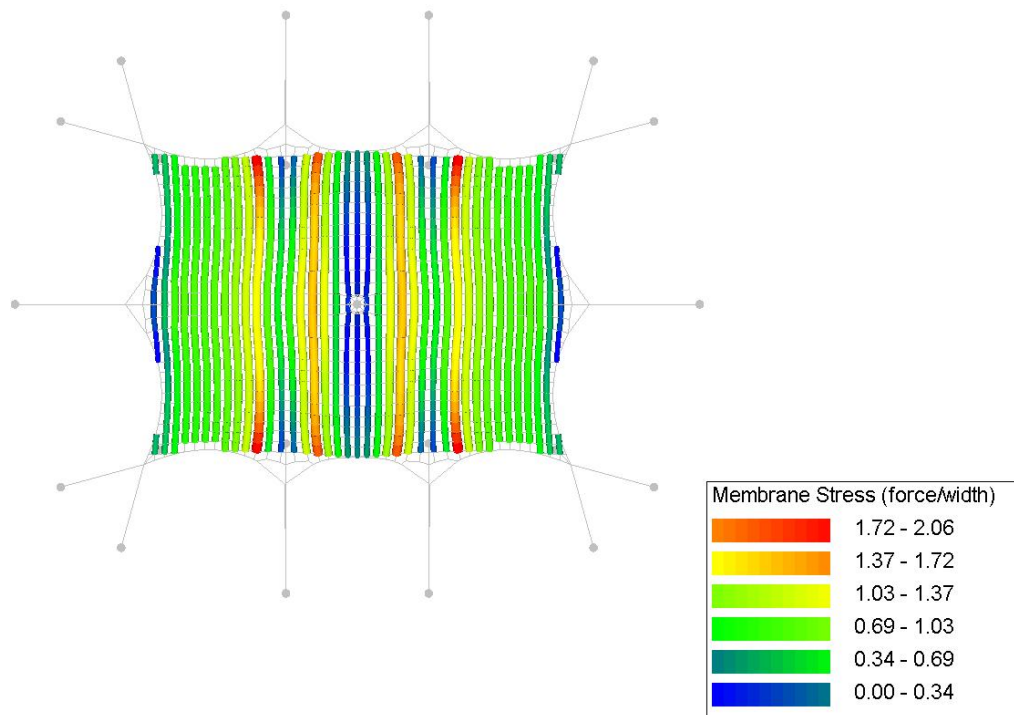


Annex C.4.8. Reaction forces

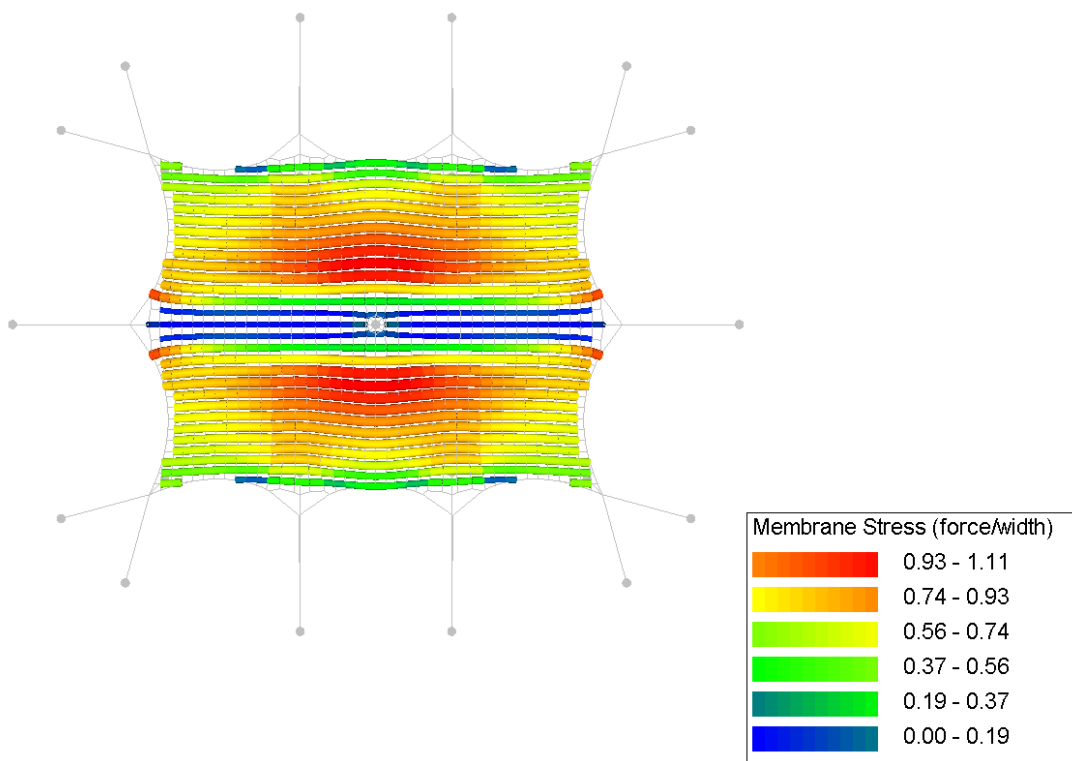


Annex C.5. CO5. Own weight + pretension + wind suction reduced [NL]

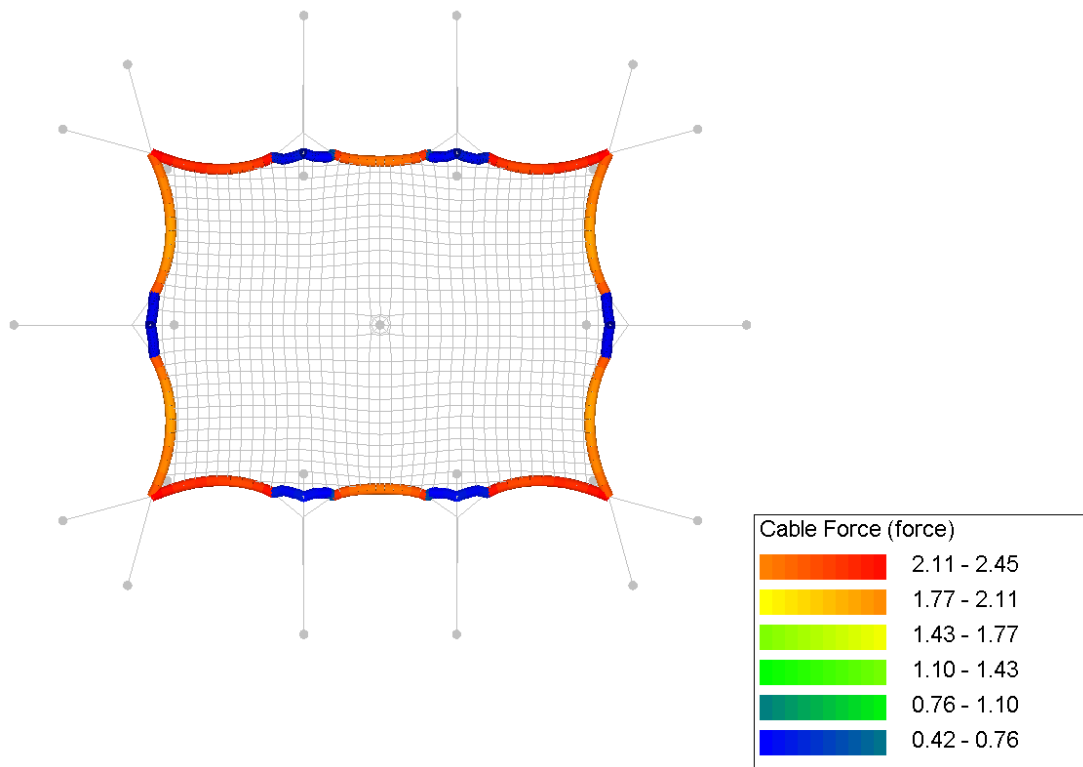
Annex C.5.1. Membrane stress (warp)



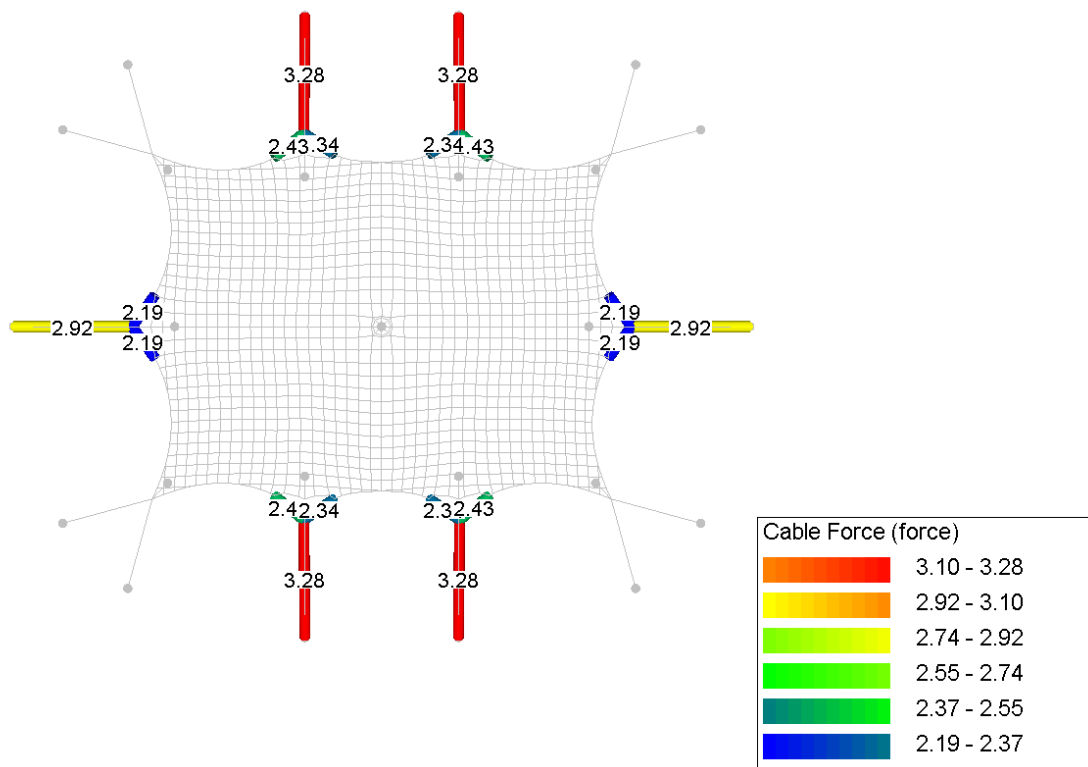
Annex C.5.2. Membrane stress (weft)



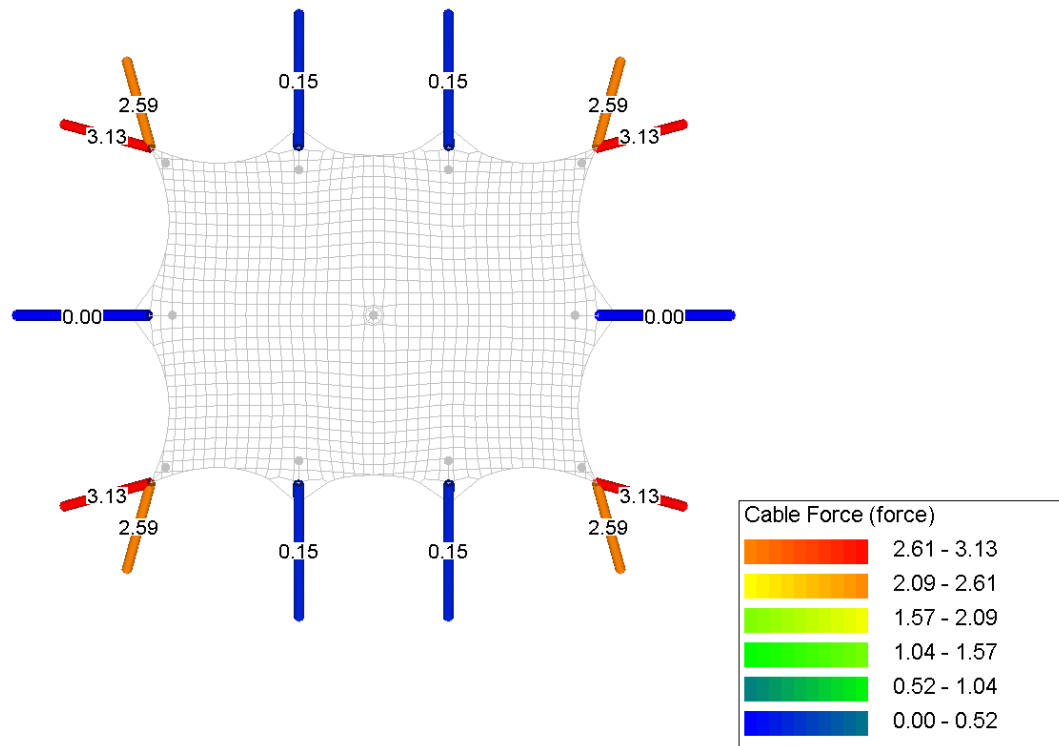
Annex C.5.3. Circumference rope forces



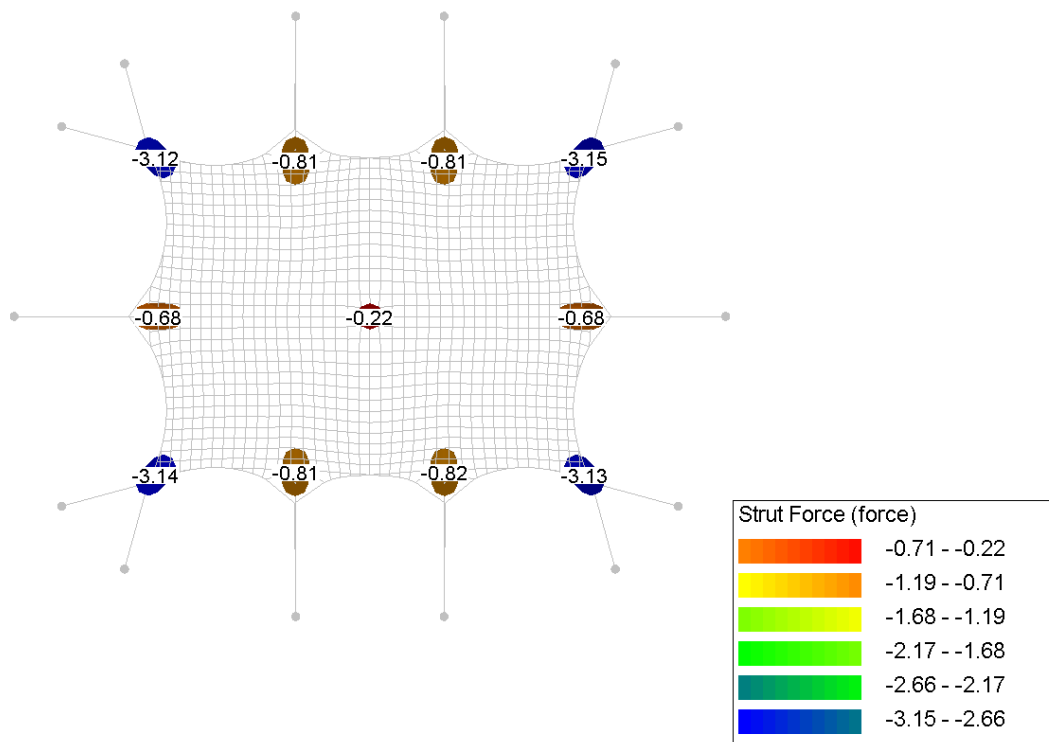
Annex C.5.4. Rope forces



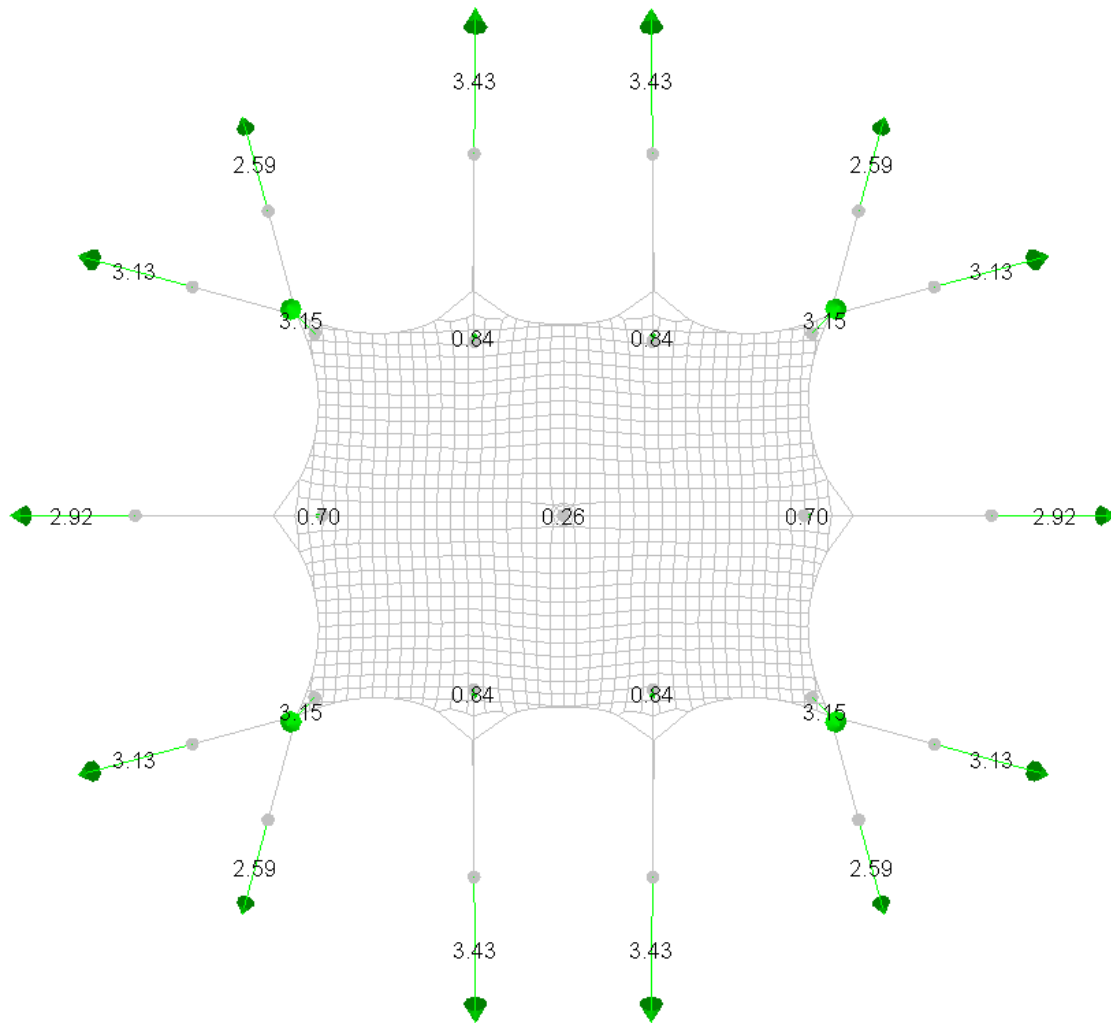
Annex C.5.5. Belt forces



Annex C.5.6. Poles forces

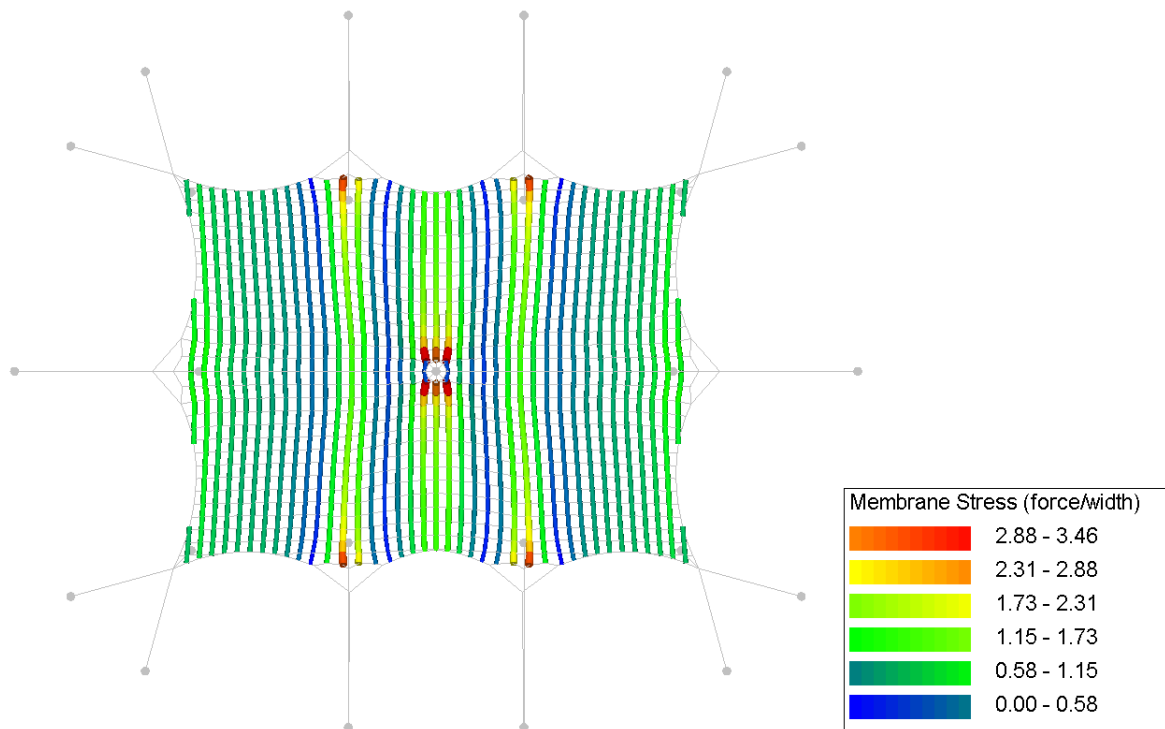


Annex C.5.7. Reaction forces

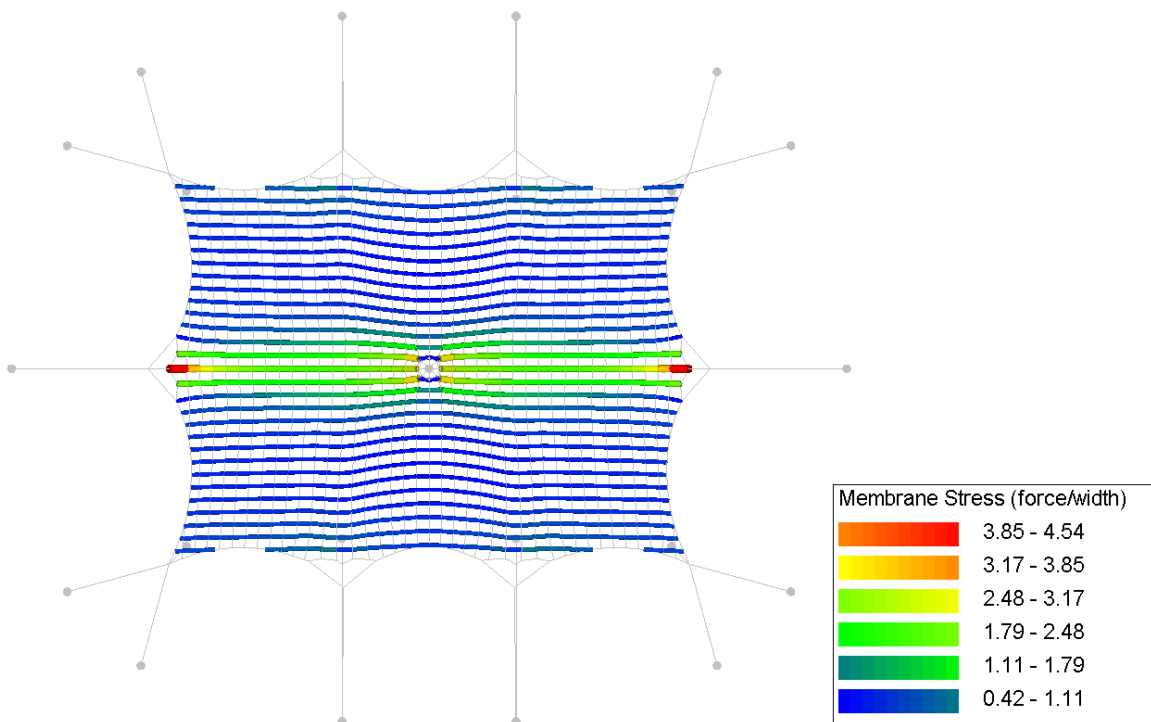


Annex C.6. CO6. Own weight + pretension + wind pressure + snow [FR]

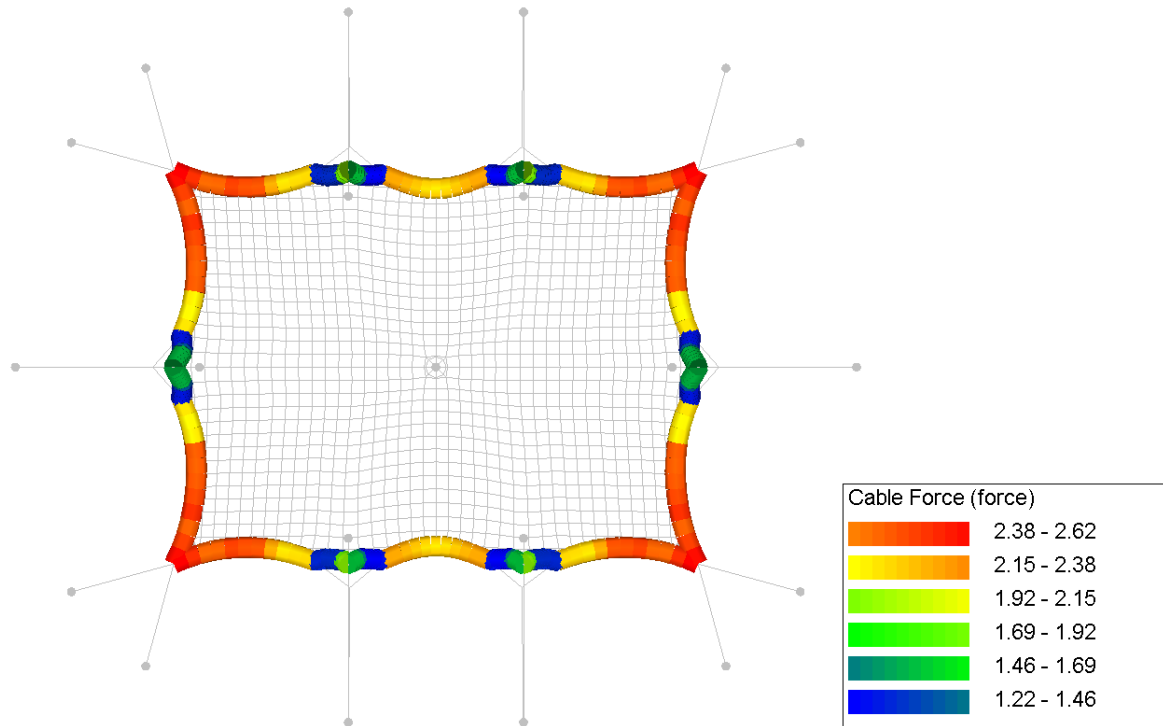
Annex C.6.1. Membrane stress (warp)



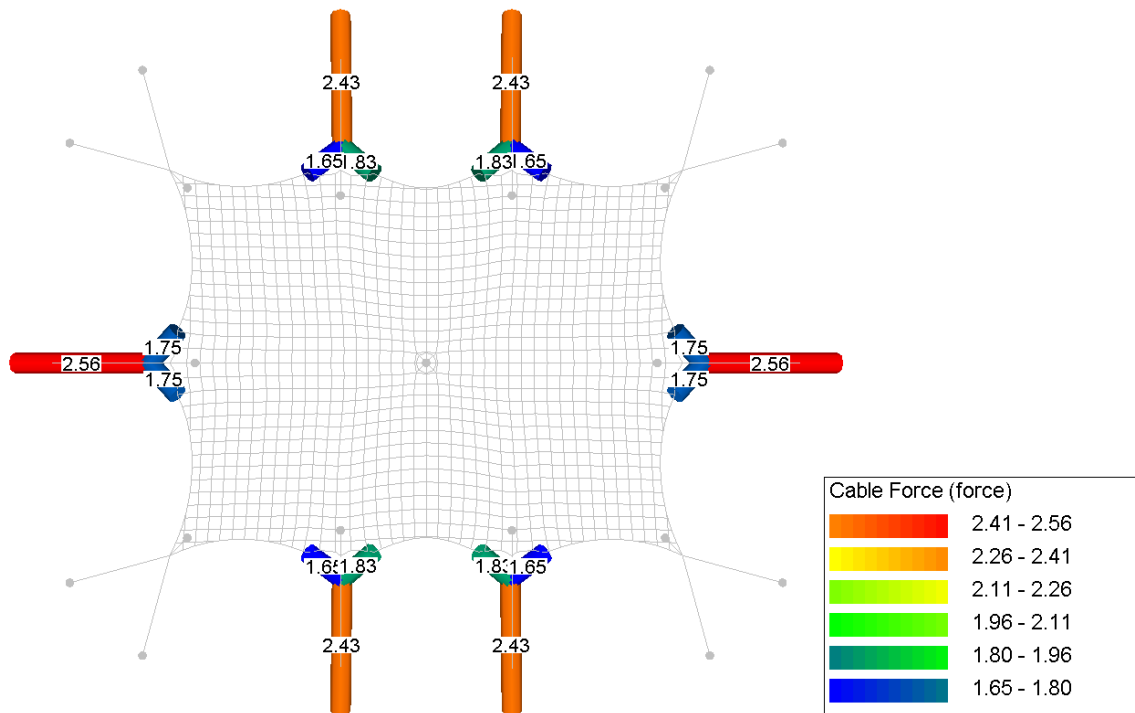
Annex C.6.2. Membrane stress (weft)



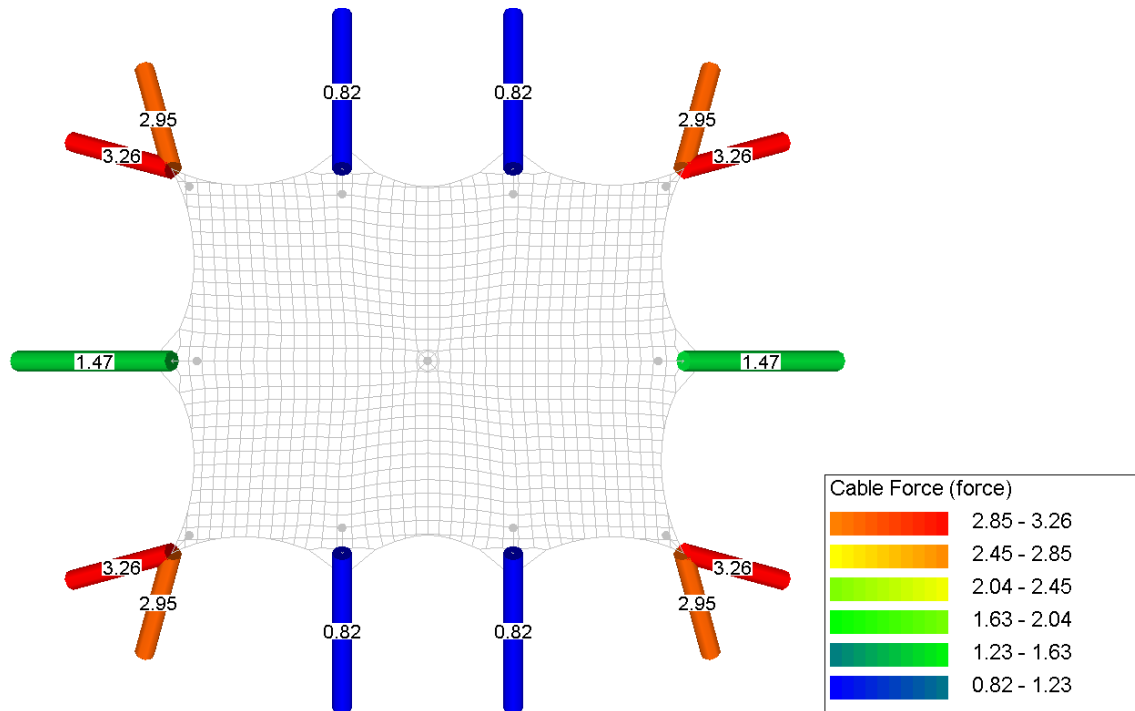
Annex C.6.3. Circumference rope forces



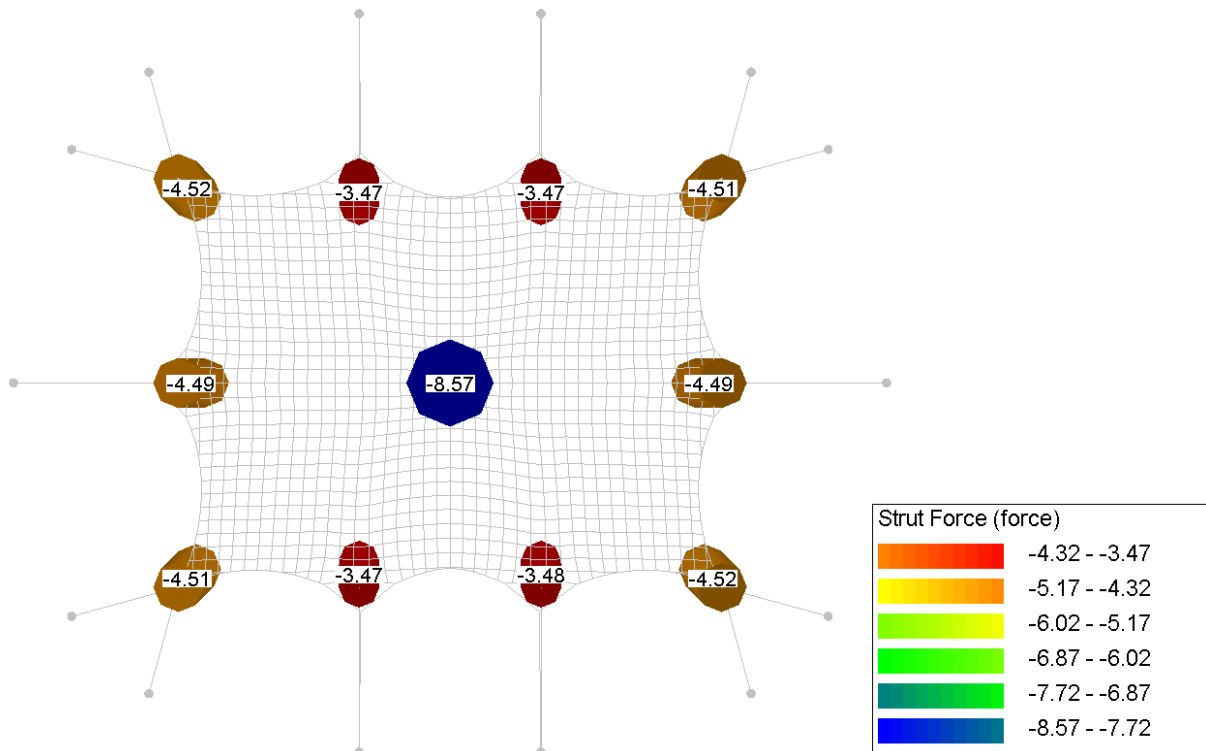
Annex C.6.4. Guy rope forces



Annex C.6.5. Belt forces

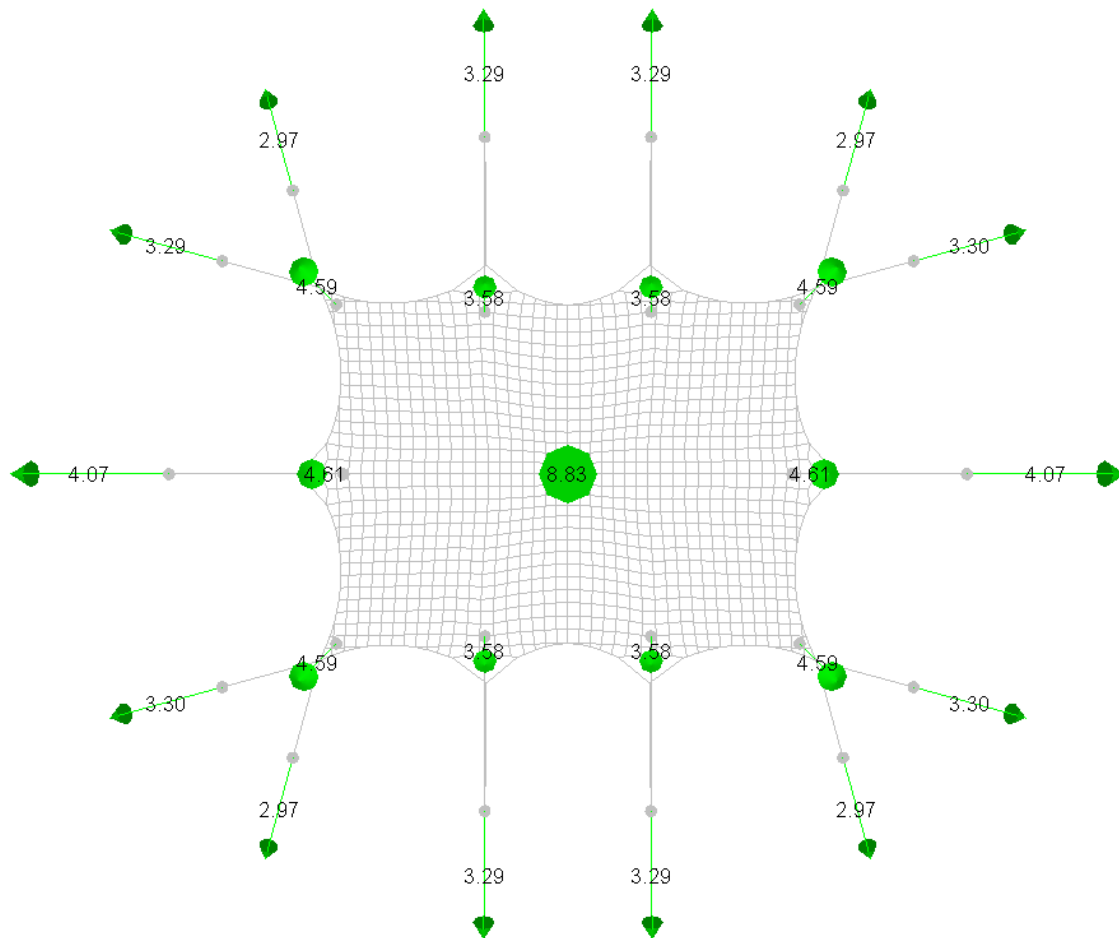


Annex C.6.6. Poles forces



Annex C.6.7. Reaction forces

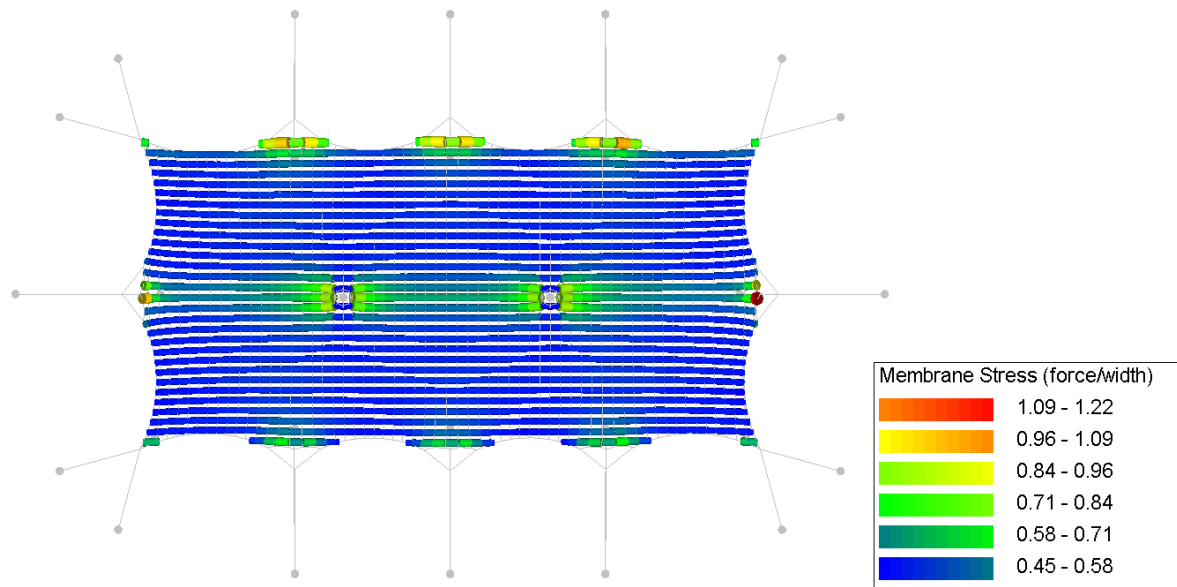
Increased snow load of $0.1 \times 1.5/1.35 = 0.1083 \text{ kN/m}^2$ is applied for ballast and anchor calculation



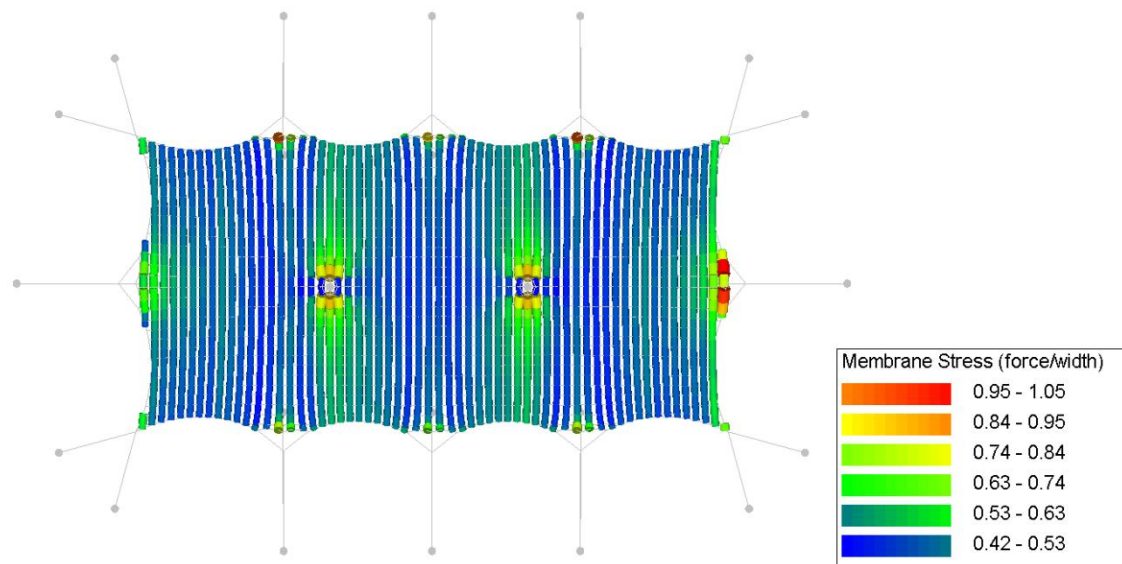
Annex D. Results per load combination – 7.5x15m model

Annex D.1. CO1. Own weight + pretension

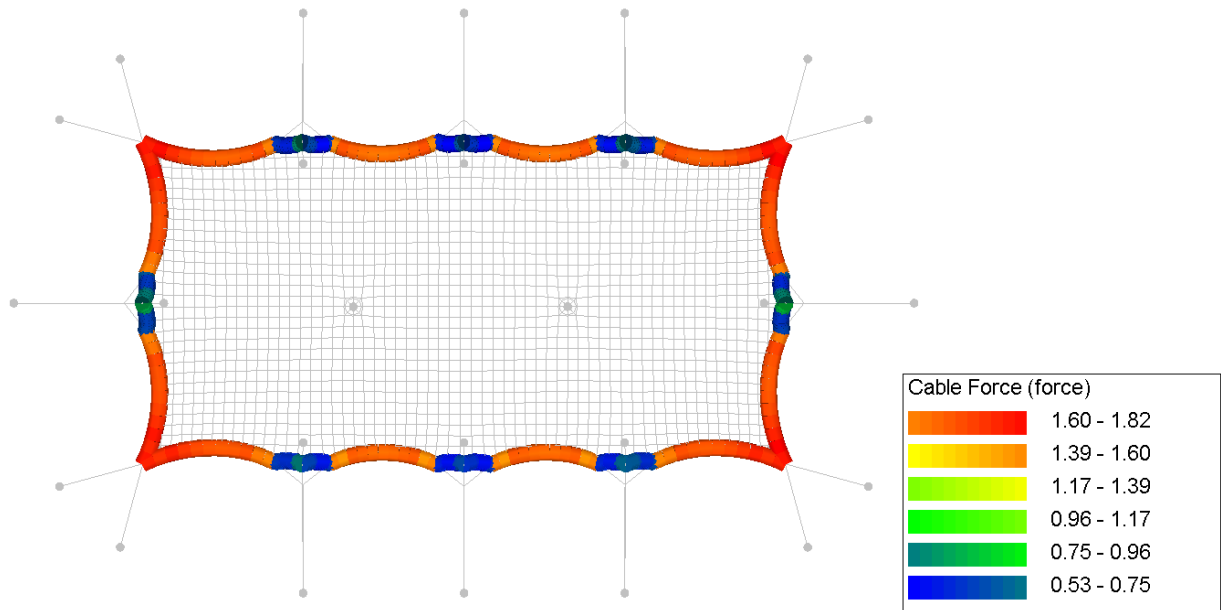
Annex D.1.1. Membrane stress (warp)



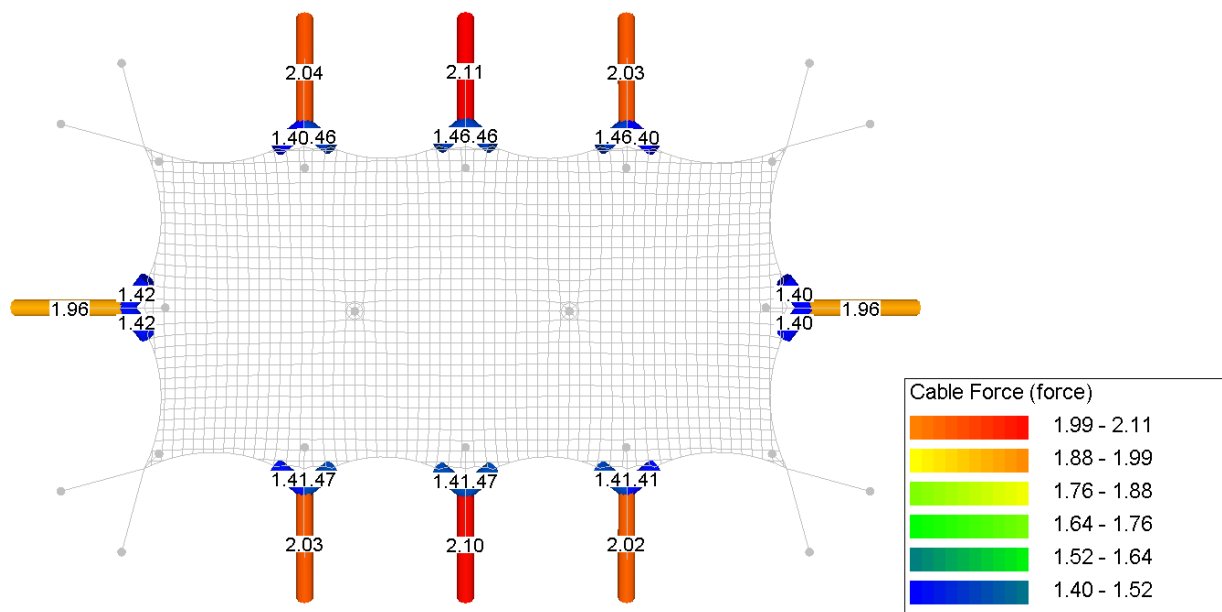
Annex D.1.2. Membrane stress (weft)



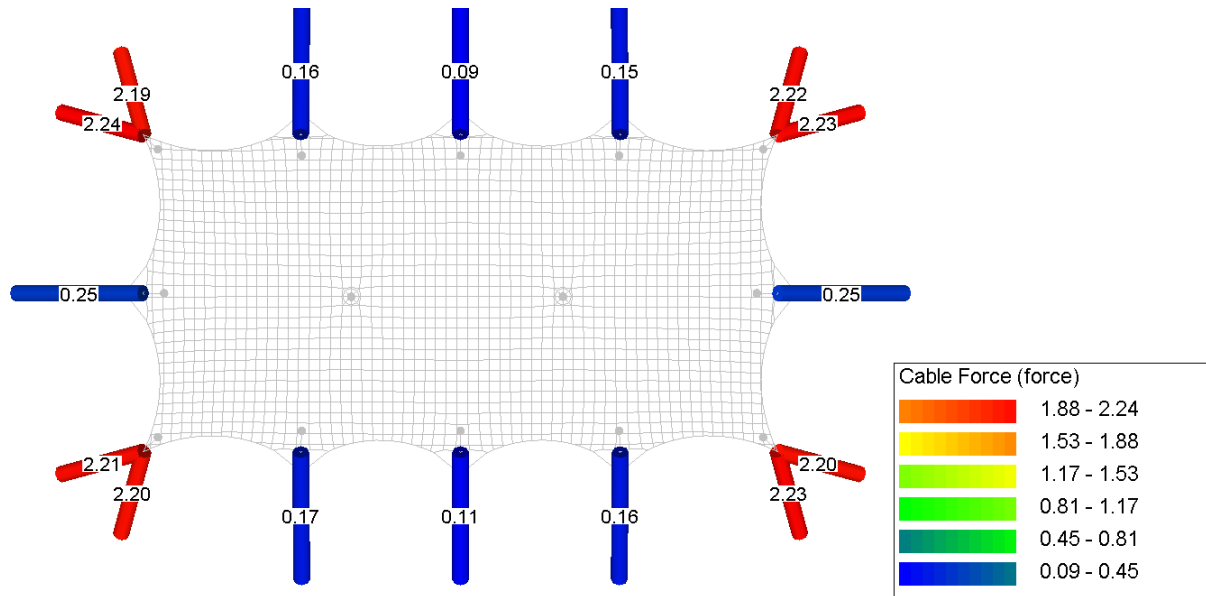
Annex D.1.3. Circumference rope forces



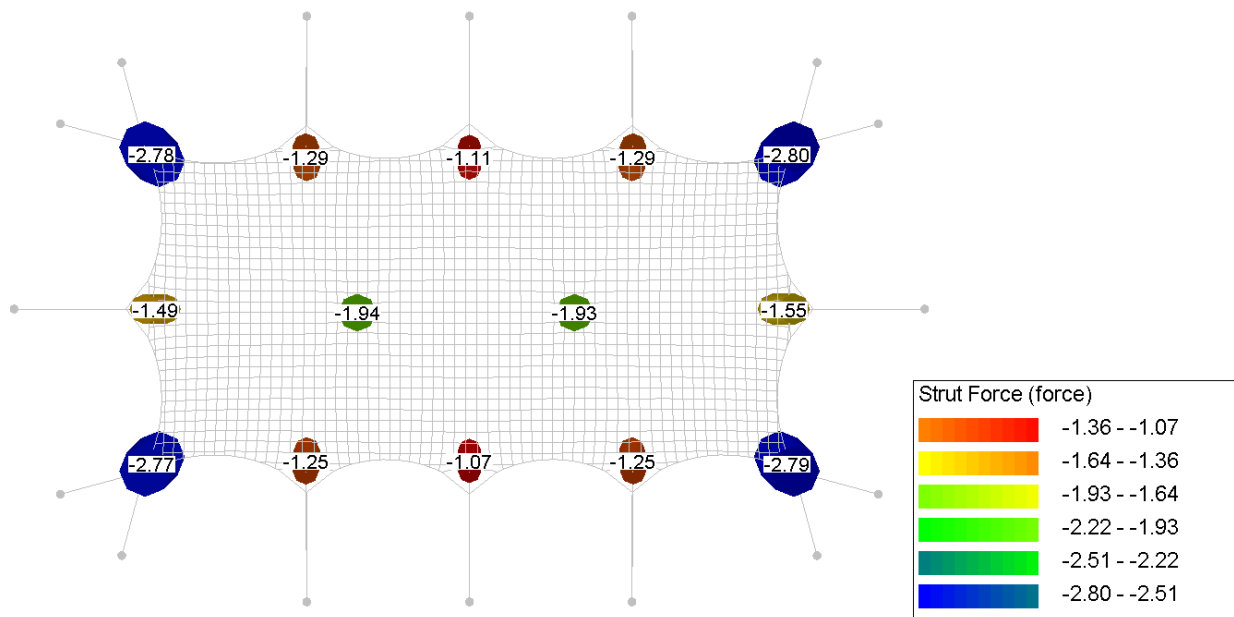
Annex D.1.4. Rope forces



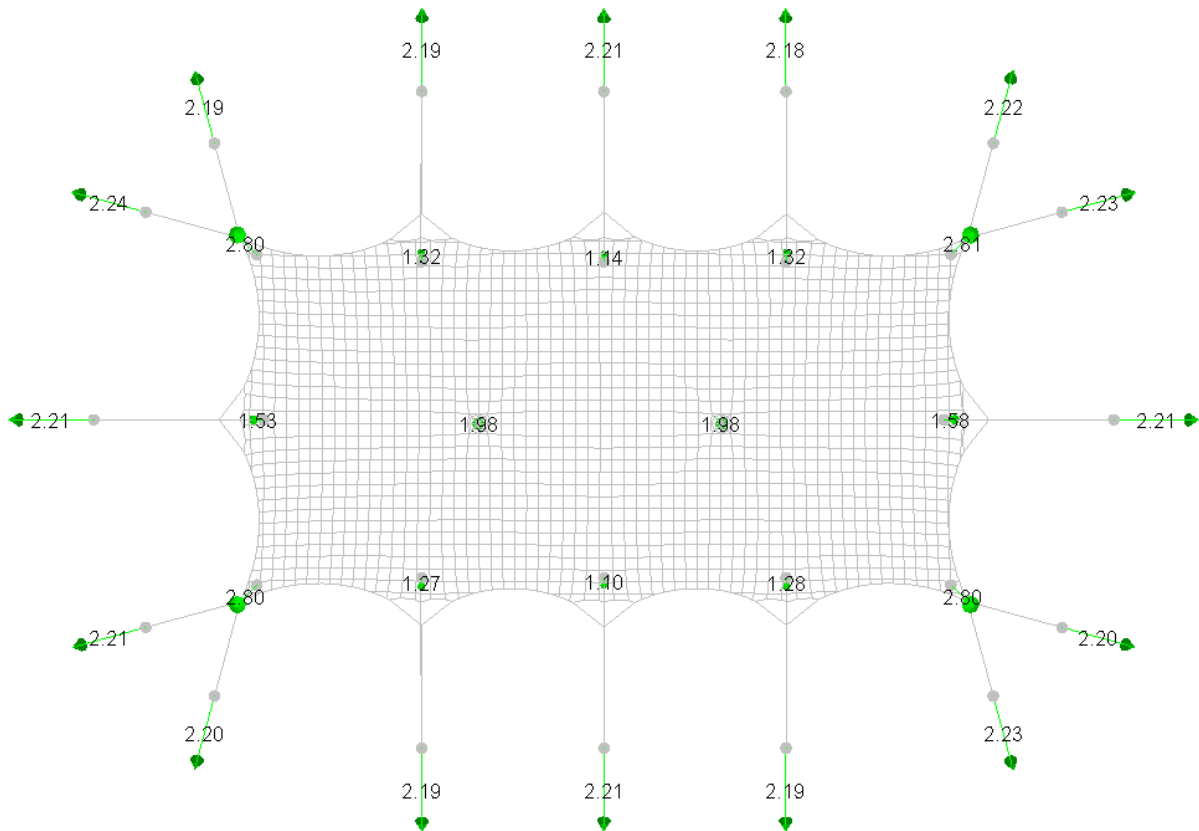
Annex D.1.5. Belt forces



Annex D.1.6. Pole forces

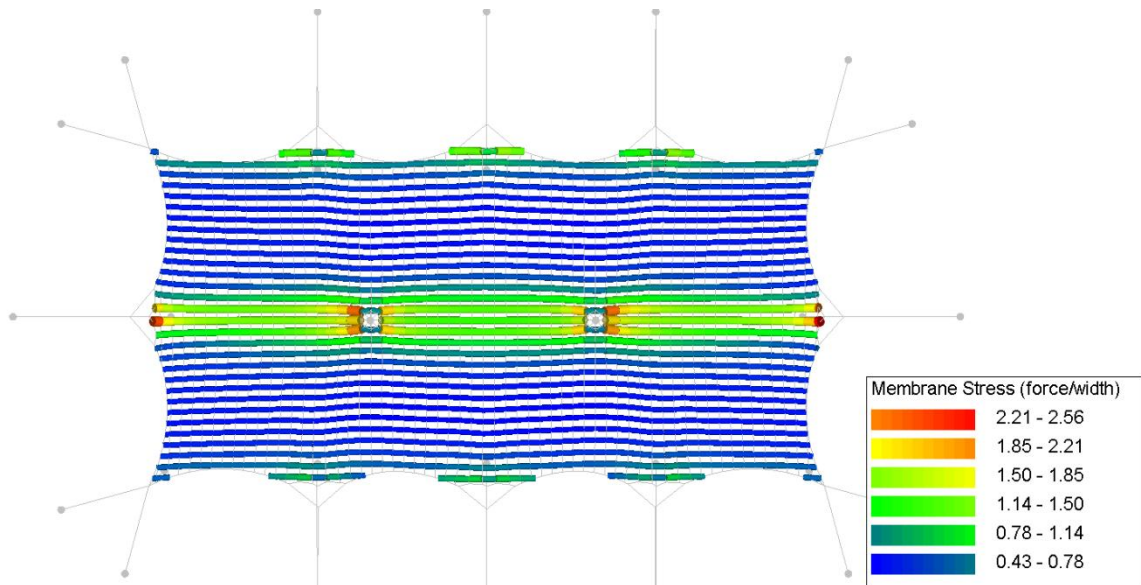


Annex D.1.7. Reaction forces

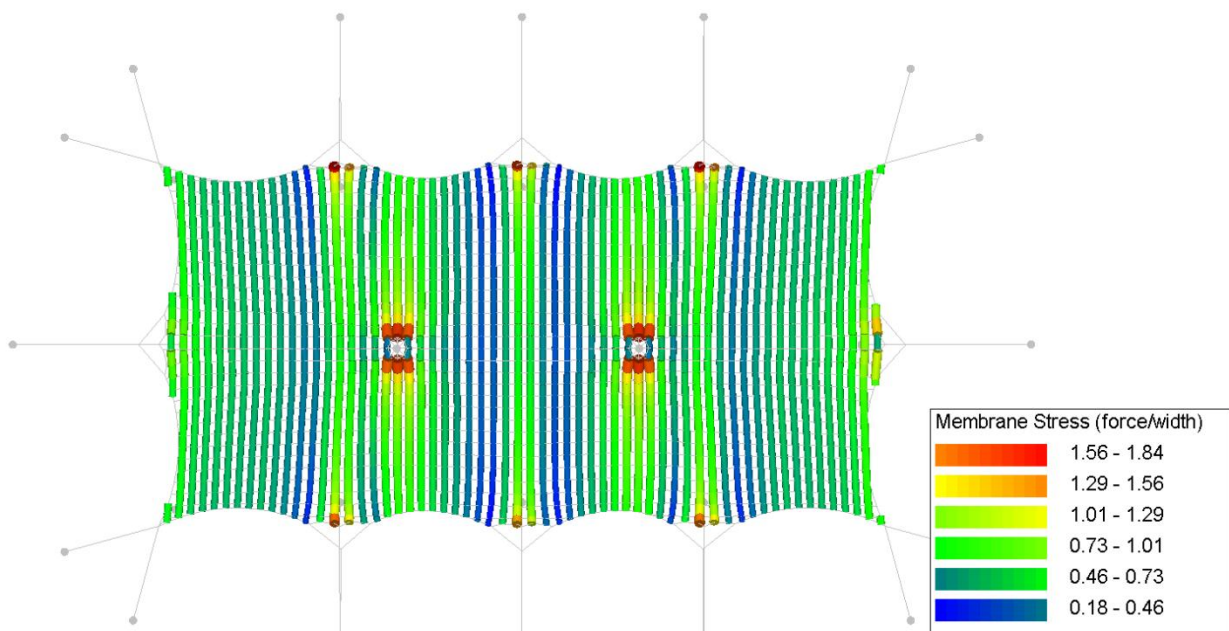


Annex D.2. CO2. Own weight + pretension + snow

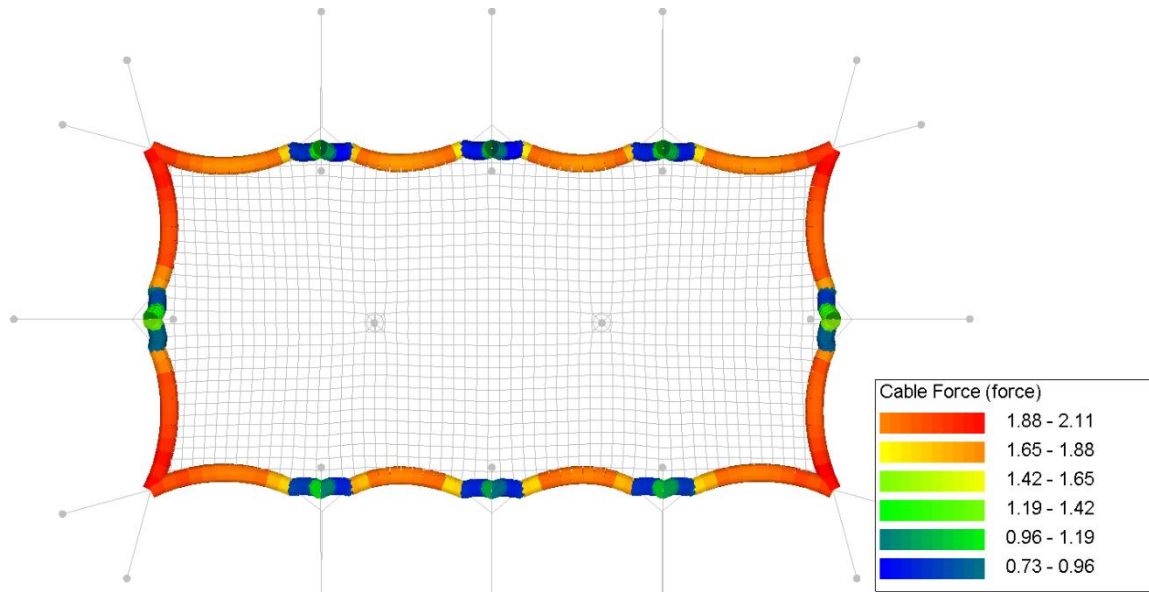
Annex D.2.1. Membrane stress (warp)



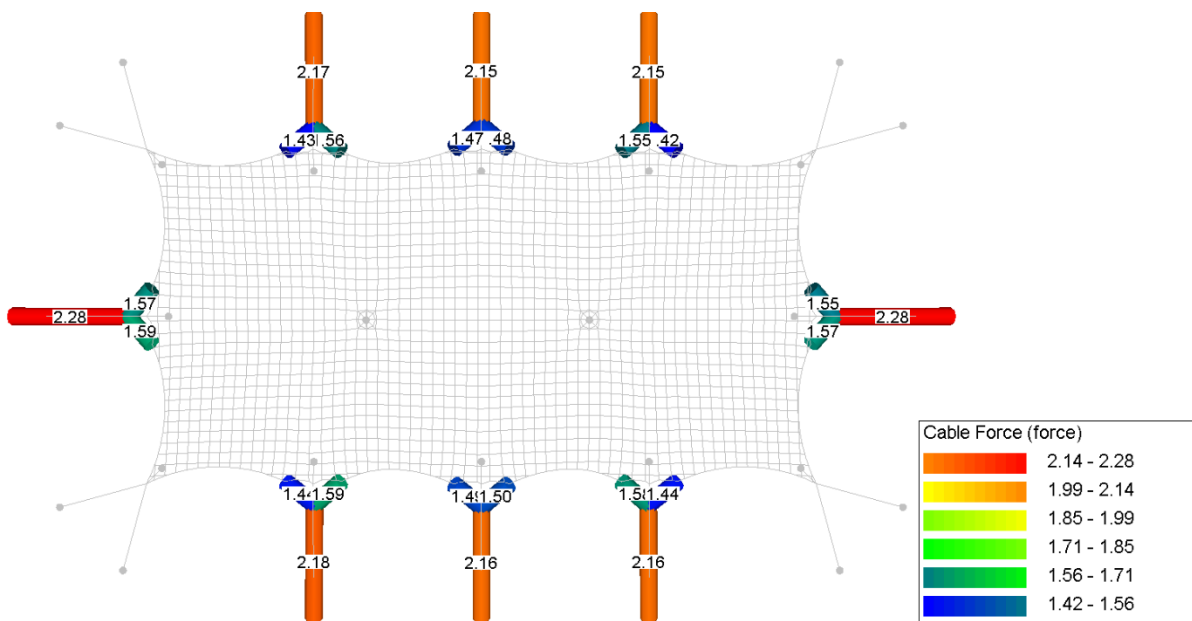
Annex D.2.2. Membrane stress (weft)



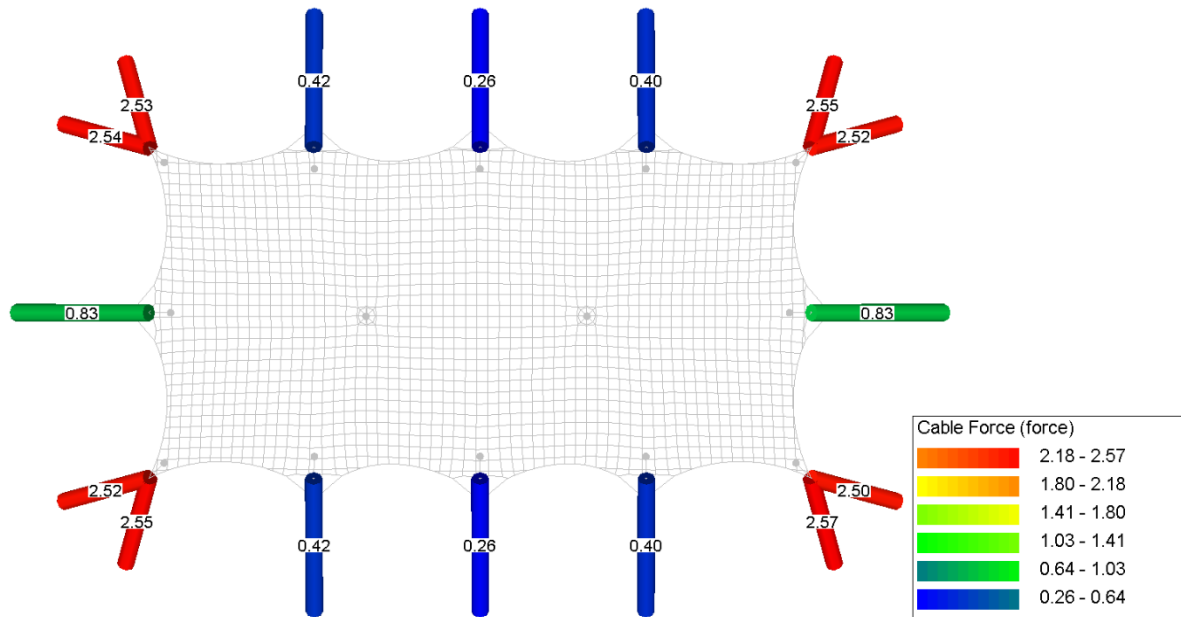
Annex D.2.3. Circumference rope forces



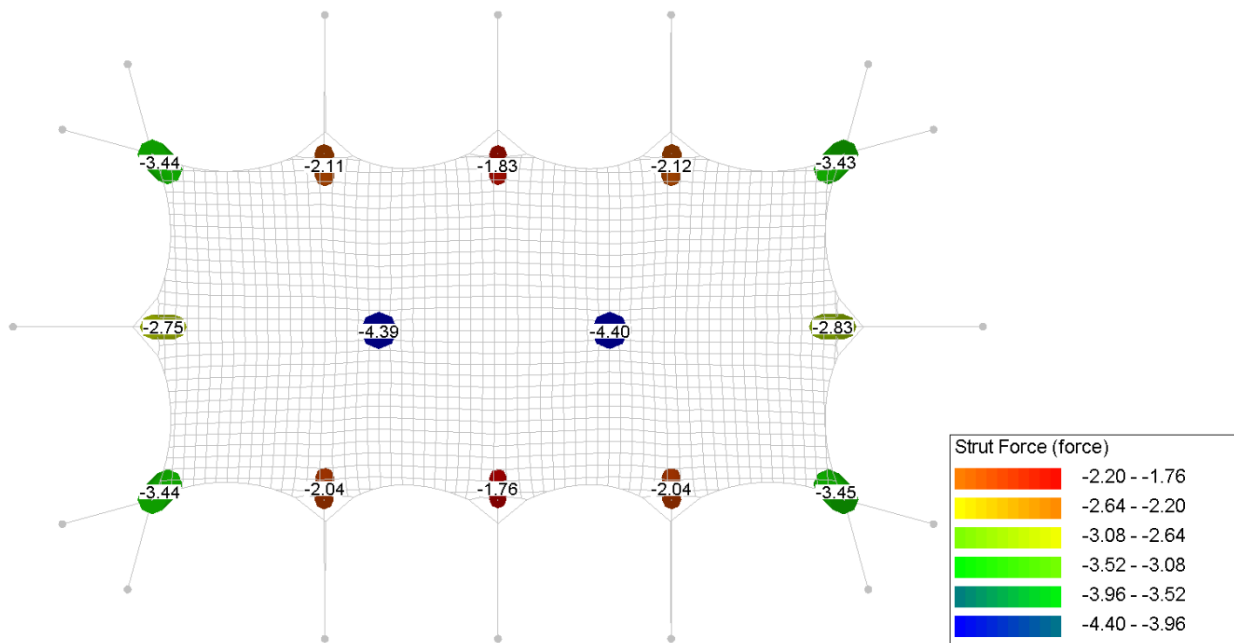
Annex D.2.4. Rope forces



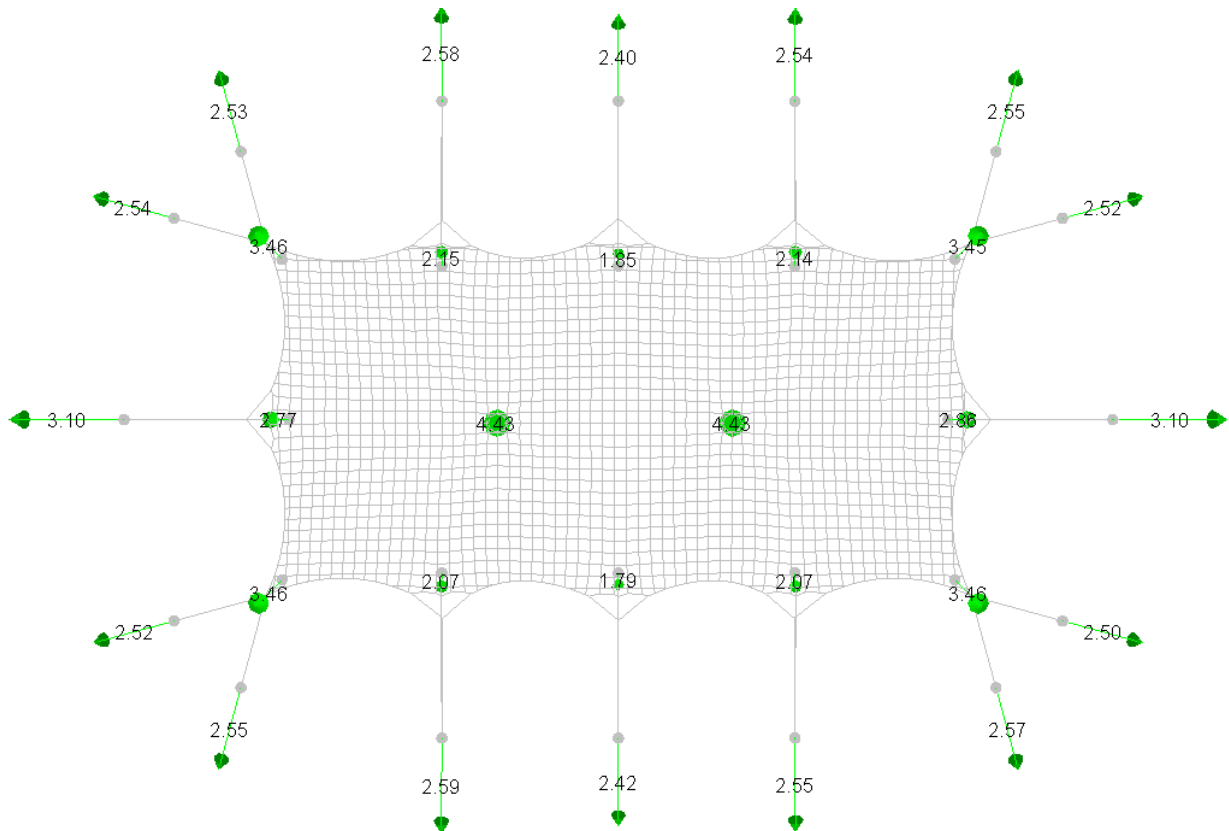
Annex D.2.5. Belt forces



Annex D.2.6. Pole forces

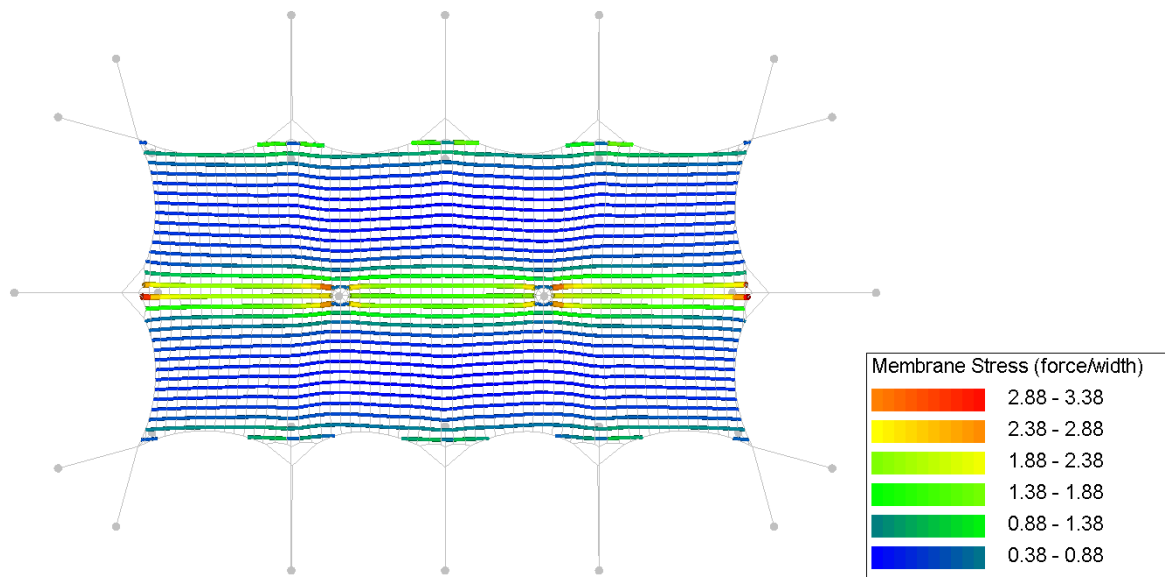


Annex D.2.7. Anchor forces

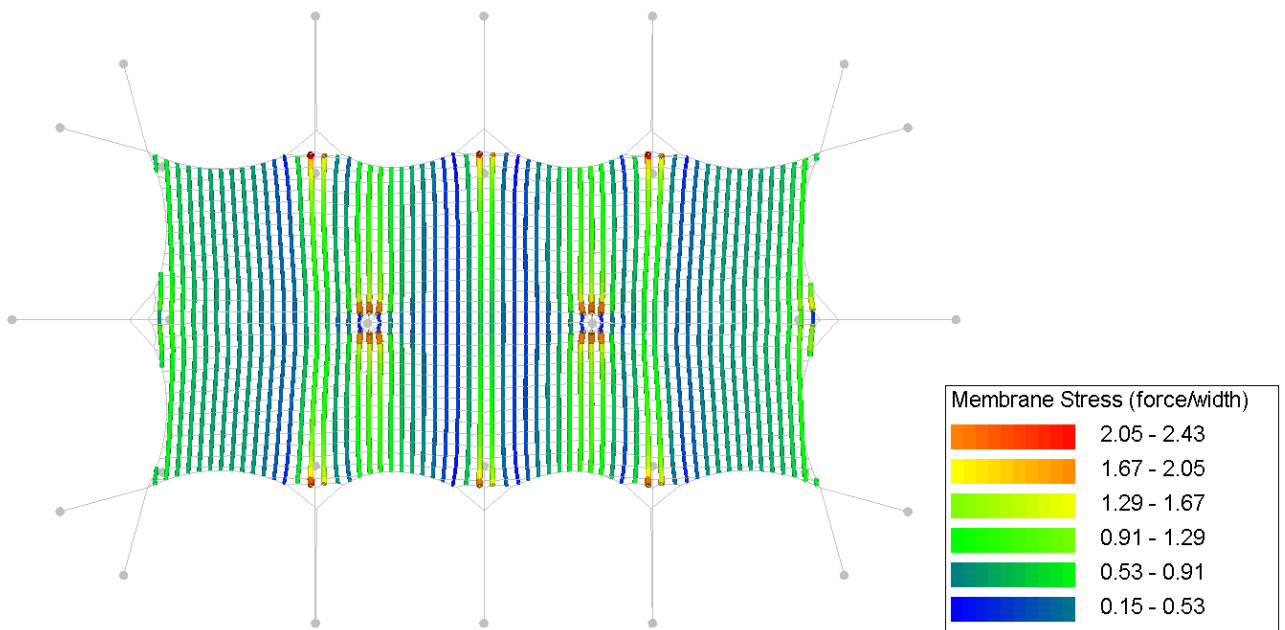


Annex D.3. CO3. Own weight + pretension + wind pressure

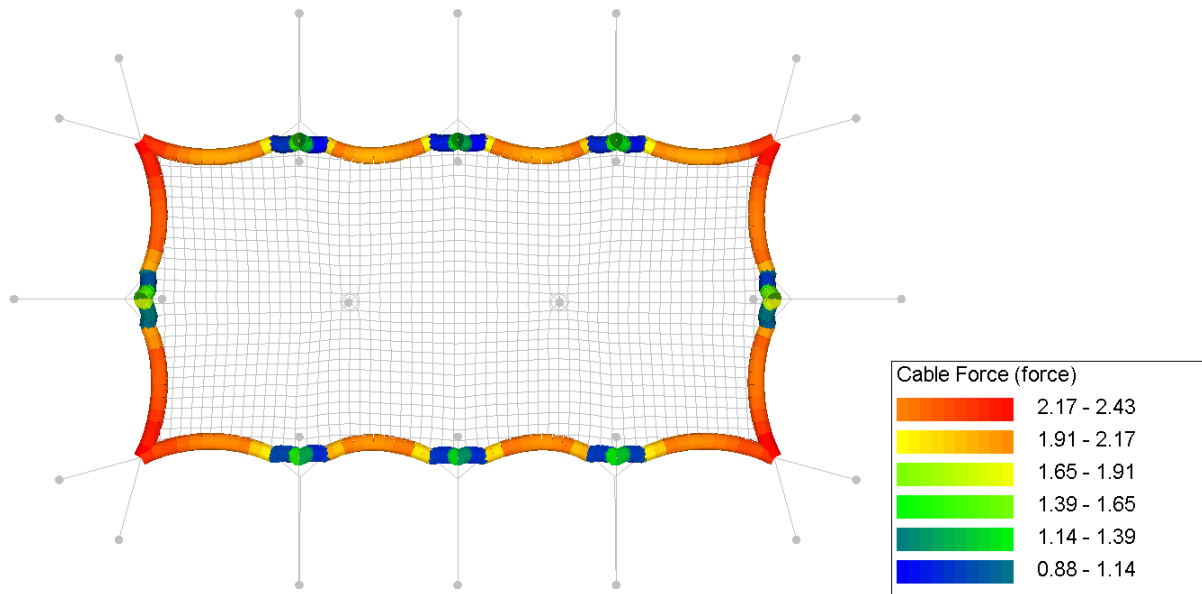
Annex D.3.1. Membrane stress (warp)



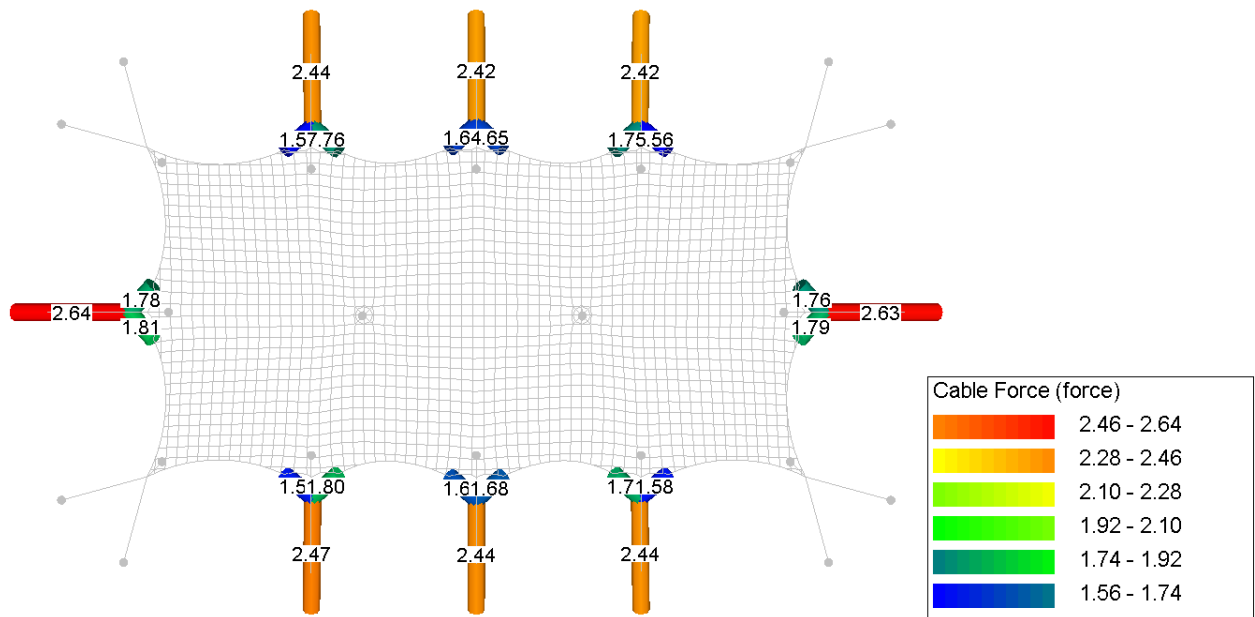
Annex D.3.2. Membrane stress (weft)



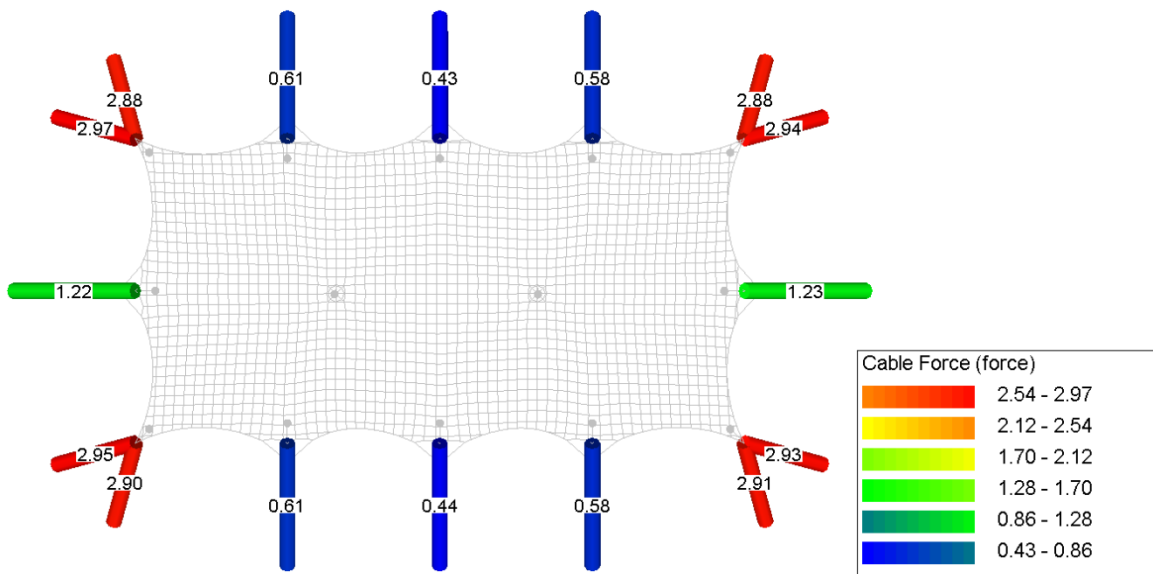
Annex D.3.3. Circumference rope forces



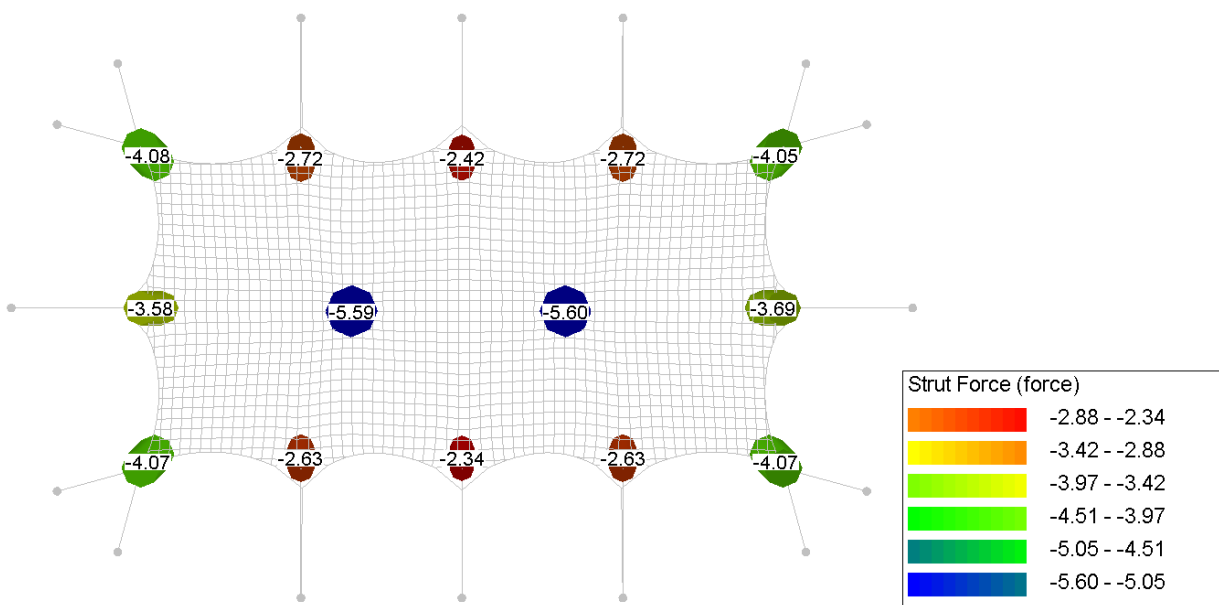
Annex D.3.4. Rope forces



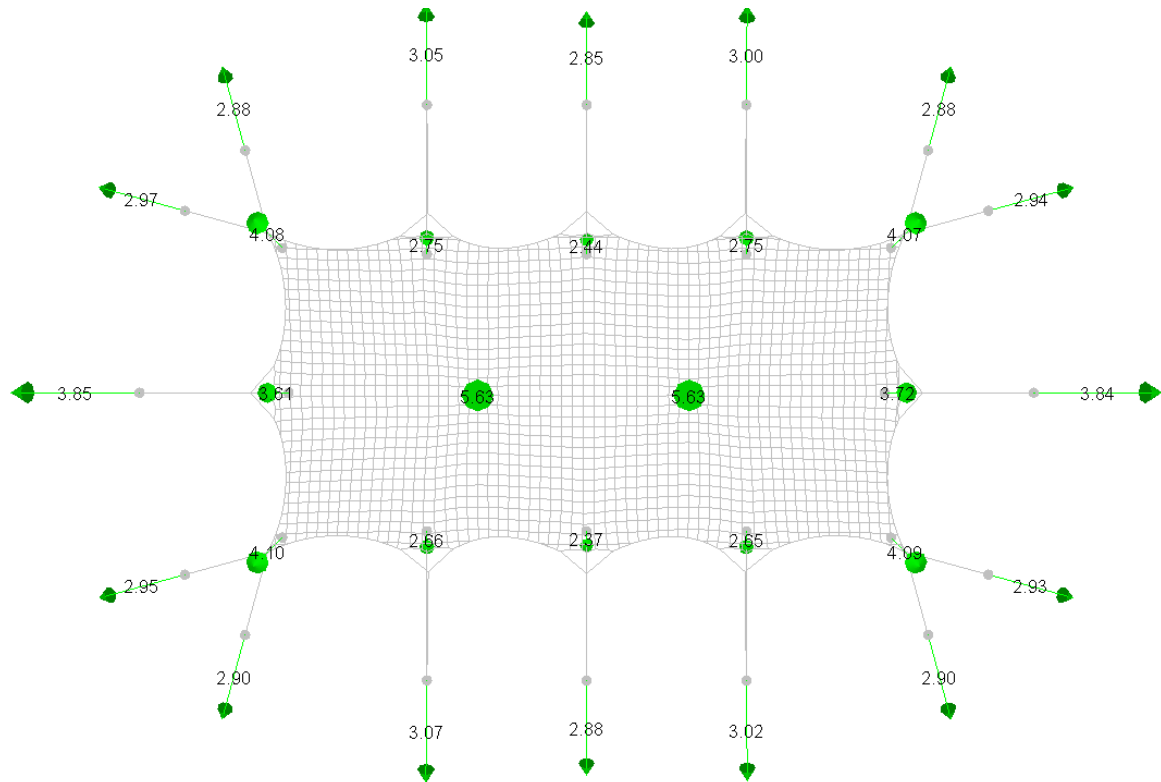
Annex D.3.5. Belt forces



Annex D.3.6. Poles forces

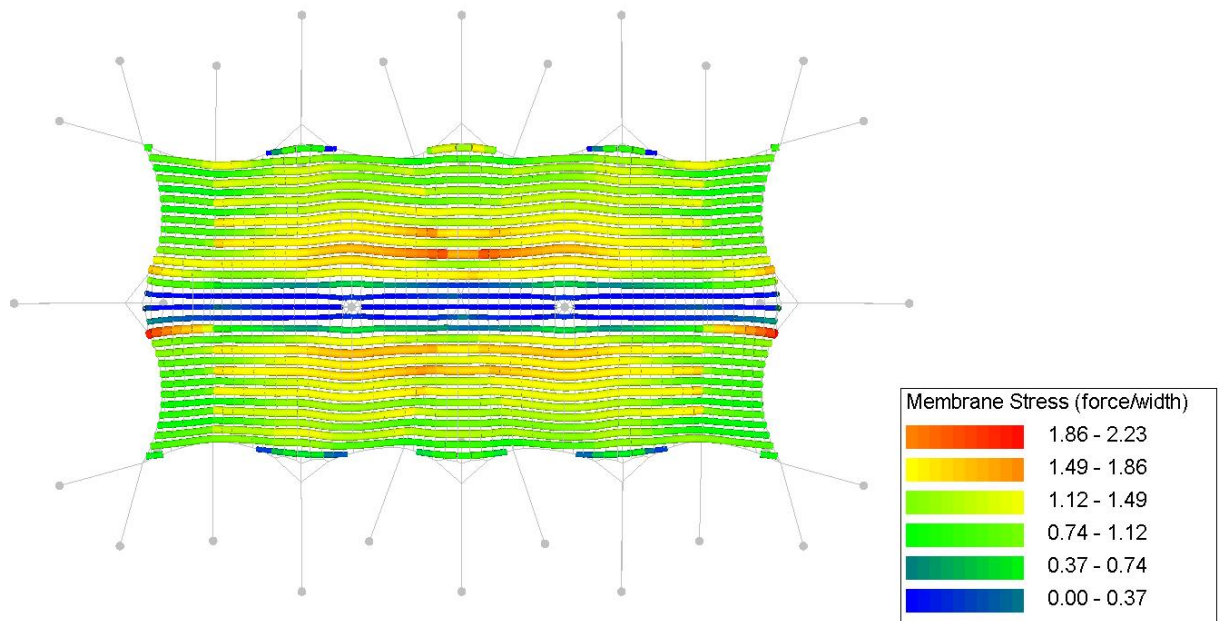


Annex D.3.7. Reaction forces

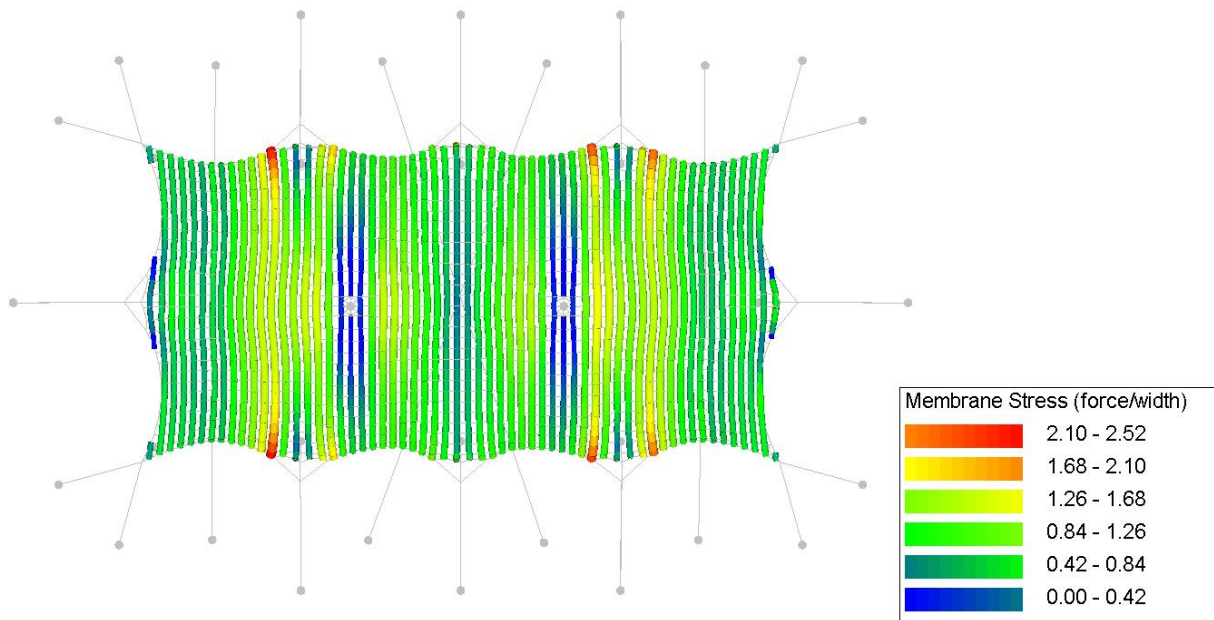


Annex D.4. CO4. Own weight + pretension + wind suction

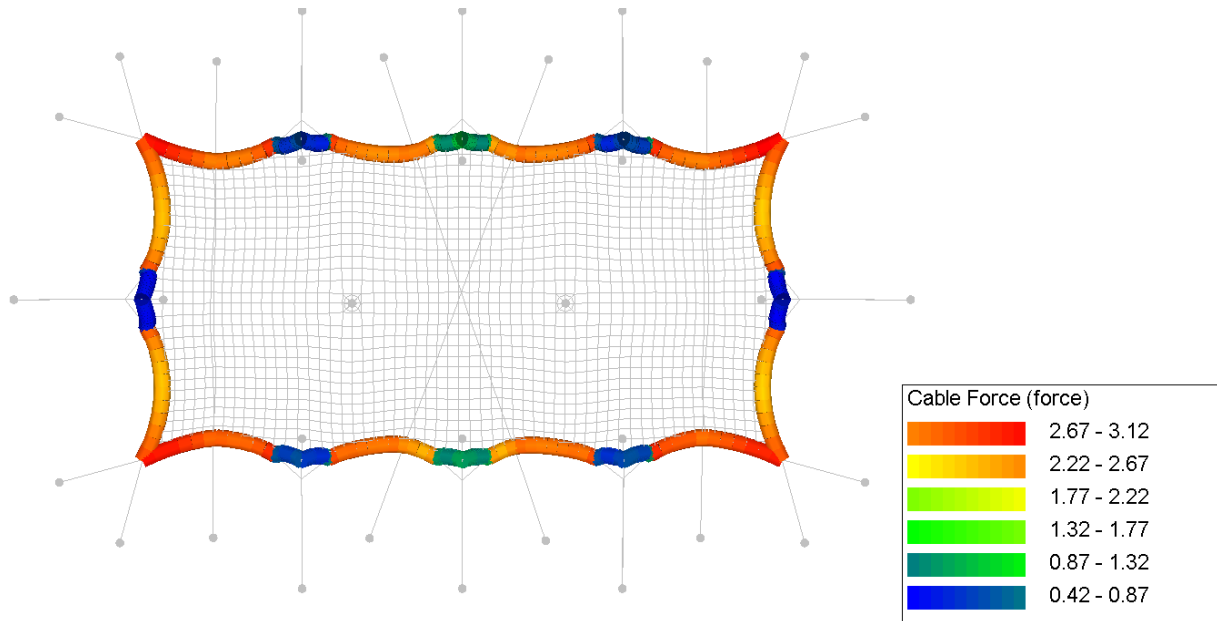
Annex D.4.1. Membrane stress (warp)



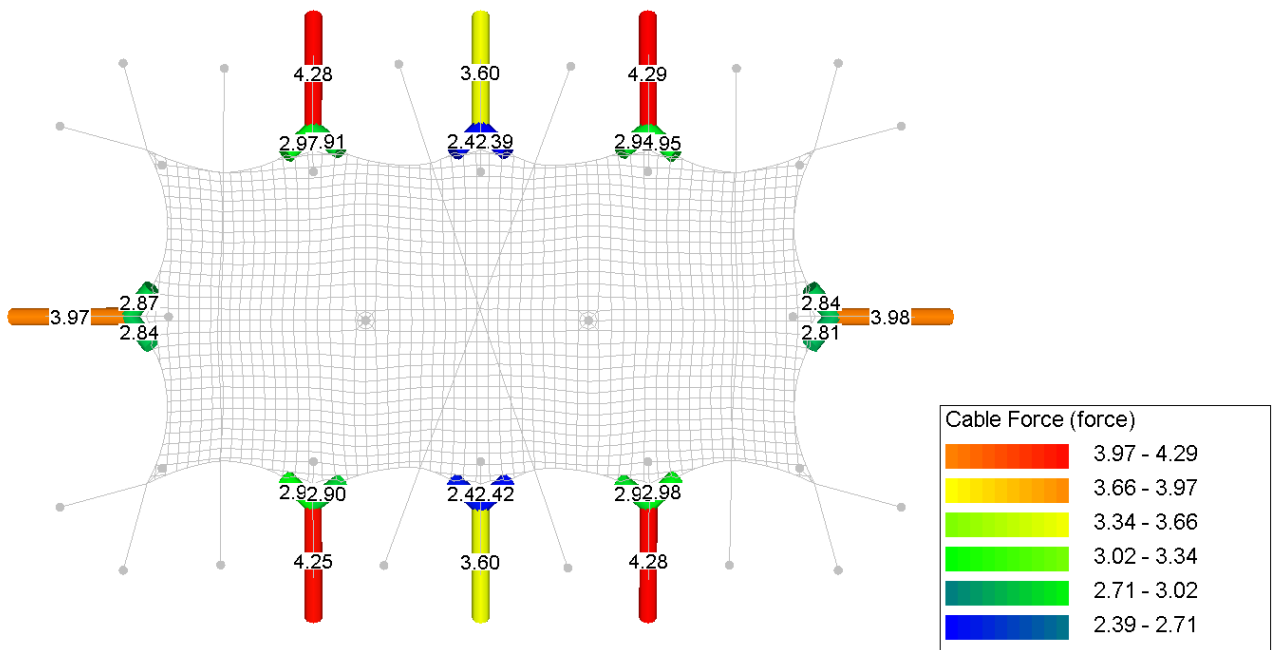
Annex D.4.2. Membrane stress (weft)



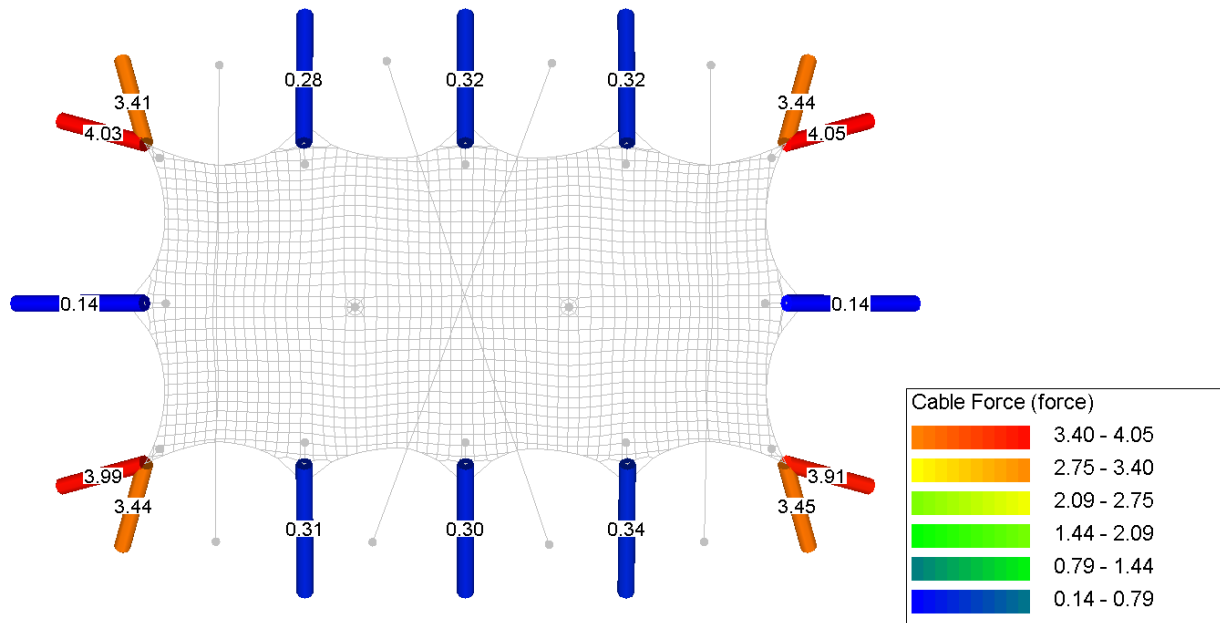
Annex D.4.3. Circumference rope forces



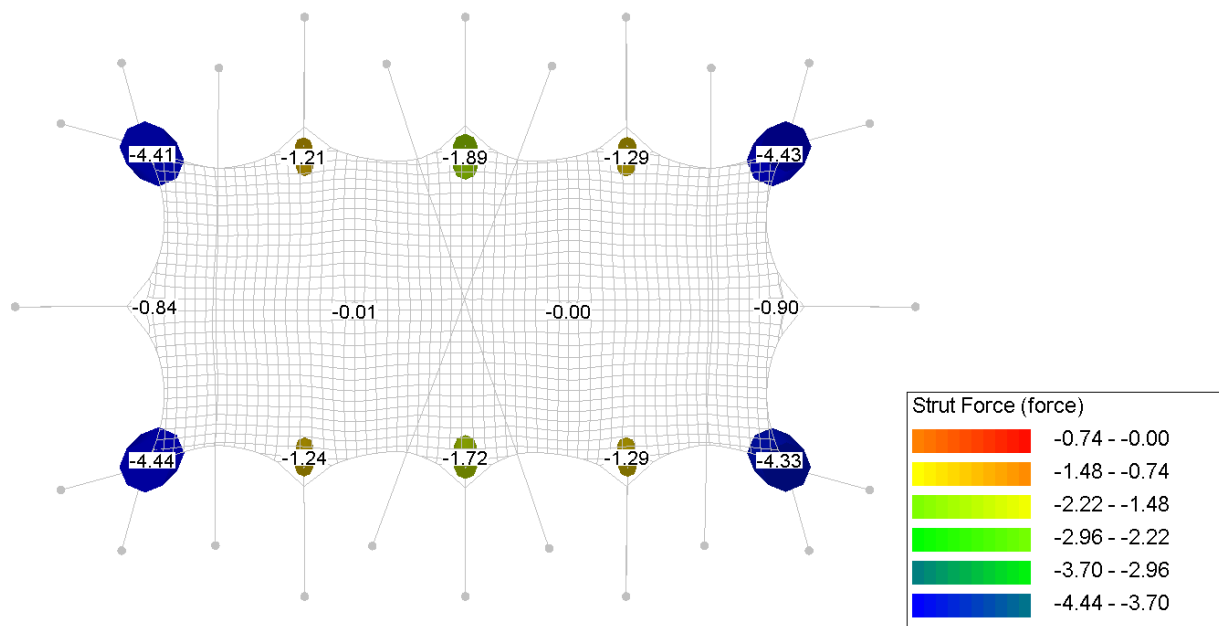
Annex D.4.4. Guy rope forces



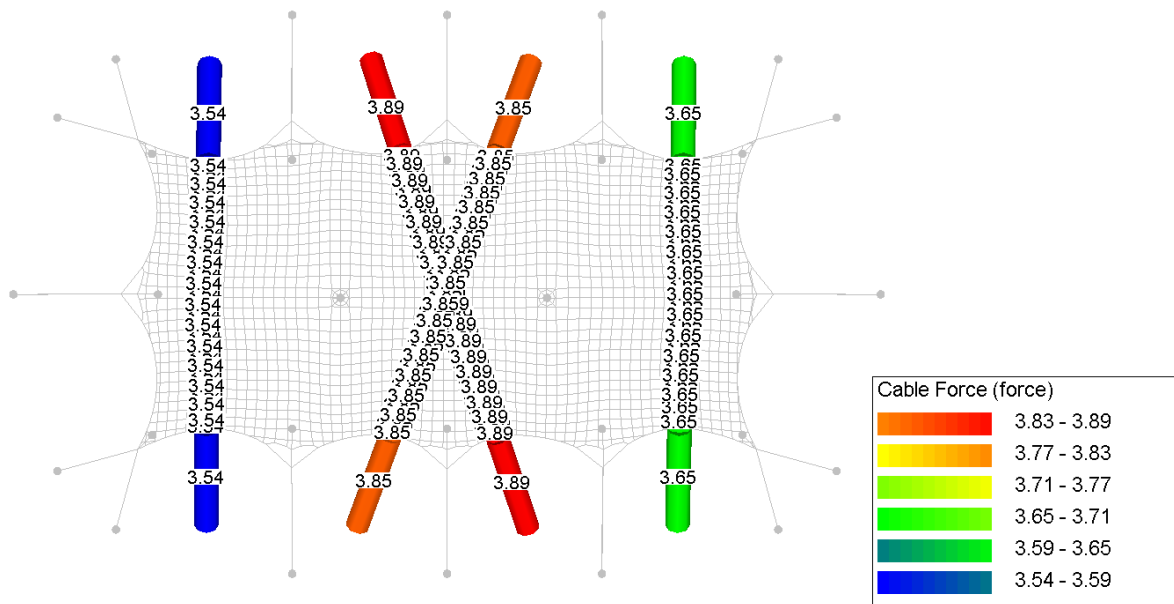
Annex D.4.5. Belt forces



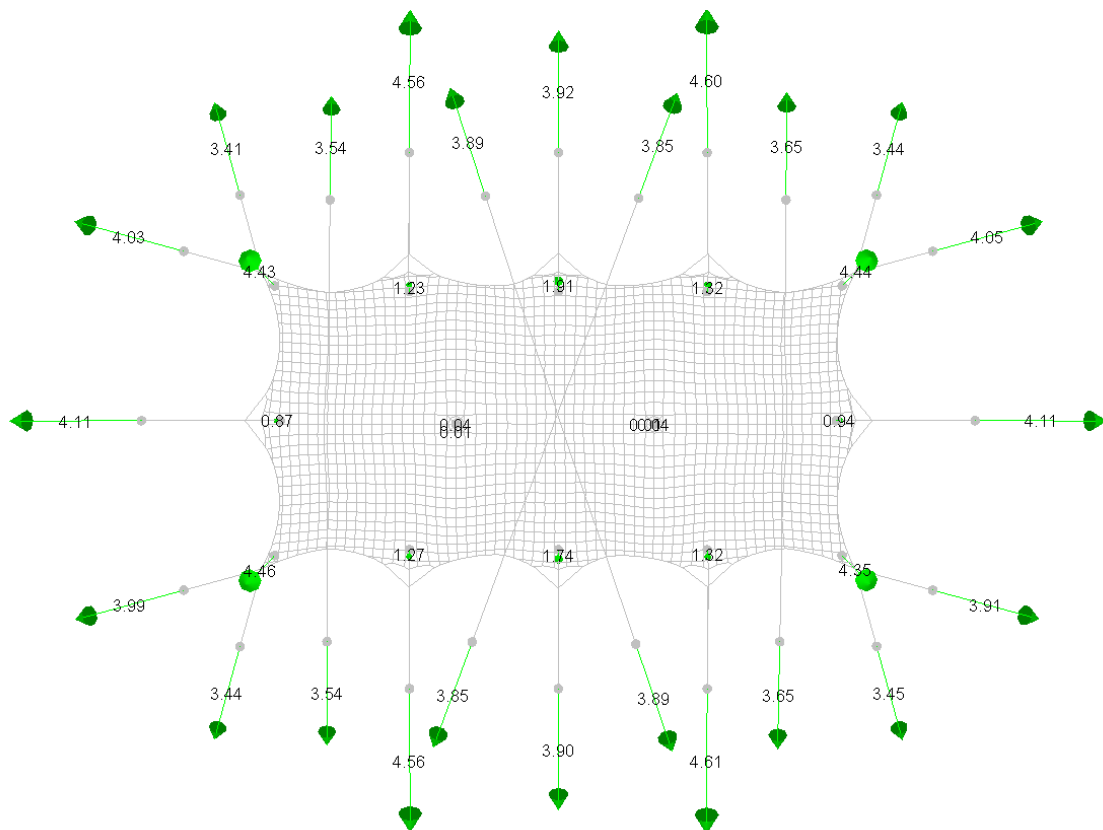
Annex D.4.6. Poles forces



Annex D.4.7. Storm belt forces

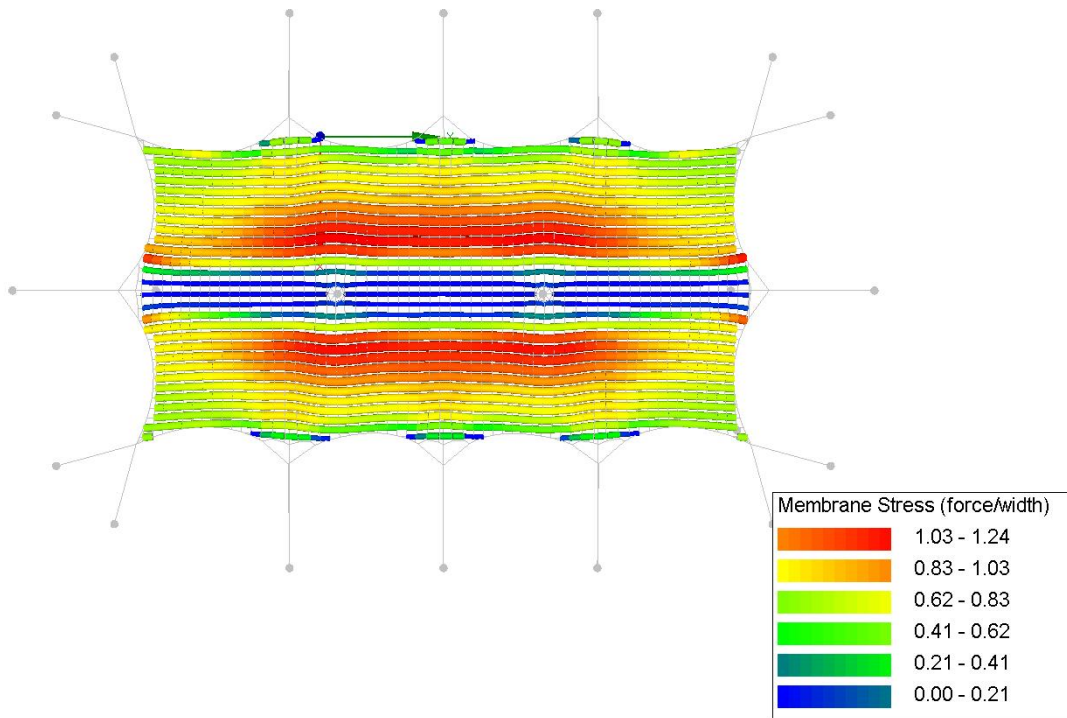


Annex D.4.8. Reaction forces

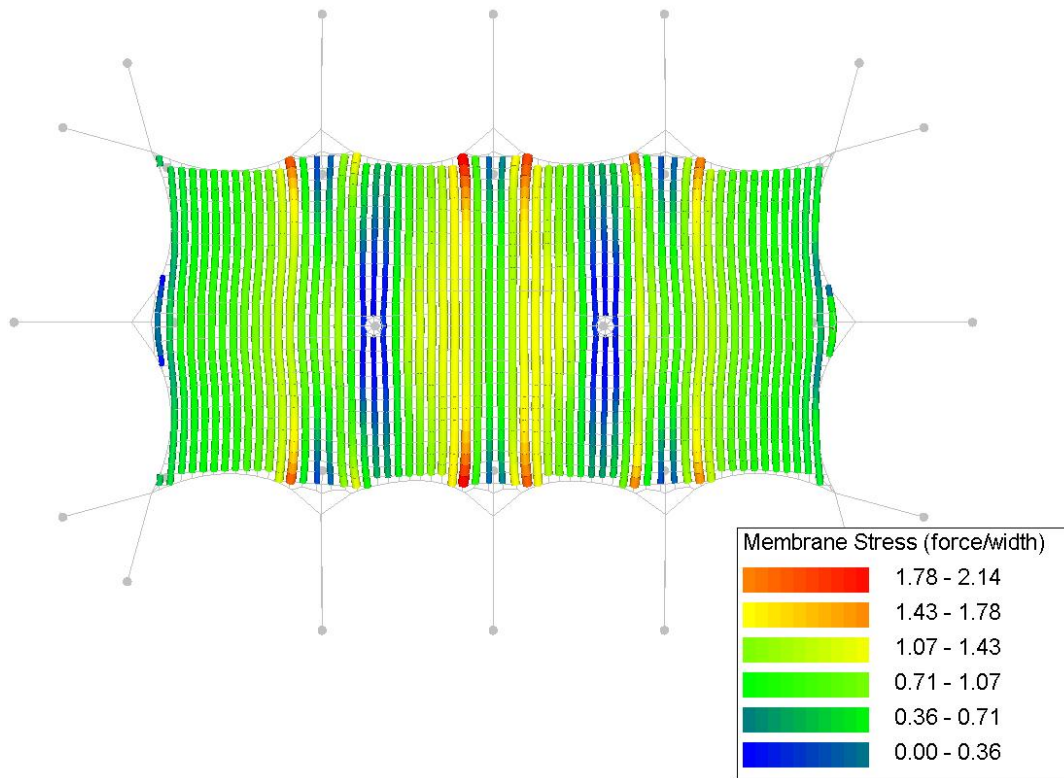


Annex D.5. CO5. Own weight + pretension + wind suction reduced [NL]

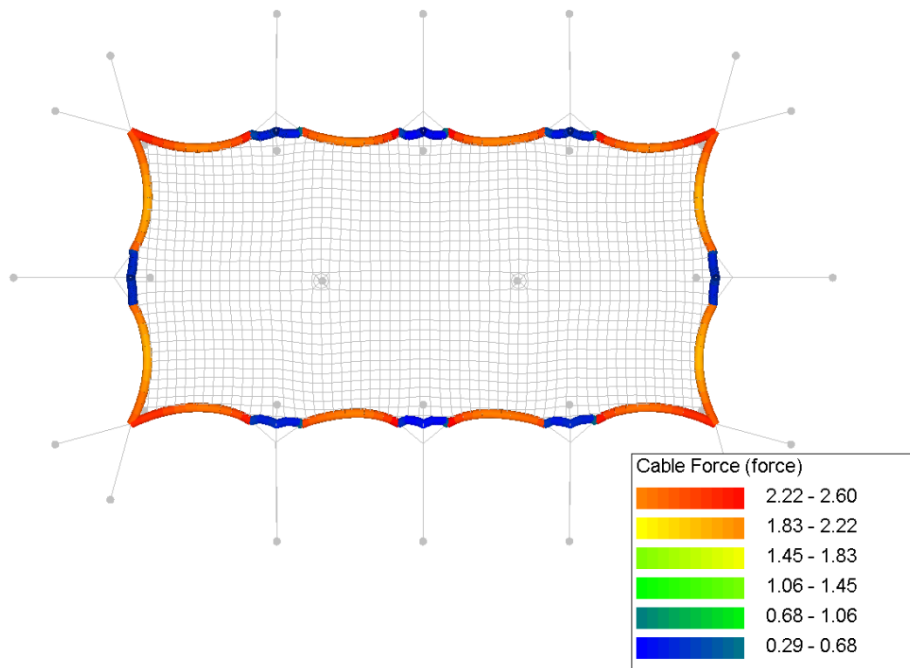
Annex D.5.1. Membrane stress (warp)



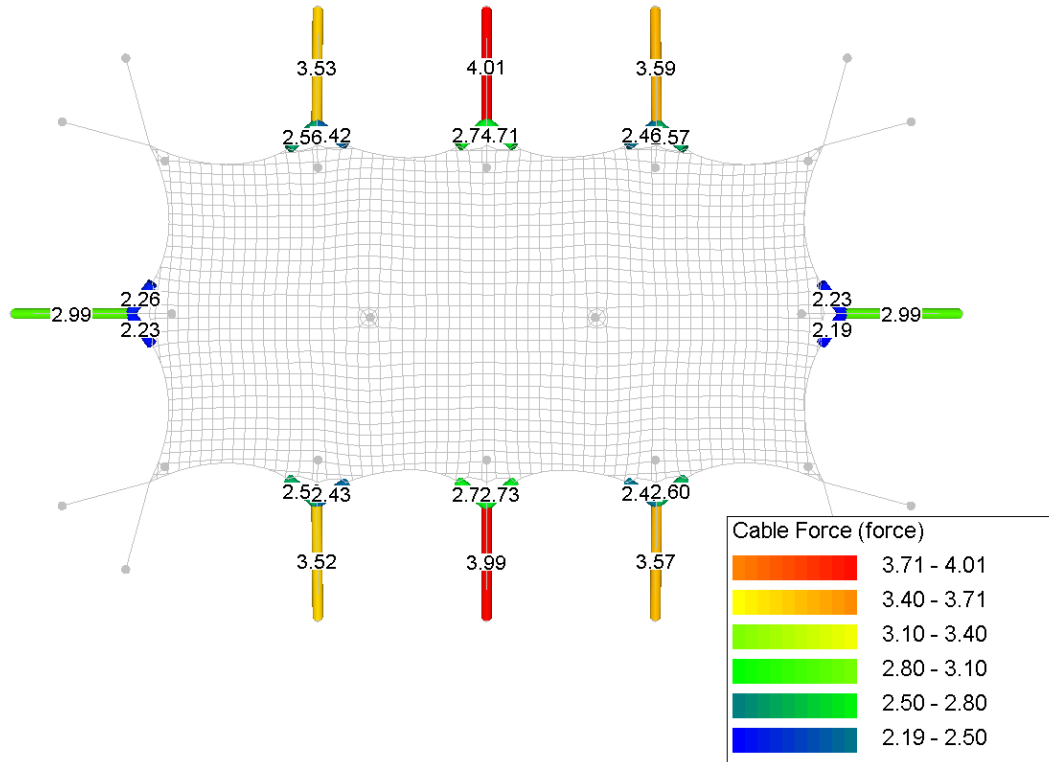
Annex D.5.2. Membrane stress (weft)



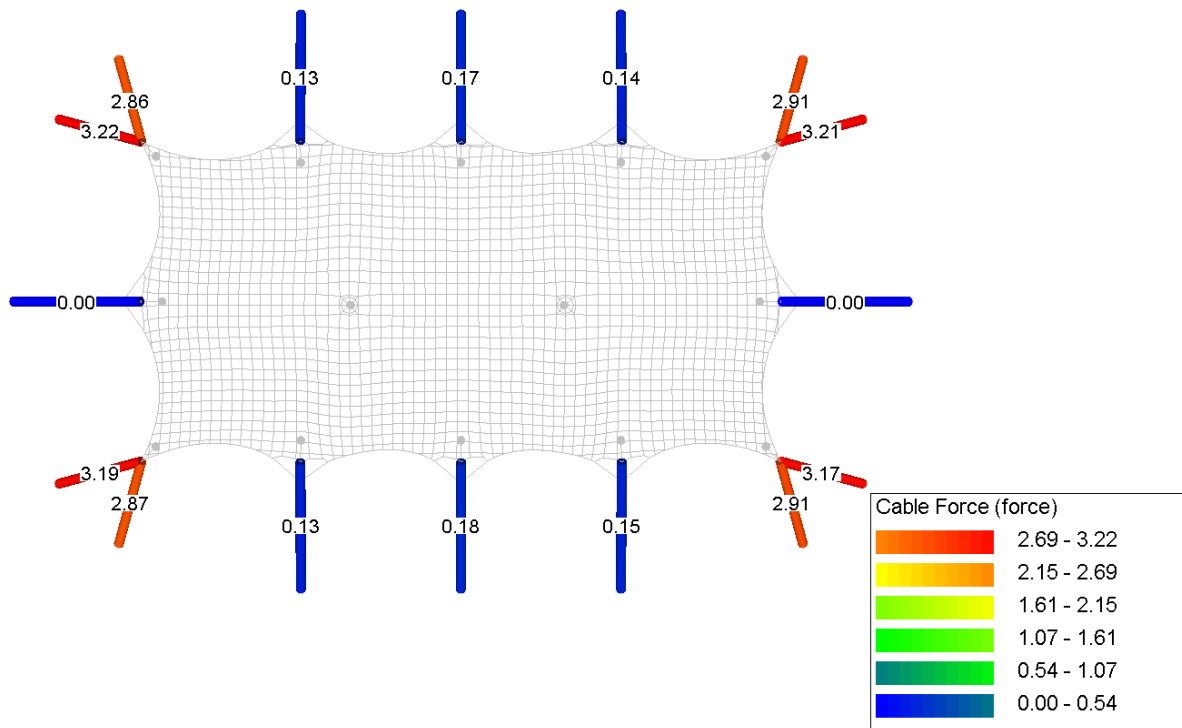
Annex D.5.3. Circumference rope forces



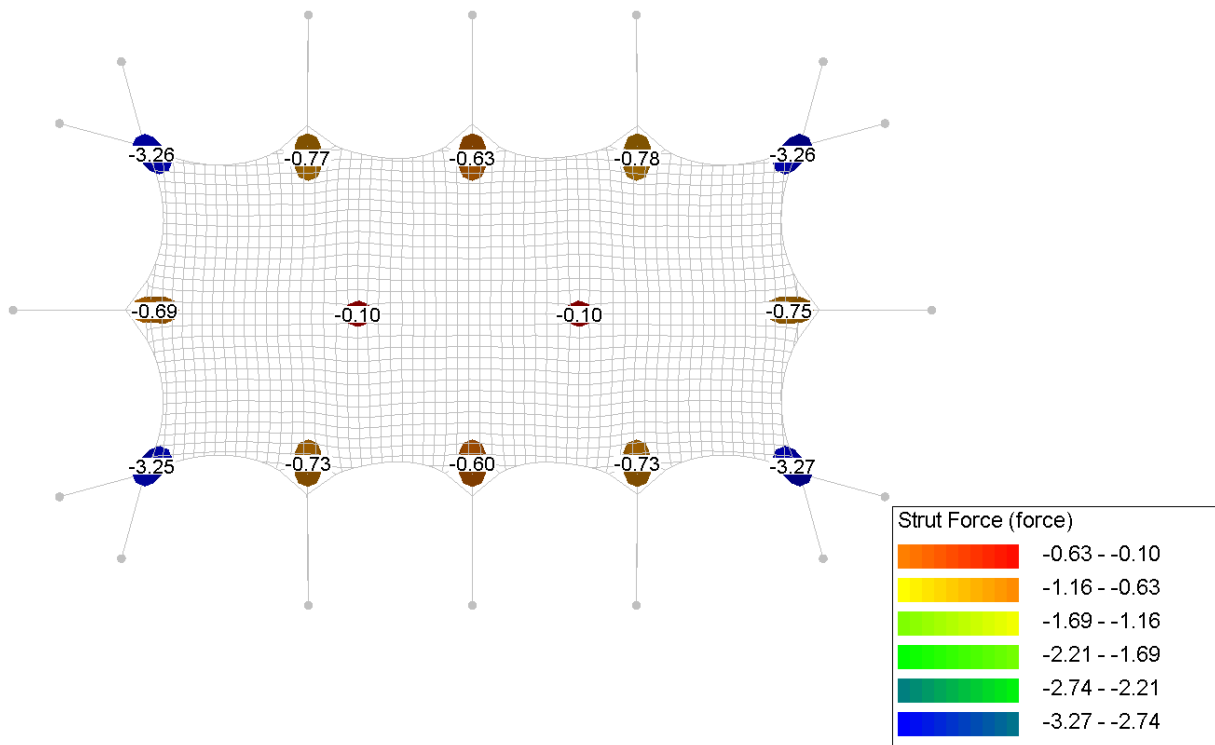
Annex D.5.4. Rope forces



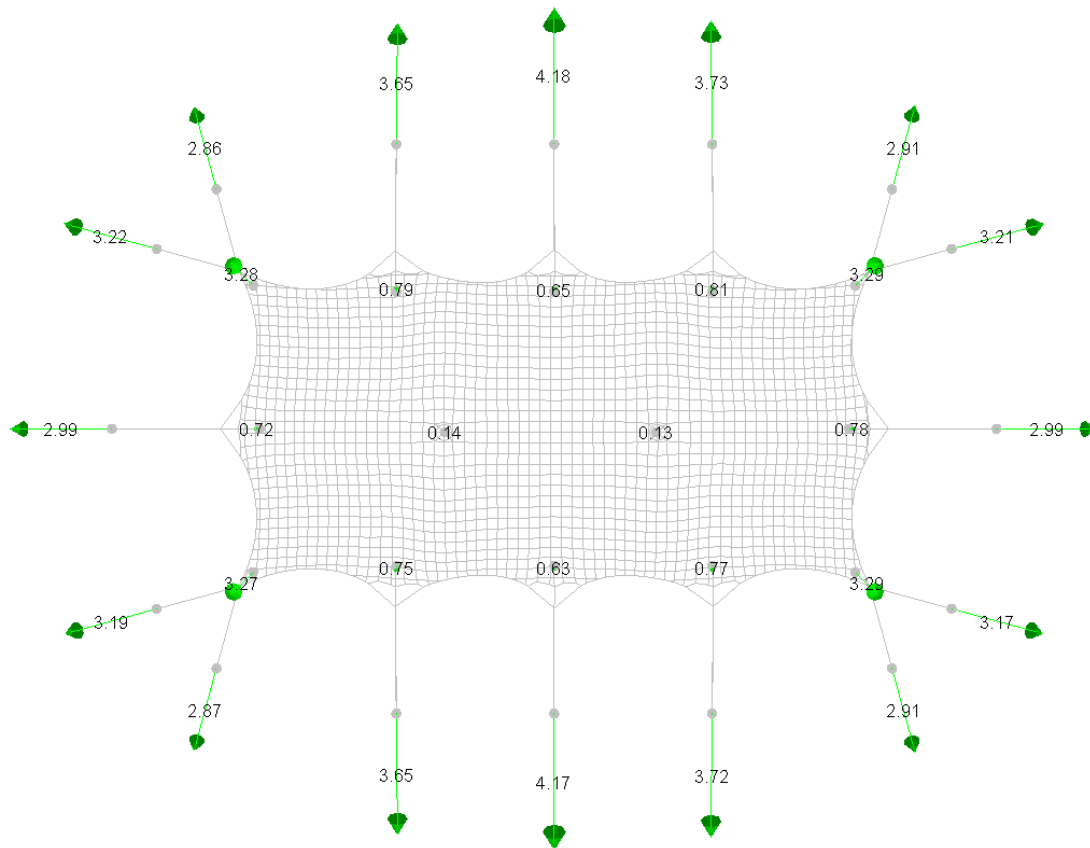
Annex D.5.5. Belt forces



Annex D.5.6. Poles forces

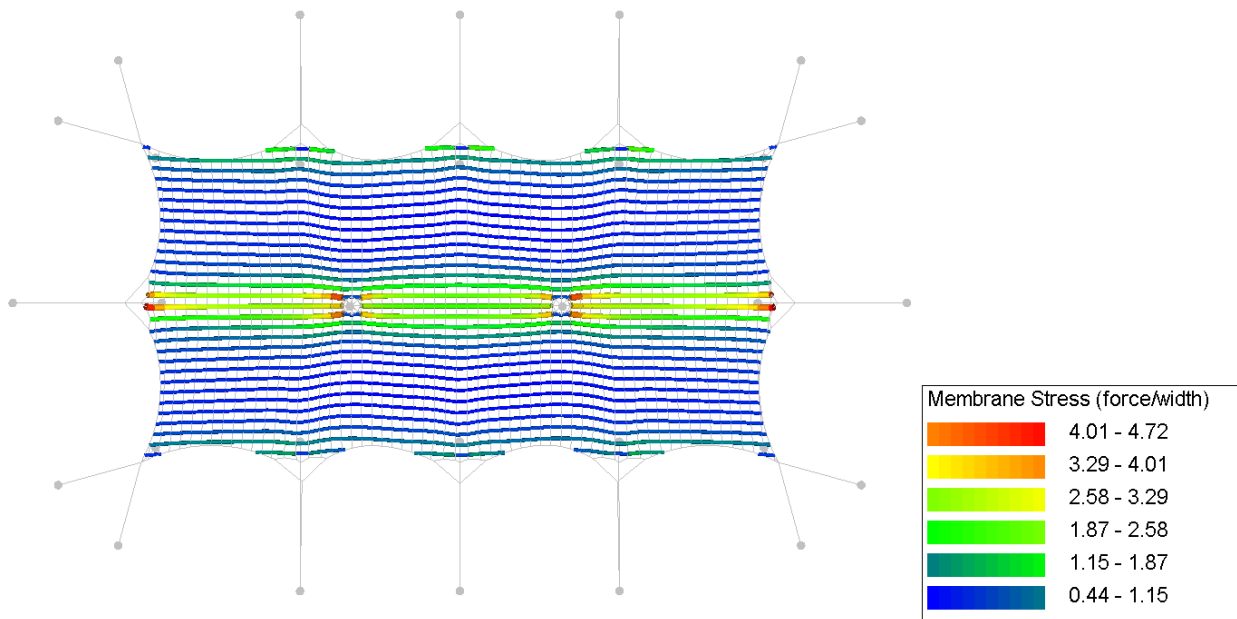


Annex D.5.7. Reaction forces

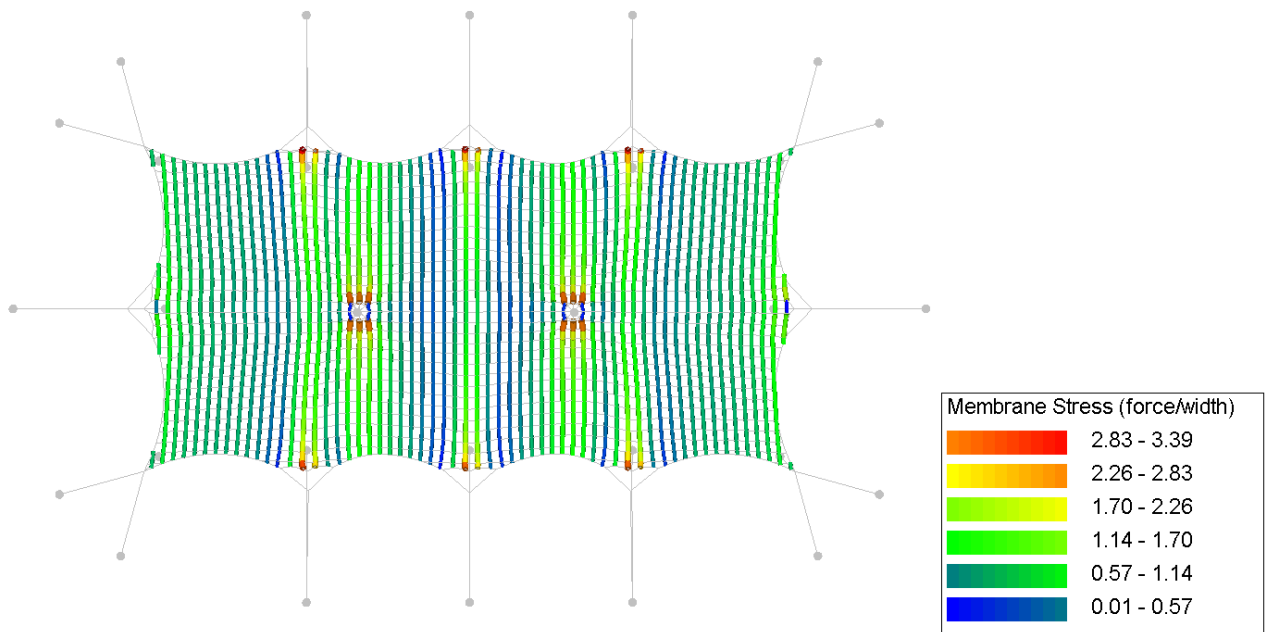


Annex D.6. CO6. Own weight + pretension + wind pressure + snow [FR]

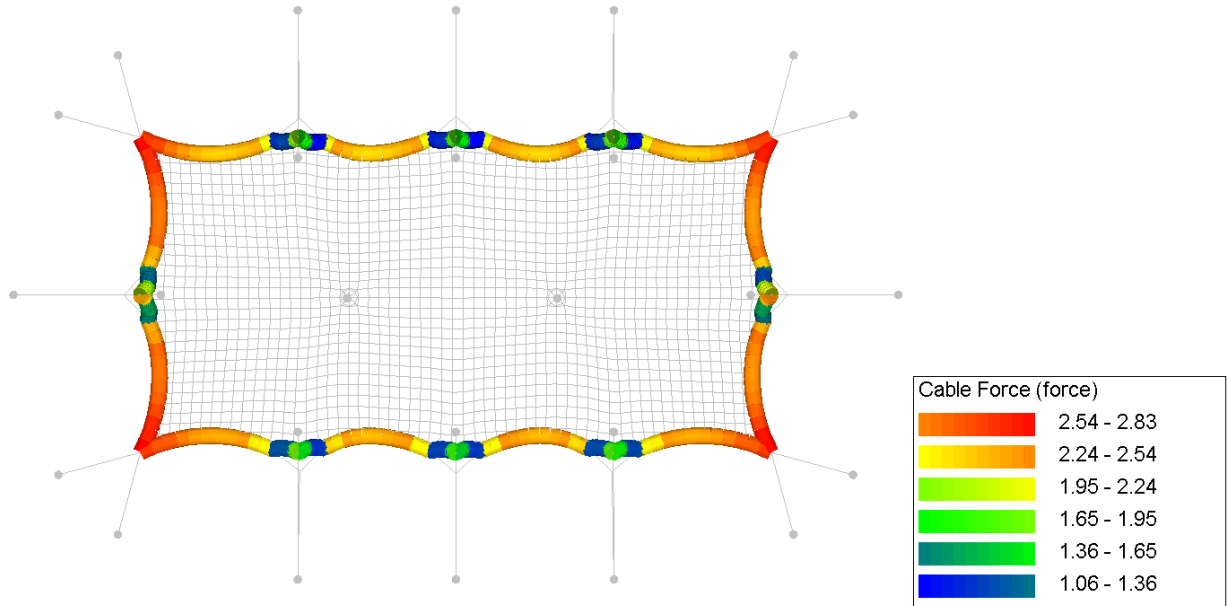
Annex D.6.1. Membrane stress (warp)



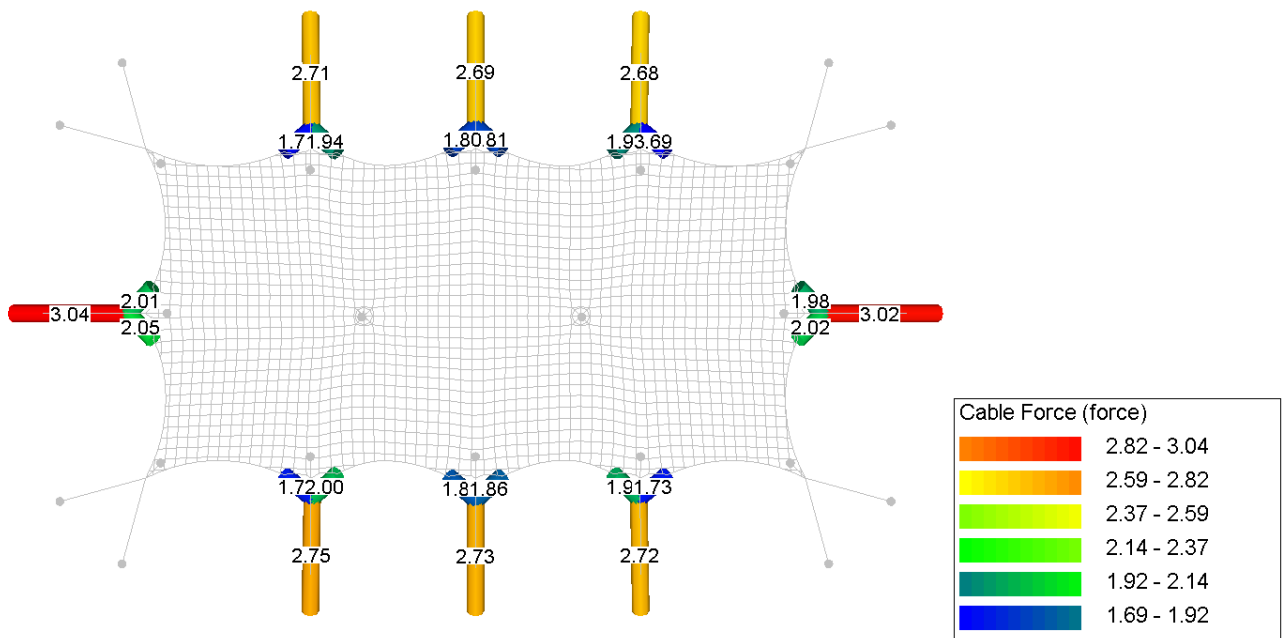
Annex D.6.2. Membrane stress (weft)



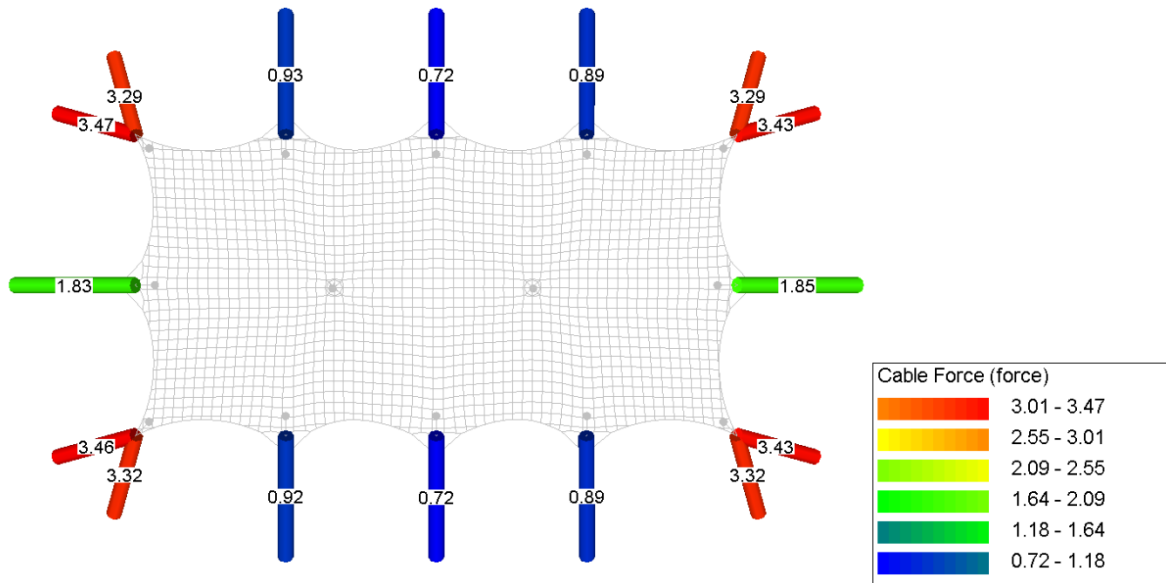
Annex D.6.3. Circumference rope forces



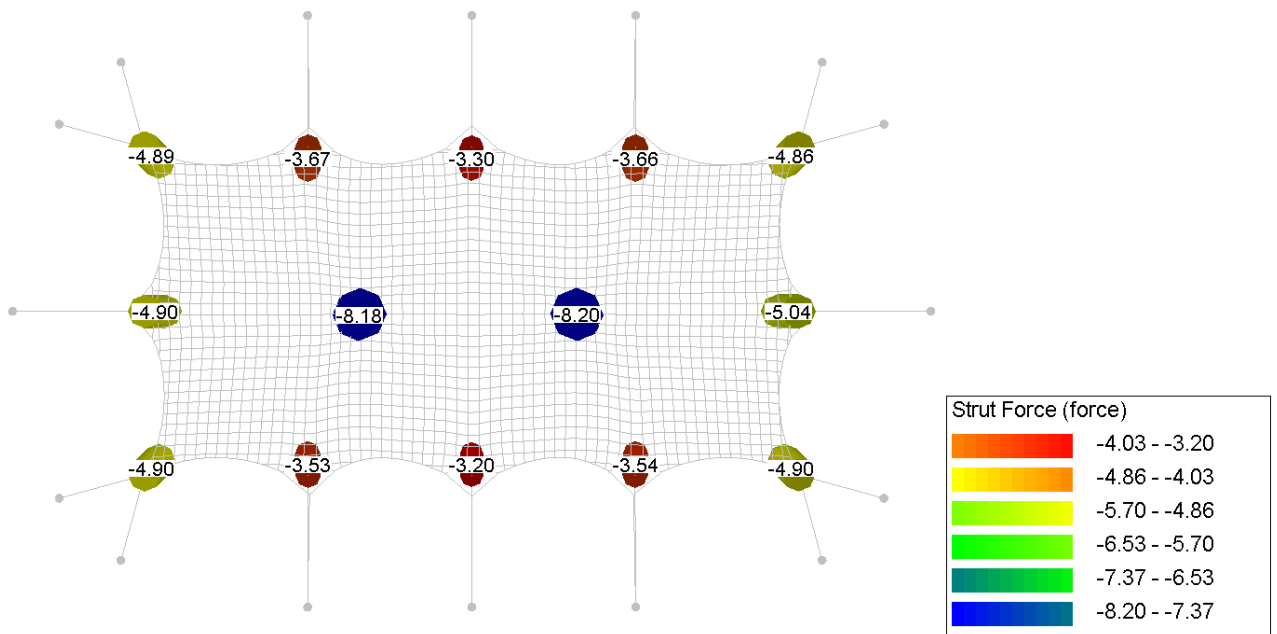
Annex D.6.4. Guy rope forces



Annex D.6.5. Belt forces

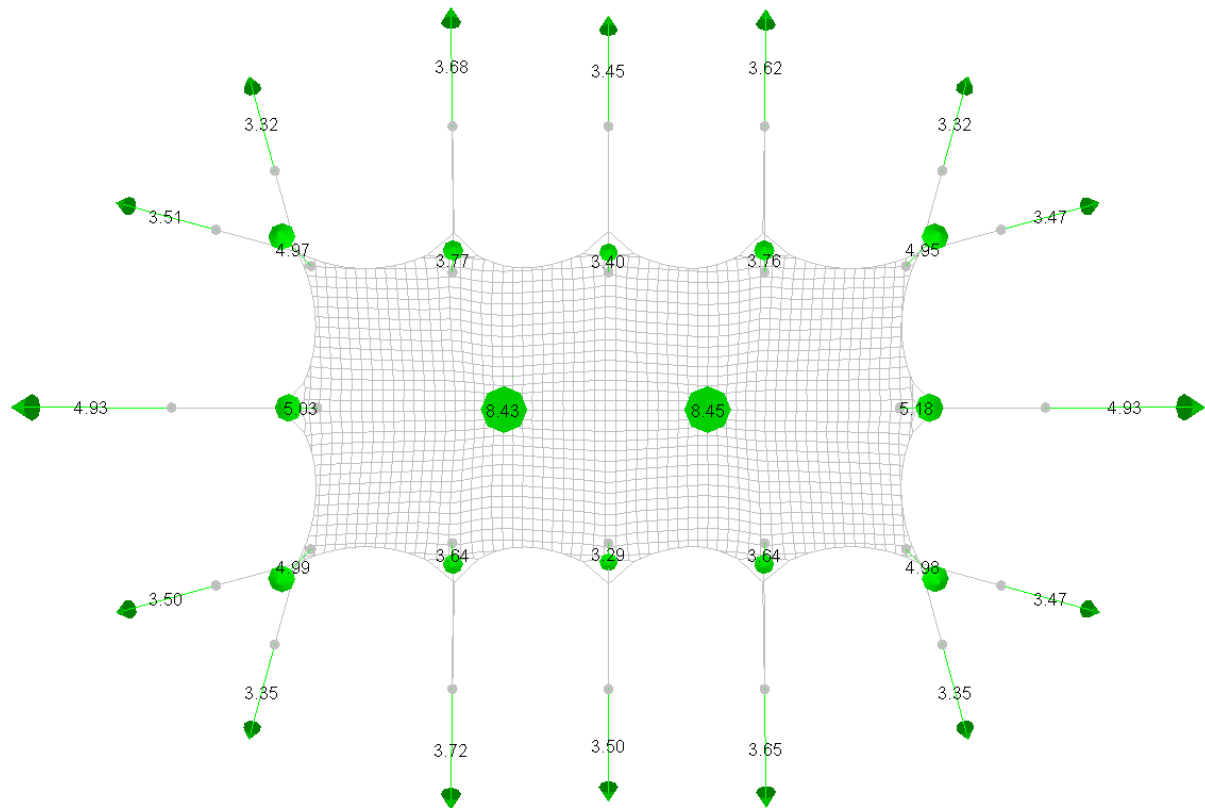


Annex D.6.6. Poles forces



Annex D.6.7. Reaction forces

Increased snow load of $0.1 \times 1.5/1.35 = 0.1083 \text{ kN/m}^2$ is applied for ballast and anchor calculation



Annex E. Member checks

Annex E.1. Main pole 4.0m – Wood, Spruce C18

Woodtype		Spruce	
Strenght type		C18	
Material factor	V_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	95 mm	
Length (buckling)	l_{bucy}	4 m	
Effective area	A	7088.218425 mm ²	
Moment of innertia	I_y	3998198.205 mm ⁴	
Elastic modules	W_{ely}	84172.59379 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	23.8 mm	
Slenderness	λ_y	168.4	
Relative slenderness	$\lambda_{rel,y}$	2.94	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	5.07	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.109	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	9.20 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cd}	1.30 N/mm ²	
Bending stress	σ_{cd}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cd}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{md}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.96	NEN-EN 1995-1-1:2005 equ. 6.23

Annex E.2. Side pole 3m – Wood, Spruce C18

Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	75 mm	
Length (buckling)	l_{buck}	3 m	
Effective area	A	4417.864669 mm ²	
Moment of inertia	I_y	1553155.548 mm ⁴	
Elastic modules	W_{ely}	41417.48127 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	18.8 mm	
Slenderness	λ_y	160.0	
Relative slenderness	$\lambda_{rel,y}$	2.79	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.64	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.120	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	5.87 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cdk}	1.33 N/mm ²	
Bending stress	σ_{cdk}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cdk}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{cdk}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.89	NEN-EN 1995-1-1:2005 equ. 6.23

Annex E.3. Corner pole 2.5m - Wood, C18

Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	70 mm	
Length (buckling)	l_{bucy}	2.5 m	
Effective area	A	3848.451001 mm ²	
Moment of innertia	I_y	1178588.119 mm ⁴	
Elastic modules	W_{ely}	33673.94626 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	17.5 mm	
Slenderness	λ_y	142.9	
Relative slenderness	$\lambda_{rel,y}$	2.49	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	3.82	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.149	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	7.00 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cdk}	1.82 N/mm ²	
Bending stress	σ_{cdk}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cdk}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{mod}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.98	NEN-EN 1995-1-1:2005 equ. 6.23

Annex E.4. Main pole 4.0m – Wood, Spruce C18 – for snow and wind combination [FR]

Woodtype		Spruce	
Strenght type		C18	
Material factor	V_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	100 mm	
Length (buckling)	l_{bucy}	4 m	
Effective area	A	7853.981634 mm ²	
Moment of inertia	I_y	4908738.521 mm ⁴	
Elastic modules	W_{ely}	98174.77042 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	25.0 mm	
Slenderness	λ_y	160.0	
Relative slenderness	$\lambda_{rel,y}$	2.79	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.64	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.120	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	11.91 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cdk}	1.52 N/mm ²	
Bending stress	σ_{cdk}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cdk}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{mdk}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		1.02	NEN-EN 1995-1-1:2005 equ. 6.23

Annex E.5. Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR]

Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	80 mm	
Length (buckling)	l_{bucy}	3 m	
Effective area	A	5026.548246 mm ²	
Moment of inertia	I_y	2010619.298 mm ⁴	
Elastic modules	$W_{el,y}$	50265.48246 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	20.0 mm	
Slenderness	λ_y	150.0	
Relative slenderness	$\lambda_{rel,y}$	2.62	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.15	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.136	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	7.14 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cdk}	1.42 N/mm ²	
Bending stress	σ_{cdk}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cdk}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{m0d}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.84	NEN-EN 1995-1-1:2005 equ. 6.23

Annex E.6. Corner pole 2.5m – Wood, Spruce C18 – for snow and wind combination [FR]

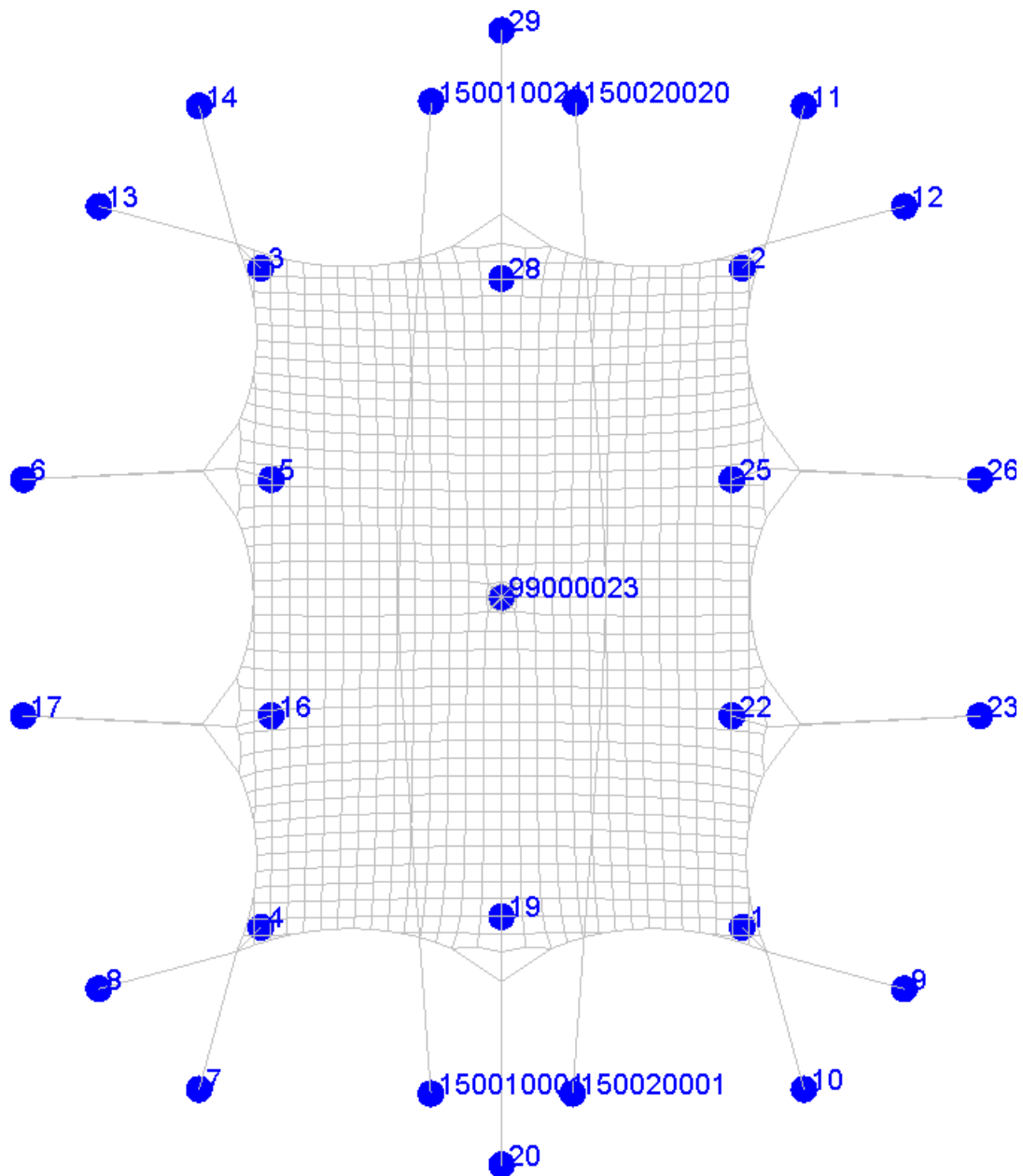
Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	70 mm	
Length (buckling)	l_{bucy}	2.5 m	
Effective area	A	3848.451001 mm ²	
Moment of inertia	I_y	1178588.119 mm ⁴	
Elastic modules	$W_{el,y}$	33673.94626 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	17.5 mm	
Slenderness	λ_y	142.9	
Relative slenderness	$\lambda_{rel,y}$	2.49	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	3.82	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.149	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	6.95 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cd}	1.81 N/mm ²	
Bending stress	σ_{cd}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cd}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{md}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.97	NEN-EN 1995-1-1:2005 equ. 6.23

Annex F. Reaction forces model 7.5x10m

Annex F.1. Point numbers



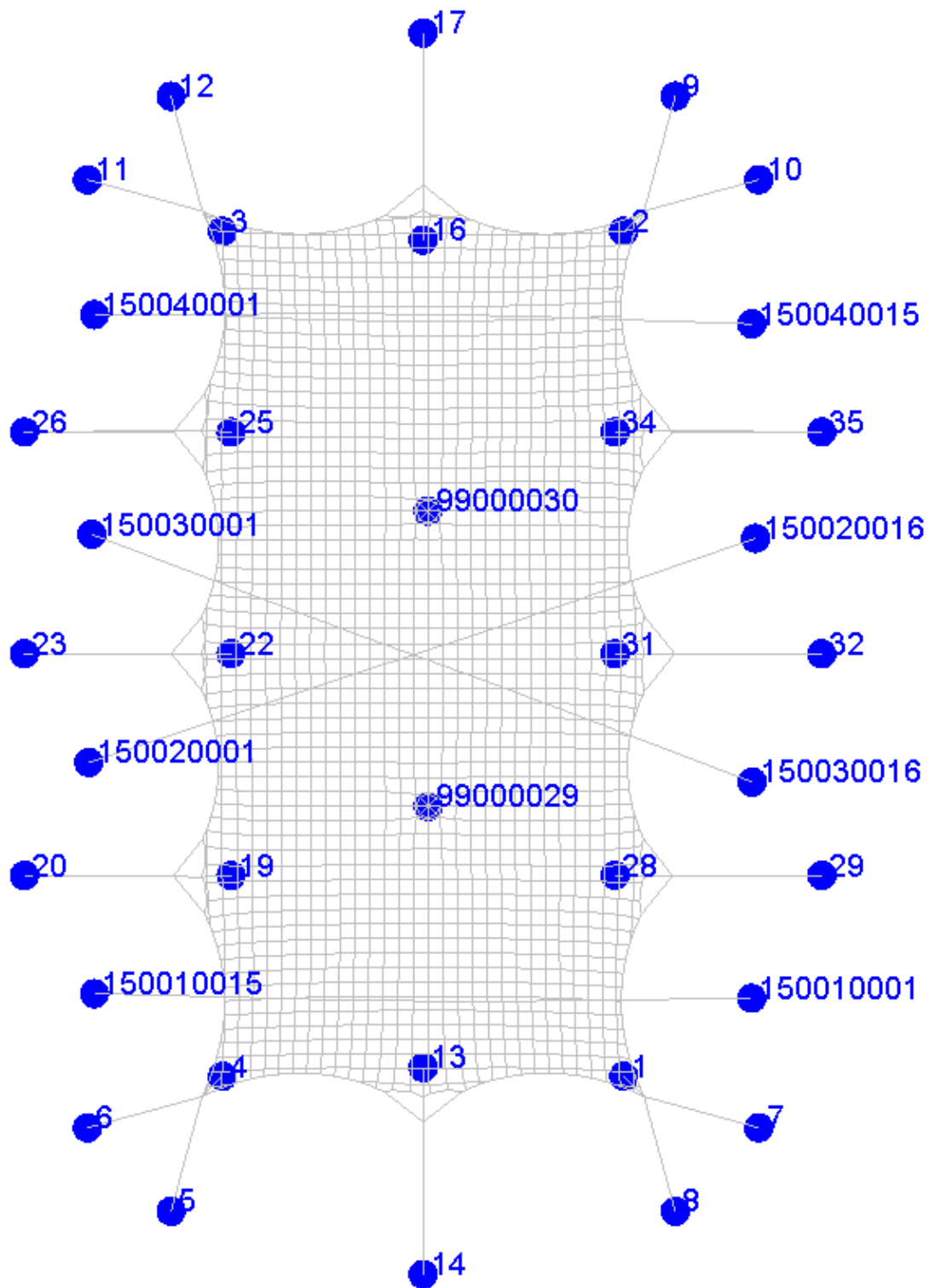
Annex F.2. Reaction forces

Reaction forces are shown in kilonewtons [kN] without any safety factor (1.35, 1.5, 1.2). A load combination especially for anchors and ballast calculation for snow and wind combined is CO6a.

Att	Node	Koor X Koor Y		CO1			CO2			CO3			CO4			CO5			CO6			CO6a		
				Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz
Tie	6	-3.00	7.17	-1.55	0.00	-1.19	-1.80	0.01	-1.41	-2.10	0.01	-1.65	-3.32	-0.15	-2.52	-2.73	-0.03	-2.07	-2.54	0.02	-2.02	-2.58	0.02	-2.05
Tie	17	-3.00	3.83	-1.55	0.00	-1.19	-1.80	-0.01	-1.41	-2.10	-0.01	-1.65	-3.32	0.15	-2.52	-2.73	0.03	-2.07	-2.54	-0.02	-2.02	-2.58	-0.02	-2.05
Tie	20	3.75	-2.50	0.00	-1.75	-1.31	0.00	-2.20	-1.71	0.00	-2.60	-2.04	0.00	-2.80	-1.95	0.00	-2.36	-1.72	0.00	-3.15	-2.50	0.00	-3.19	-2.53
Tie	23	10.50	3.83	1.55	0.00	-1.19	1.80	-0.01	-1.41	2.10	-0.01	-1.65	3.31	0.15	-2.51	2.73	0.03	-2.07	2.54	-0.02	-2.02	2.58	-0.02	-2.05
Tie	26	10.50	7.17	1.55	0.00	-1.19	1.80	0.01	-1.41	2.10	0.01	-1.65	3.31	-0.15	-2.51	2.73	-0.03	-2.07	2.54	0.02	-2.02	2.58	0.02	-2.05
Tie	29	3.75	13.50	0.00	1.75	-1.31	0.00	2.20	-1.71	0.00	2.60	-2.04	0.00	2.80	-1.95	0.00	2.36	-1.72	0.00	3.15	-2.50	0.00	3.19	-2.53
Tie	7	-0.52	-1.43	-0.41	-1.51	-1.56	-0.46	-1.70	-1.75	-0.52	-1.93	-1.99	-0.64	-2.33	-2.40	-0.58	-2.14	-2.21	-0.61	-2.23	-2.30	-0.61	-2.26	-2.32
Tie	8	-1.93	-0.02	-1.40	-0.38	-1.44	-1.57	-0.43	-1.62	-1.76	-0.48	-1.82	-2.47	-0.67	-2.54	-1.77	-0.49	-1.83	-2.01	-0.55	-2.08	-2.03	-0.56	-2.10
Tie	9	9.43	-0.02	1.40	-0.38	-1.44	1.57	-0.43	-1.62	1.76	-0.48	-1.82	2.47	-0.67	-2.54	1.77	-0.49	-1.83	2.01	-0.55	-2.08	2.03	-0.56	-2.10
Tie	10	8.02	-1.43	0.41	-1.51	-1.56	0.46	-1.70	-1.75	0.52	-1.93	-1.99	0.64	-2.33	-2.40	0.58	-2.14	-2.21	0.61	-2.23	-2.30	0.61	-2.26	-2.32
Tie	11	8.02	12.43	0.41	1.51	-1.56	0.46	1.70	-1.75	0.52	1.93	-1.99	0.64	2.32	-2.39	0.58	2.14	-2.21	0.61	2.23	-2.30	0.61	2.26	-2.32
Tie	12	9.43	11.02	1.40	0.38	-1.44	1.57	0.43	-1.62	1.76	0.48	-1.82	2.47	0.67	-2.54	1.77	0.49	-1.83	2.01	0.55	-2.08	2.03	0.56	-2.10
Tie	13	-1.93	11.02	-1.40	0.38	-1.44	-1.57	0.43	-1.62	-1.76	0.48	-1.82	-2.46	0.67	-2.53	-1.77	0.49	-1.83	-2.01	0.55	-2.08	-2.03	0.56	-2.10
Tie	14	-0.52	12.43	-0.41	1.51	-1.56	-0.46	1.70	-1.75	-0.52	1.93	-1.99	-0.64	2.33	-2.40	-0.58	2.14	-2.21	-0.61	2.23	-2.30	-0.61	2.26	-2.32
Pole	1	7.15	0.85	0.45	-0.45	2.63	0.54	-0.54	3.18	0.63	-0.62	3.69	0.64	-0.65	3.83	0.52	-0.52	3.07	0.75	-0.74	4.41	0.76	-0.75	4.47
Pole	2	7.15	10.15	0.45	0.45	2.63	0.54	0.54	3.18	0.63	0.62	3.69	0.64	0.65	3.83	0.52	0.51	3.06	0.75	0.74	4.41	0.76	0.75	4.47
Pole	3	0.35	10.15	-0.45	0.45	2.63	-0.54	0.54	3.18	-0.63	0.62	3.69	-0.64	0.64	3.83	-0.52	0.52	3.07	-0.75	0.74	4.41	-0.76	0.75	4.47
Pole	4	0.35	0.85	-0.45	-0.45	2.63	-0.54	-0.54	3.18	-0.63	-0.62	3.69	-0.64	-0.65	3.83	-0.52	-0.51	3.06	-0.75	-0.74	4.41	-0.76	-0.75	4.47
Pole	5	0.50	7.17	-0.20	0.01	1.25	-0.33	0.01	2.01	-0.42	0.01	2.55	-0.09	0.03	0.58	-0.13	0.00	0.83	-0.56	0.00	3.46	-0.58	0.00	3.53
Pole	16	0.50	3.83	-0.20	-0.01	1.25	-0.33	-0.01	2.01	-0.42	-0.01	2.55	-0.09	-0.03	0.58	-0.13	0.00	0.83	-0.56	0.00	3.46	-0.58	0.00	3.53
Pole	19	3.75	1.00	0.00	-0.25	1.51	0.00	-0.42	2.60	0.00	-0.54	3.31	0.00	-0.38	2.31	0.00	-0.11	0.69	0.00	-0.72	4.46	0.00	-0.73	4.56
Pole	22	7.00	3.83	0.20	-0.01	1.25	0.33	-0.01	2.01	0.42	-0.01	2.55	0.09	-0.03	0.58	0.13	0.00	0.83	0.56	0.00	3.46	0.58	0.00	3.53
Pole	25	7.00	7.17	0.20	0.01	1.25	0.33	0.01	2.01	0.42	0.01	2.55	0.09	0.03	0.58	0.13	0.00	0.83	0.56	0.00	3.46	0.58	0.00	3.53
Pole	28	3.75	10.00	0.00	0.25	1.51	0.00	0.42	2.60	0.00	0.54	3.31	0.00	0.38	2.28	0.00	0.11	0.69	0.00	0.72	4.46	0.00	0.73	4.56
Pole	99000023	3.75	5.50	0.00	0.00	2.22	0.00	0.00	4.76	0.00	0.00	5.95	0.00	0.00	0.10	0.00	0.00	0.26	0.00	0.00	8.61	0.00	0.00	8.83
Stormbelt	150010001	2.75	-1.50										0.18	-3.13	-2.87									
Stormbelt	150010021	2.76	12.50										0.18	3.13	-2.87									
Stormbelt	150020001	4.75	-1.49										-0.18	-3.10	-2.84									
Stormbelt	150020020	4.79	12.49										-0.17	3.11	-2.83									

Annex G. Reaction forces model 7.5x15m

Annex G.1. Point numbers



Annex G.2. Reaction forces

Reaction forces are shown in kilonewtons [kN] without any safety factor (1.35, 1.5, 1.2). A load combination especially for anchors and ballast calculation for snow and wind combined is CO6a.

Att	Node	Koor X Koor Y		CO1			CO2			CO3			CO4			CO5			CO6			CO6a		
				Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz
Tie	14	3.75	-7.50	0.00	-1.77	-1.32	0.00	-2.45	-1.90	0.00	-3.03	-2.37	0.01	-3.33	-2.41	0.00	-2.42	-1.76	-0.01	-3.80	-3.02	-0.01	-3.86	-3.07
Tie	17	3.75	13.50	0.00	1.76	-1.33	0.00	2.44	-1.91	0.00	3.02	-2.38	0.00	3.32	-2.42	0.00	2.41	-1.77	-0.01	3.79	-3.02	-0.01	3.85	-3.08
Tie	20	-3.00	-0.75	-1.73	0.01	-1.35	-2.02	-0.01	-1.60	-2.38	-0.02	-1.90	-3.61	0.02	-2.78	-2.88	0.03	-2.25	-2.83	-0.04	-2.27	-2.87	-0.04	-2.30
Tie	23	-3.00	3.00	-1.71	0.00	-1.39	-1.86	0.00	-1.52	-2.21	0.00	-1.80	-3.08	0.01	-2.43	-3.24	0.00	-2.63	-2.63	0.00	-2.16	-2.67	0.00	-2.19
Tie	26	-3.00	6.75	-1.72	-0.01	-1.34	-1.99	0.01	-1.58	-2.35	0.02	-1.86	-3.65	-0.01	-2.81	-2.94	-0.03	-2.29	-2.79	0.03	-2.23	-2.82	0.04	-2.26
Tie	29	10.50	-0.75	1.73	0.00	-1.35	2.03	-0.01	-1.60	2.41	-0.02	-1.90	3.63	0.02	-2.77	2.88	0.03	-2.24	2.87	-0.04	-2.29	2.91	-0.04	-2.32
Tie	32	10.50	3.00	1.72	0.00	-1.39	1.88	0.00	-1.52	2.24	0.00	-1.81	3.07	0.00	-2.40	3.24	0.00	-2.62	2.68	0.00	-2.18	2.71	0.00	-2.21
Tie	35	10.50	6.75	1.72	-0.01	-1.34	2.01	0.01	-1.58	2.37	0.02	-1.87	3.67	-0.02	-2.80	2.95	-0.03	-2.28	2.82	0.04	-2.24	2.86	0.04	-2.27
Tie	5	-0.52	-6.43	-0.42	-1.53	-1.58	-0.47	-1.74	-1.79	-0.55	-2.03	-2.09	-0.75	-2.76	-2.83	-0.60	-2.21	-2.27	-0.65	-2.37	-2.44	-0.66	-2.40	-2.47
Tie	6	-1.93	-5.02	-1.49	-0.41	-1.54	-1.73	-0.47	-1.79	-1.97	-0.54	-2.03	-2.33	-0.64	-2.41	-1.96	-0.54	-2.02	-2.25	-0.62	-2.32	-2.27	-0.62	-2.34
Tie	7	9.43	-5.02	1.50	-0.41	-1.55	1.74	-0.47	-1.80	1.98	-0.54	-2.04	2.35	-0.65	-2.42	1.96	-0.54	-2.02	2.27	-0.62	-2.34	2.29	-0.63	-2.36
Tie	8	8.02	-6.43	0.41	-1.51	-1.56	0.47	-1.72	-1.77	0.55	-2.02	-2.08	0.74	-2.73	-2.81	0.59	-2.18	-2.25	0.65	-2.37	-2.44	0.65	-2.40	-2.47
Tie	9	8.02	12.43	0.41	1.50	-1.55	0.47	1.71	-1.76	0.55	2.01	-2.07	0.73	2.68	-2.76	0.59	2.17	-2.24	0.64	2.35	-2.42	0.65	2.38	-2.45
Tie	10	9.43	11.02	1.53	0.41	-1.57	1.76	0.48	-1.81	1.99	0.54	-2.05	2.36	0.65	-2.43	1.99	0.54	-2.05	2.27	0.62	-2.34	2.29	0.63	-2.36
Tie	11	-1.93	11.02	-1.52	0.41	-1.56	-1.74	0.47	-1.80	-1.97	0.54	-2.03	-2.35	0.65	-2.42	-1.99	0.54	-2.05	-2.25	0.62	-2.32	-2.27	0.62	-2.34
Tie	12	-0.52	12.43	-0.41	1.52	-1.57	-0.47	1.72	-1.78	-0.55	2.01	-2.07	-0.75	2.77	-2.85	-0.60	2.19	-2.26	-0.64	2.35	-2.42	-0.65	2.37	-2.44
Pole	1	7.15	-4.15	0.46	-0.46	2.72	0.57	-0.57	3.36	0.68	-0.67	3.98	0.73	-0.72	4.33	0.54	-0.54	3.18	0.81	-0.81	4.79	0.82	-0.82	4.86
Pole	2	7.15	10.15	0.46	0.47	2.73	0.57	0.57	3.36	0.67	0.67	3.98	0.71	0.70	4.23	0.54	0.54	3.20	0.81	0.80	4.78	0.82	0.81	4.84
Pole	3	0.35	10.15	-0.47	0.47	2.73	-0.57	0.57	3.36	-0.67	0.67	3.96	-0.73	0.72	4.32	-0.54	0.54	3.20	-0.80	0.80	4.76	-0.81	0.81	4.82
Pole	4	0.35	-4.15	-0.47	-0.46	2.72	-0.57	-0.57	3.36	-0.67	-0.67	3.97	-0.73	-0.72	4.31	-0.54	-0.54	3.20	-0.81	-0.80	4.77	-0.82	-0.81	4.84
Pole	13	3.75	-4.00	0.00	-0.25	1.51	0.00	-0.44	2.73	0.00	-0.58	3.56	0.00	-0.14	0.86	0.00	-0.11	0.71	0.00	-0.78	4.86	0.00	-0.80	4.97
Pole	16	3.75	10.00	0.00	0.25	1.56	0.00	0.46	2.82	0.00	0.59	3.67	0.00	0.15	0.93	0.00	0.12	0.77	0.00	0.80	5.01	0.00	0.82	5.12
Pole	19	0.50	-0.75	-0.21	-0.01	1.30	-0.35	-0.01	2.12	-0.44	0.00	2.72	-0.20	-0.01	1.22	-0.13	0.00	0.78	-0.59	0.01	3.64	-0.60	0.01	3.72
Pole	22	0.50	3.00	-0.18	0.00	1.12	-0.30	0.00	1.83	-0.39	0.00	2.41	-0.31	0.00	1.89	-0.10	0.00	0.64	-0.53	0.00	3.28	-0.55	0.00	3.35
Pole	25	0.50	6.75	-0.21	0.01	1.30	-0.35	0.01	2.11	-0.44	0.00	2.71	-0.21	0.01	1.30	-0.13	0.01	0.79	-0.59	-0.01	3.64	-0.60	-0.01	3.71
Pole	28	7.00	-0.75	0.20	-0.01	1.26	0.33	-0.01	2.04	0.43	0.00	2.62	0.20	-0.01	1.25	0.12	0.00	0.74	0.57	0.01	3.52	0.58	0.01	3.59
Pole	31	7.00	3.00	0.18	0.00	1.09	0.29	0.00	1.77	0.38	0.00	2.33	0.28	0.00	1.71	0.10	0.00	0.62	0.52	0.00	3.18	0.53	0.00	3.25
Pole	34	7.00	6.75	0.20	0.01	1.26	0.33	0.01	2.04	0.43	0.00	2.62	0.21	0.01	1.30	0.12	0.01	0.76	0.57	-0.01	3.52	0.58	-0.01	3.59
Pole	99000029	3.83	0.42	0.00	0.00	1.98	-0.01	0.01	4.43	-0.02	0.03	5.63	0.00	0.00	0.04	0.00	0.00	0.14	-0.04	0.08	8.22	-0.04	0.08	8.43
Pole	99000030	3.83	5.42	0.00	0.00	1.98	-0.01	0.00	4.43	-0.02	-0.01	5.63	0.00	0.00	0.04	0.00	0.00	0.13	-0.04	-0.04	8.24	-0.04	-0.04	8.45
Stormbelt	150010001	9.31	-2.83										2.62	0.00	-2.38									
Stormbelt	150010015	-1.81	-2.75										-2.62	0.04	-2.37									
Stormbelt	150020001	-1.91	1.16										-2.72	-0.89	-2.63									
Stormbelt	150020016	9.38	4.95										2.71	0.92	-2.63									
Stormbelt	150030001	-1.86	5.02										-2.65	0.98	-2.60									
Stormbelt	150030016	9.32	0.83										2.66	-0.98	-2.60									
Stormbelt	150040001	-1.81	8.73										-2.71	0.02	-2.45									
Stormbelt	150040015	9.32	8.58										2.71	-0.06	-2.45									